Human Factors and Cognitive Psychology

Learning Objectives



Explain what human factors means in computing



Describe how cognitive processes affect interaction with systems



Identify **design problems caused by cognitive overload**



Apply psychology-based principles to improve interface design

Why Psychology Matters in HCI

Human Factors + Cognitive Psychology help us design systems that match **how people actually think**, not how engineers *wish* they thought.

What Are Human Factors?

Human Factors = designing systems that fit **human abilities and limitations**

It studies:

- Physical abilities (vision, motor skills)
- Mental abilities (memory, attention, decision-making)
- Emotional responses (stress, frustration, trust)

GOAL

Reduce human error and increase efficiency, safety, and satisfaction.

Cognitive Psychology

Cognitive psychology focuses on how the mind processes information.

Process	What It Means	Design Impact
Perception	How users interpret visual info	Color, contrast, layout matter
Attention	What users focus on	Avoid clutter
Memory	How users remember info	Don’t overload memory
Learning	How users get better over time	Make systems predictable
Decision-Making	How users choose actions	Clear options reduce errors

Human Memory

Type	Duration	Example	HCI Design Tip
Sensory Memory	milliseconds	Seeing a notification flash	Use visual cues effectively
Short-Term (Working) Memory	~20 seconds, 7±2 items	Remembering a verification code	Keep steps simple
Long-Term Memory	Unlimited	Remembering how to use Facebook	Use familiar patterns

Golden Rule

Don’t make users remember; make them recognize!

Cognitive Load

Cognitive Load = the total amount of mental effort and working memory resources used at any given time, impacting learning and task performance

3 Types:

- **Intrinsic Load** – Difficulty of the task itself
- **Extraneous Load** – Poor design making things harder
- **Germane Load** – Mental effort that helps learning

Today in Finance

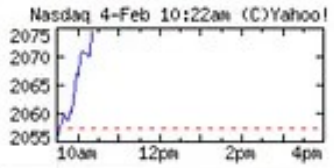
Sunday, January 30, 2005, 10:34PM ET - **U.S. Markets Closed.**

Enter Symbol(s): e.g. YHOO, ^DJI GO Symbol Lookup | Finance Search

Mutual Funds | Bill Pay | Banking | Loans | Insurance | Taxes

Market Summary

Edit



Dow	10,427.20	-40.20 (-0.38%)
Nasdaq	2,035.83	-11.32 (-0.55%)
S&P 500	1,171.36	-3.19 (-0.27%)
10-Yr Bond	4.138%	-0.069
NYSE Volume	1,642,010,000	
Nasdaq Volume	2,101,825,000	

Brokers:

Investing

Today's Markets
Overview, Indices, Actives...

Stock Research
Reports, Screener, Financials...

Financial News
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Options

Top Stories

Report: SBC to Buy AT&T for $16 Billion

Sun 9:25PM ET - Associated Press

Local phone giant SBC Communications Inc. has agreed to acquire AT&T Corp. for roughly $16 billion in cash and stock, The Wall Street Journal reported Sunday.

Most Actives, US Indices, World Indices

- **Market Overview: Sat 10:03 AM ET** - Briefing.com
- OPEC Warns Oil Prices Will Remain High - Associated Press
- Iraq Vote, Fed Meeting May Set Market Tone - Associated Press
- Tokyo Stocks Down in Morning Trading - Associated Press
- Dollar Continues to Rise in Asian Trade - Associated Press

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The psychology of splurging
Bankrate.com
Dear Debt Adviser, Do you have any statistics on spontaneous consumer-spending behavior? Why do we spend the way we do?

Money Matters

The Case For The Prenup
by Suze Orman
If you believe in insurance to protect your life, your health, your home, your car—why don't you also believe in marriage insurance?

more Money Matters Topics...

Investing Ideas

Three Stocks You Need To Know About For Monday
TradingMarkets.com
Here's your hit list for the day.

Return of The Kings
Forbes Magazine
Investors who stuck with the Internet boom are starting to make big money--finally.

Still Sweet On Dividend Stocks
BusinessWeek Online

more Investing Ideas...

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Google Search Engine

This is a demo of the Google Search Engine. Note, it is research in progress so expect some downtimes and malfunctions. You can find the older Backrub web page here.

Google is being developed by Larry Page and Sergey Brin with very talented implementation help by Scott Hassan and Alan Steremberg.



Search Stanford

10 results | clustering on | Search

Search The Web

10 results | clustering on | Search

Current Status of Google:

Web Page Statistics	
Number of Web Pages Fetched	24 million
Number of Urls Seen	76.5 million
Number of Email Addresses	1.7 million
Number of 404's	1.6 million

Storage Statistics

Attention & Distraction

Humans have **limited attention**

The brain cannot truly multitask; it just switches rapidly between tasks.

Design Implications:

- Too many popups = missed information
- Flashy ads = reduced task focus

✔ **Use visual hierarchy**
✔ **Highlight important actions**
✔ **Remove unnecessary elements**

Mental Models

A mental model = how users think a system works

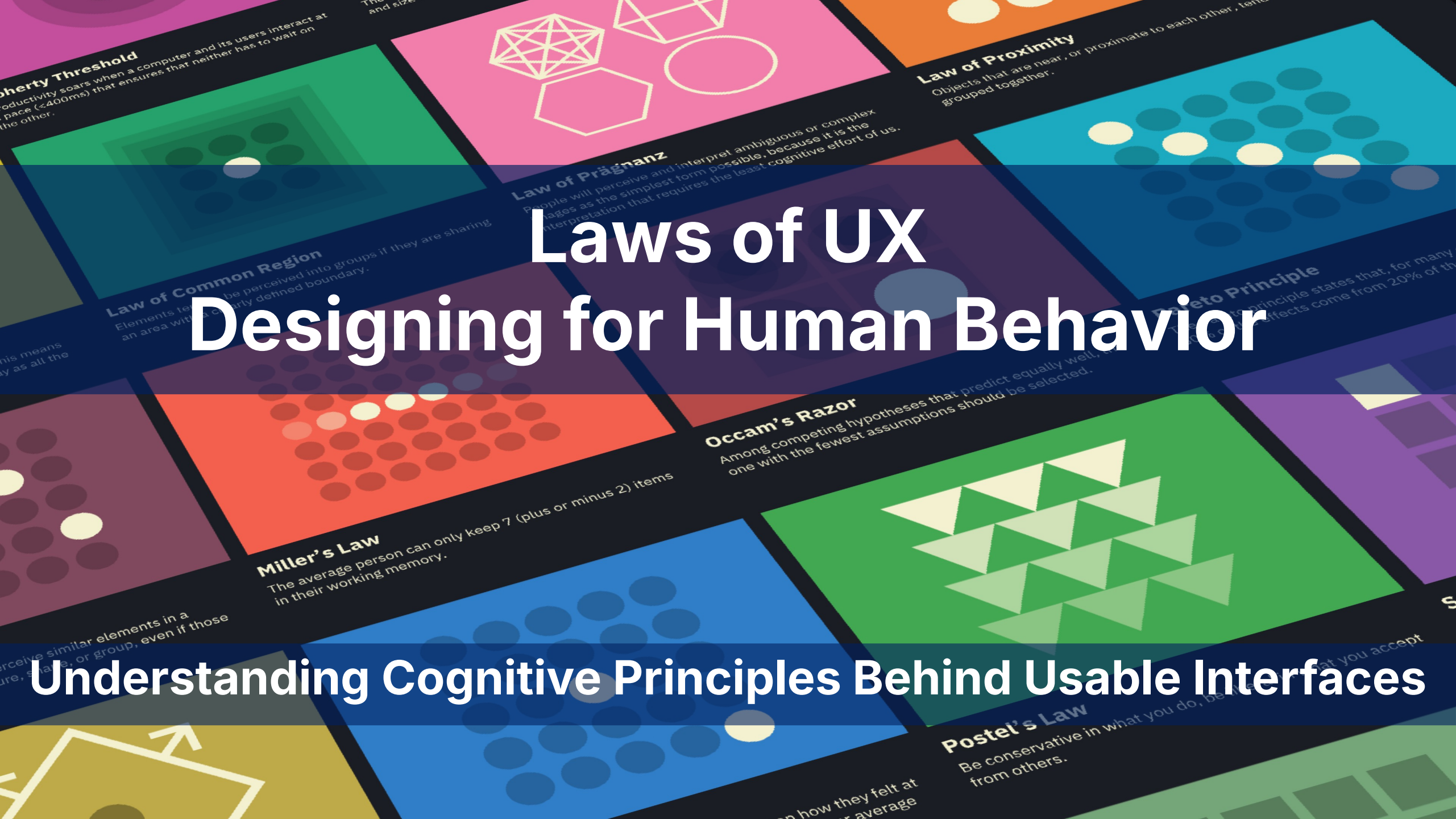




Good design matches user mental models
Bad design forces users to learn new logic

Human Error

Type	Description	Example	Design Solution
Slips	Correct intention, wrong action	Clicking "Delete" instead of "Save"	Undo, confirmation, space risky buttons apart
Mistakes	Wrong understanding	Formatting the wrong drive, thinking it's the USB	Clear labeling, warnings



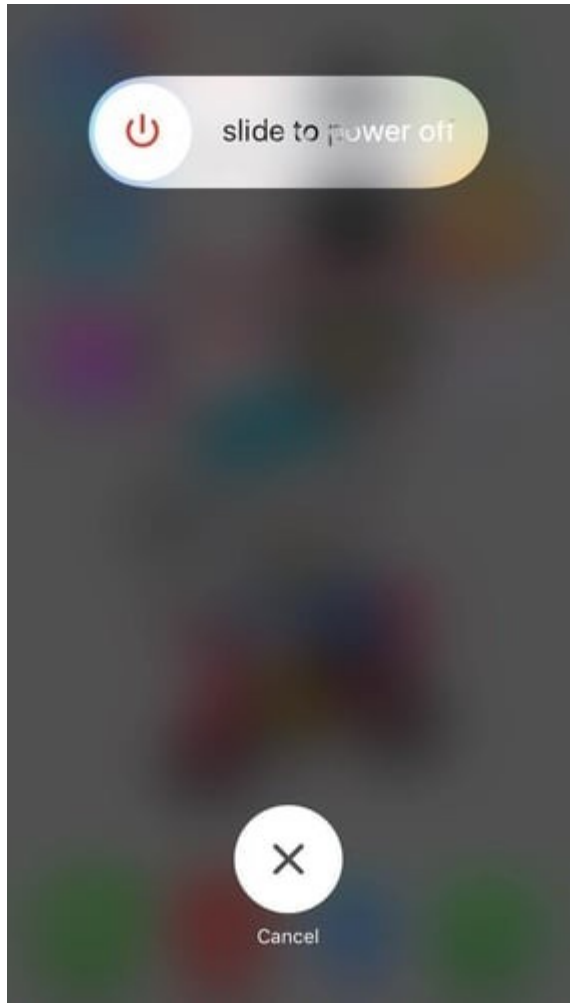
Laws of UX
Designing for Human Behavior

Understanding Cognitive Principles Behind Usable Interfaces

Fitts’s Law (Movement Time)

Time to click = depends on **distance + size of target**





Laws of UX

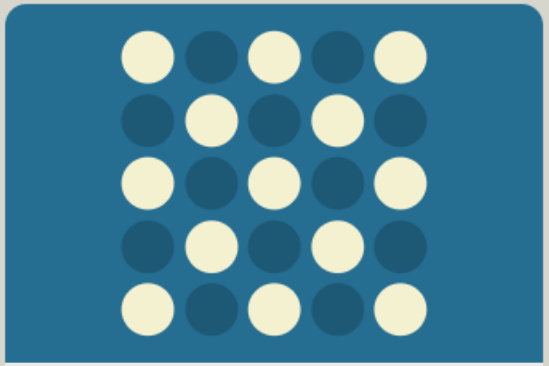
The "Laws of UX" are a collection of psychological principles, behavioral science, data science, and HCI that explain how users interact with digital interfaces





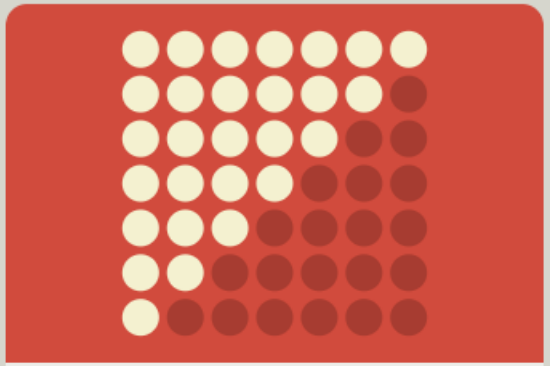
Aesthetic-Usability Effect

Users often perceive aesthetically pleasing design as design that's more usable.



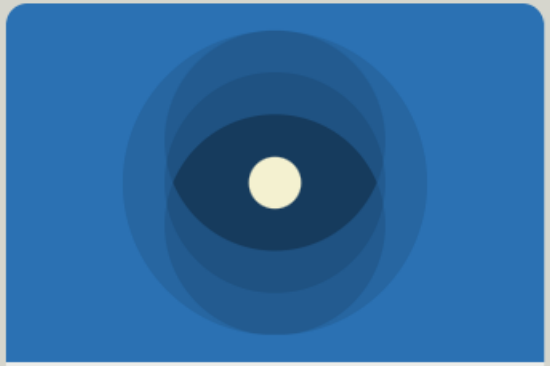
Choice Overload

The tendency for people to get overwhelmed when they are presented with a large number of options, often used interchangeably with the term paradox of choice.



Chunking

A process by which individual pieces of an information set are broken down and then grouped together in a meaningful whole.



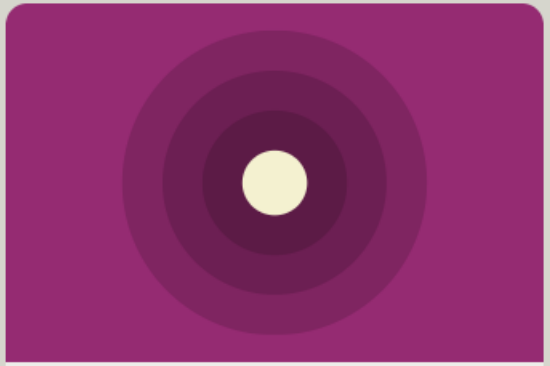
Cognitive Bias

A systematic error of thinking or rationality in judgment that influence our perception of the world and our decision-making ability.



Cognitive Load

The amount of mental resources needed to understand and interact with an interface.



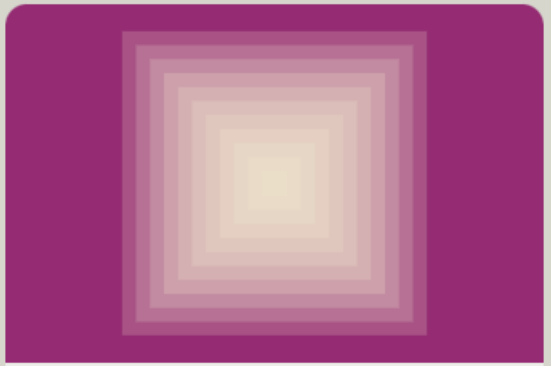
Doherty Threshold

Productivity soars when a computer and its users interact at a pace (<400ms) that ensures that neither has to wait on the other.



Fitts's Law

The time to acquire a target is a function of the distance to and size of the target.



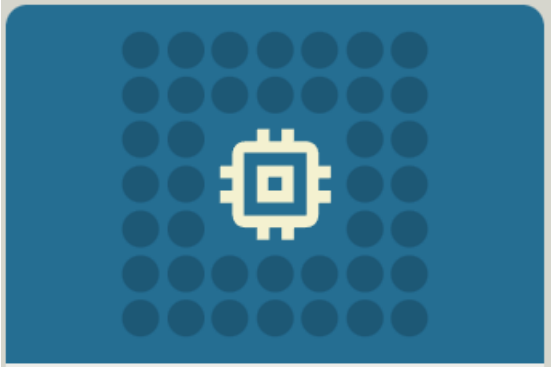
Flow

The mental state in which a person performing some activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment in the process of the activity.



Goal-Gradient Effect

The tendency to approach a goal increases with proximity to the goal.



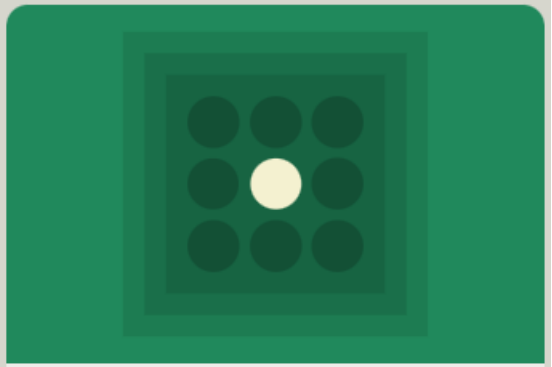
Hick's Law

The time it takes to make a decision increases with the number and complexity of choices.



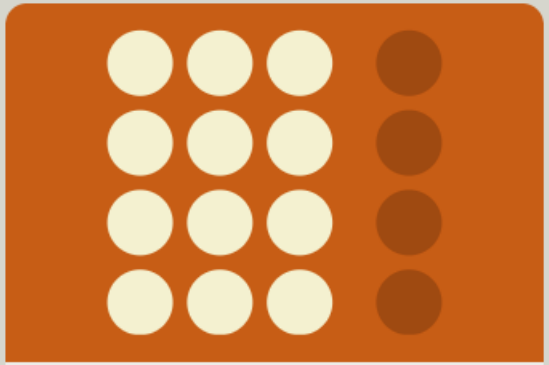
Jakob's Law

Users spend most of their time on other sites. This means that users prefer your site to work the same way as all the other sites they already know.



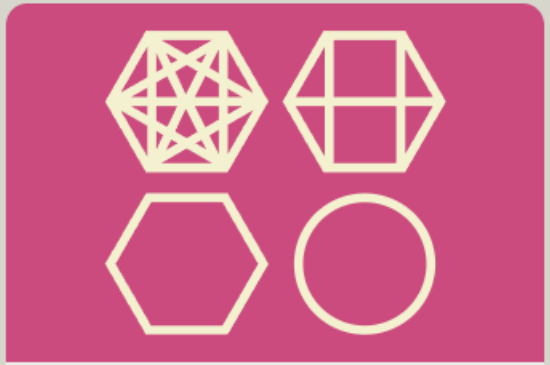
Law of Common Region

Elements tend to be perceived into groups if they are sharing an area with a clearly defined boundary.



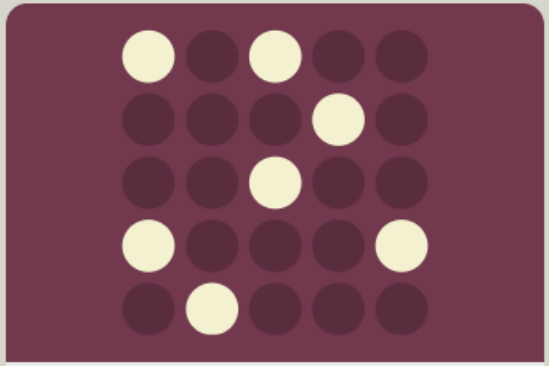
Law of Proximity

Objects that are near, or proximate to each other, tend to be grouped together.



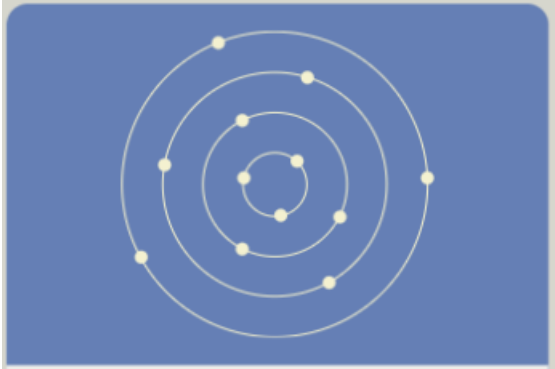
Law of Prägnanz

People will perceive and interpret ambiguous or complex images as the simplest form possible, because it is the interpretation that requires the least cognitive effort of us.



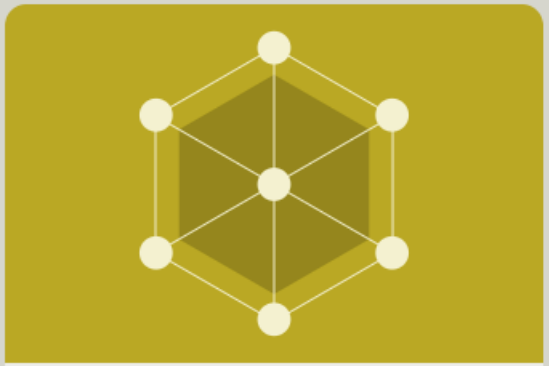
Law of Similarity

The human eye tends to perceive similar elements as a complete picture, shape, or group, even if those elements are separated.



Law of Uniform Connectedness

Elements that are visually connected are perceived as more related than elements with no connection.



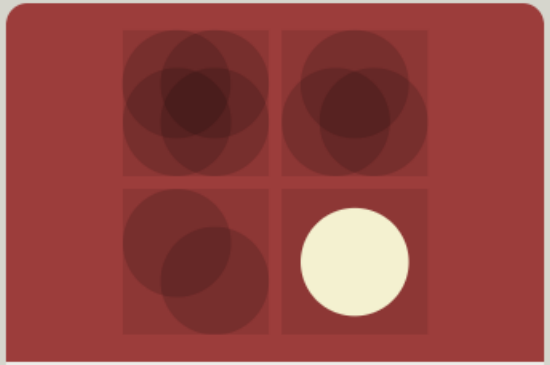
Mental Model

A compressed model based on what we think we know about a system and how it works.



Miller's Law

The average person can only keep 7 (plus or minus 2) items in their working memory.



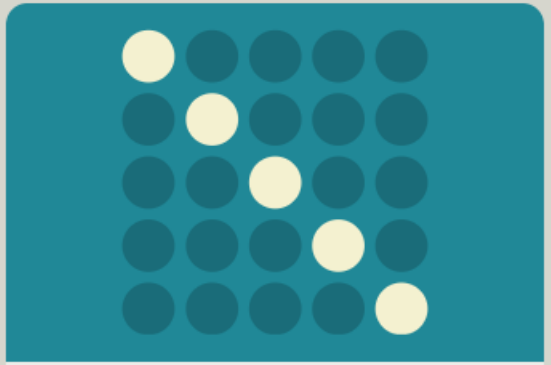
Occam's Razor

Among competing hypotheses that predict equally well, the one with the fewest assumptions should be selected.



Paradox of the Active User

Users never read manuals but start using the software immediately.



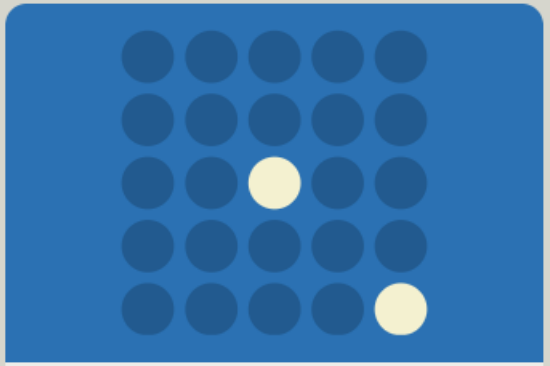
Pareto Principle

The Pareto principle states that, for many events, roughly 80% of the effects come from 20% of the causes.



Parkinson's Law

Any task will inflate until all of the available time is spent.



Peak-End Rule

People judge an experience largely based on how they felt at its peak and at its end, rather than the total sum or average of every moment of the experience.



Postel's Law

Be liberal in what you accept, and conservative in what you send.

Lesson Objectives



1. Explain key principles of the Laws of UX.



2. Identify real-world examples of UX laws in practice.



3. Apply UX laws to improve interface design.

Introduction



UX designers create experiences based on how humans perceive and interact with interfaces.



Laws of UX combine psychology, design heuristics, and human behavior.

The Laws of UX

- **Heuristics**: These laws include rules of thumb that can help with decision-making and problem-solving.

- **Gestalt principles**: These laws deal with how the human brain organizes and perceives visual elements as unified wholes.

- **Cognitive biases**: These rules are about the ingrained mental tendencies that cause systematic patterns of deviation from rational judgment.

- **Principles:** These include general guidelines and best practices borrowed from disciplines like psychology and design.

The Laws of UX

- **Heuristics**: Parkinson’s Law, Miller’s Law, Hick’s Law, Aesthetic-Usability Effect, Fitts’s Law, Jakob’s Law, Paradox of the Active User, Goal-Gradient Effect

- **Gestalt principles**: Law of Prägnanz, Law of Common Region, Law of Proximity, Law of Similarity, Law of Uniform Connectedness

- **Cognitive biases:** Peak-End Rule, Serial Position Effect, Von Restorff Effect, Zeigarnik Effect

- **Principles:** Doherty Threshold, Occam's Razor, Tesler's Law, Postel's Law

Why Laws of UX Matter

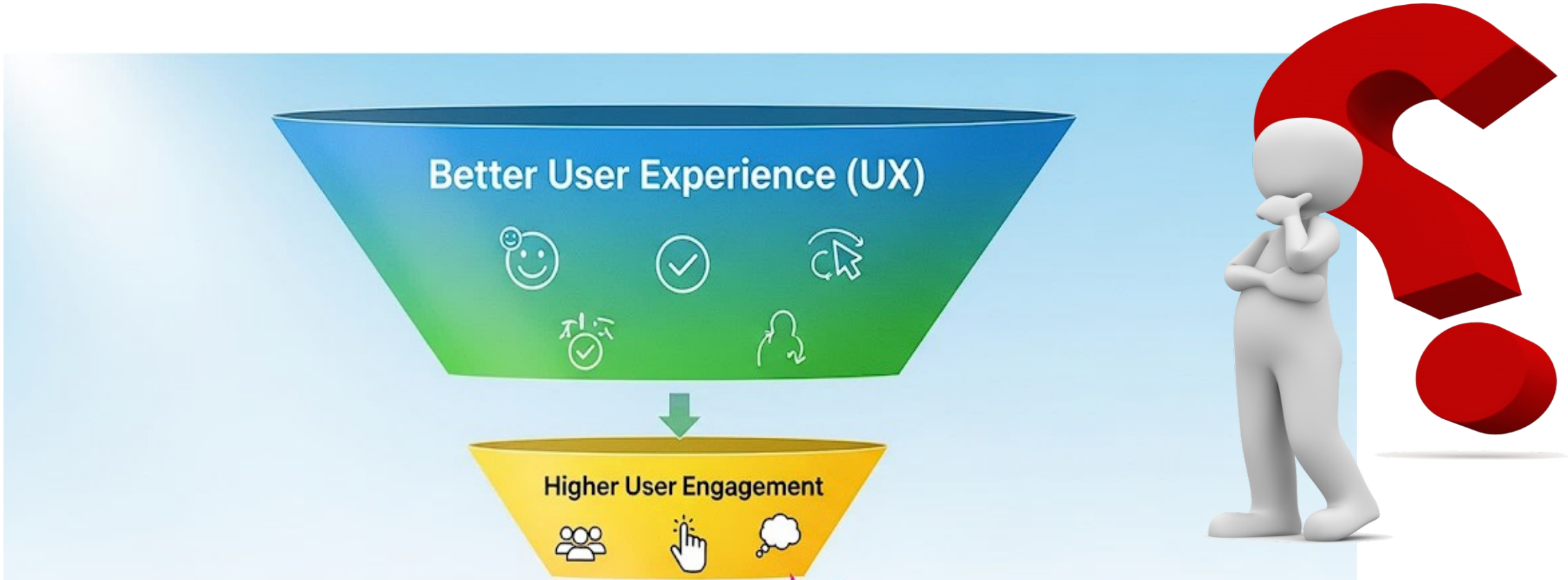


Why Laws of UX Matter

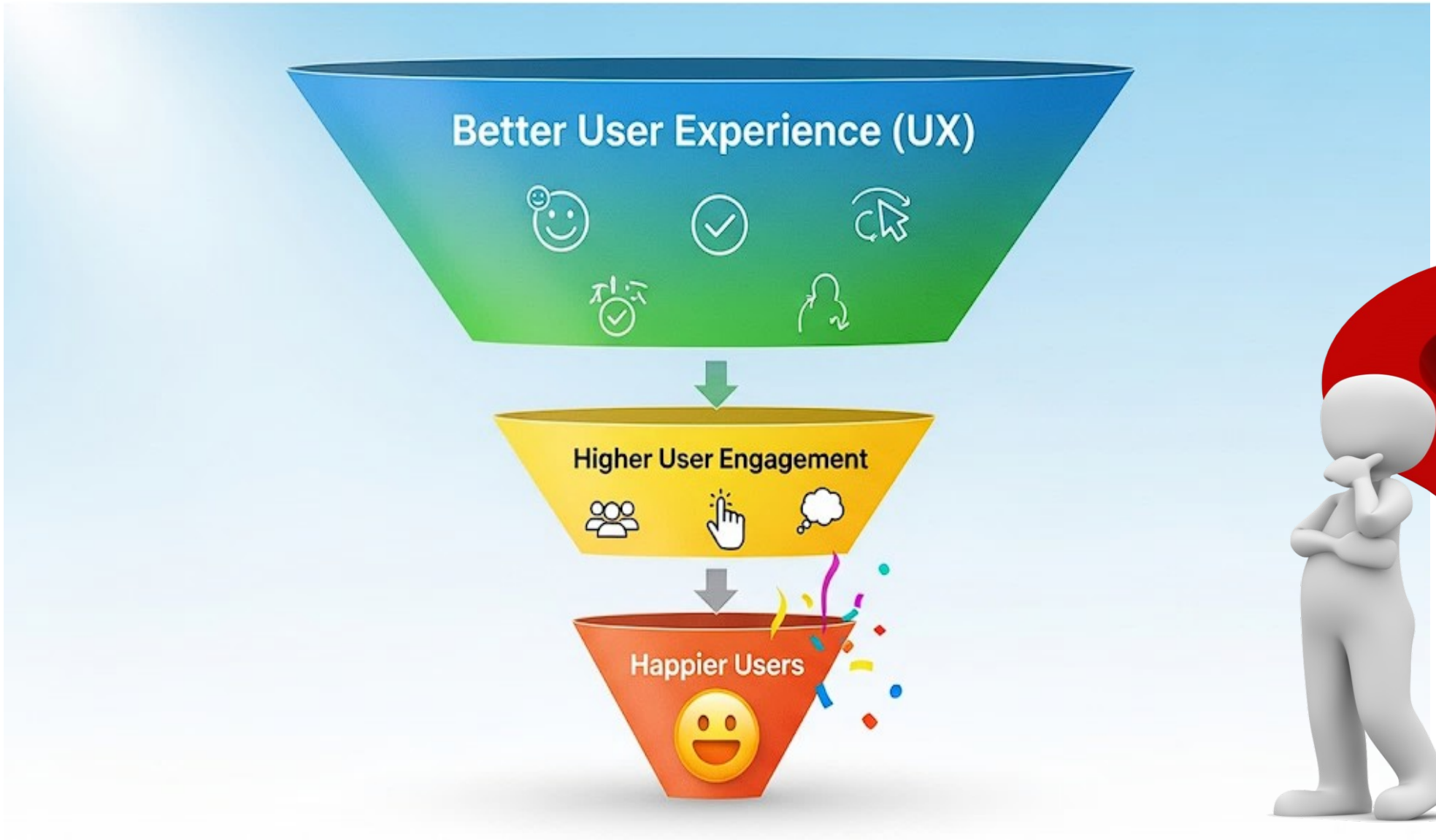




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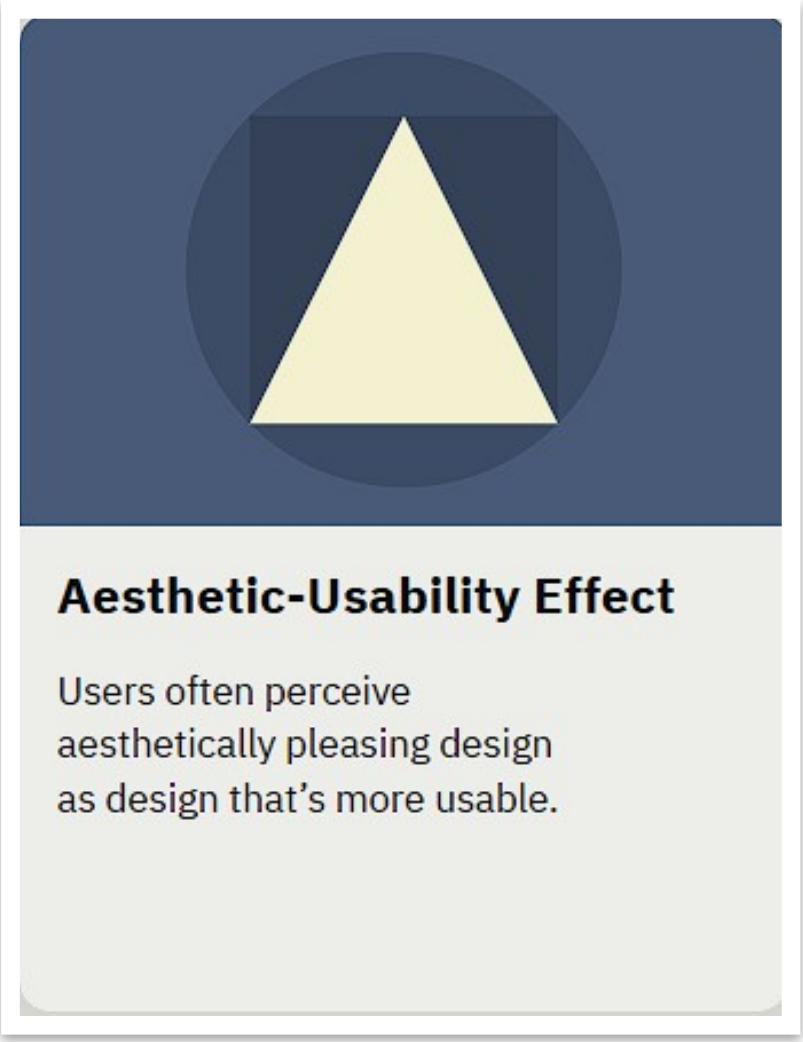


Why Laws of UX Matter





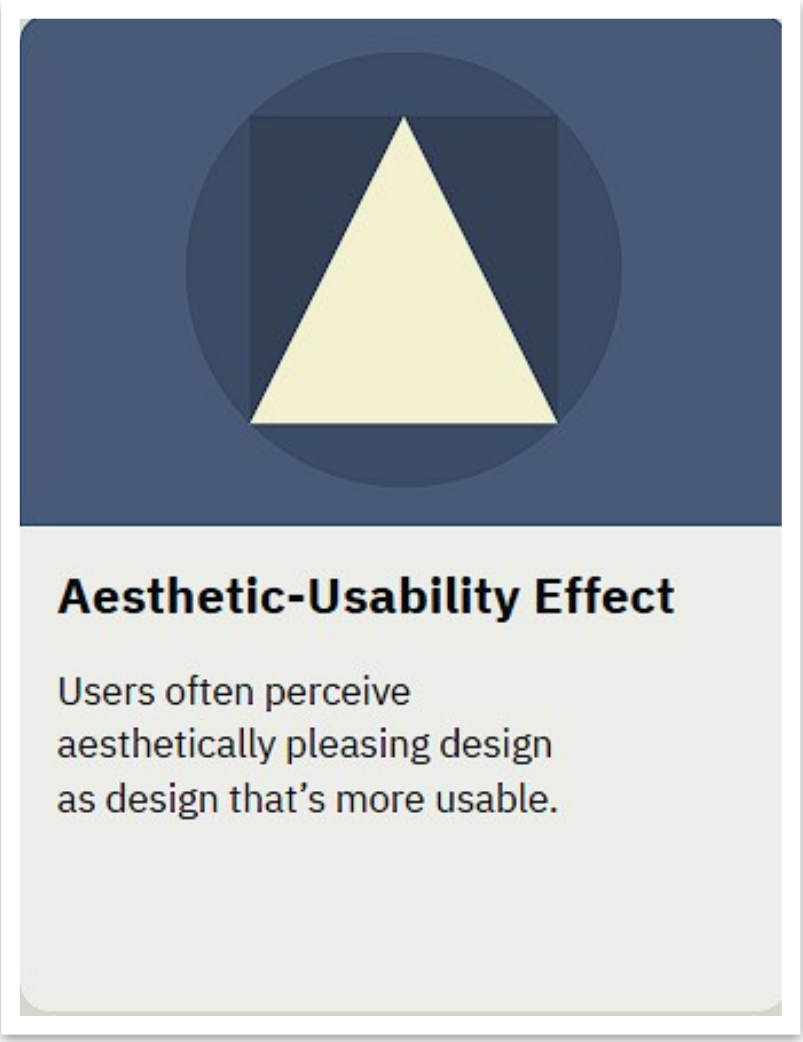
Aesthetic-Usability Effect



People tend to believe that things that look better will work better, even if they aren't more effective or efficient.

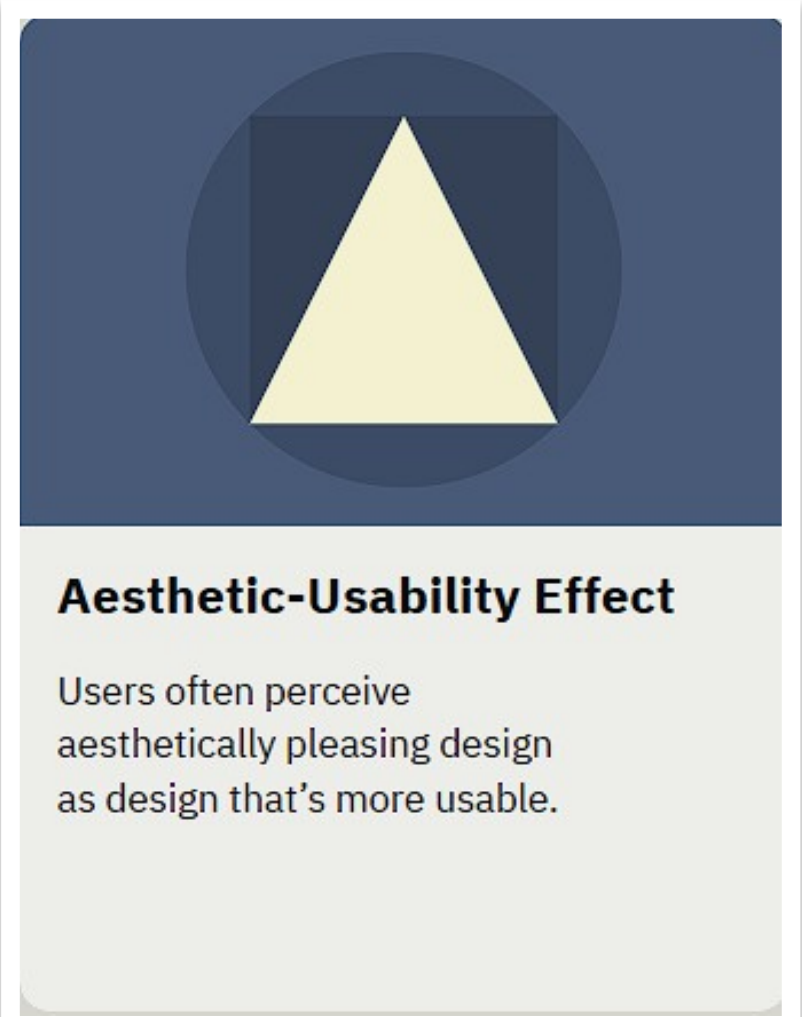
When users have a positive emotional response to visual design, it makes them more tolerant of minor usability issues.

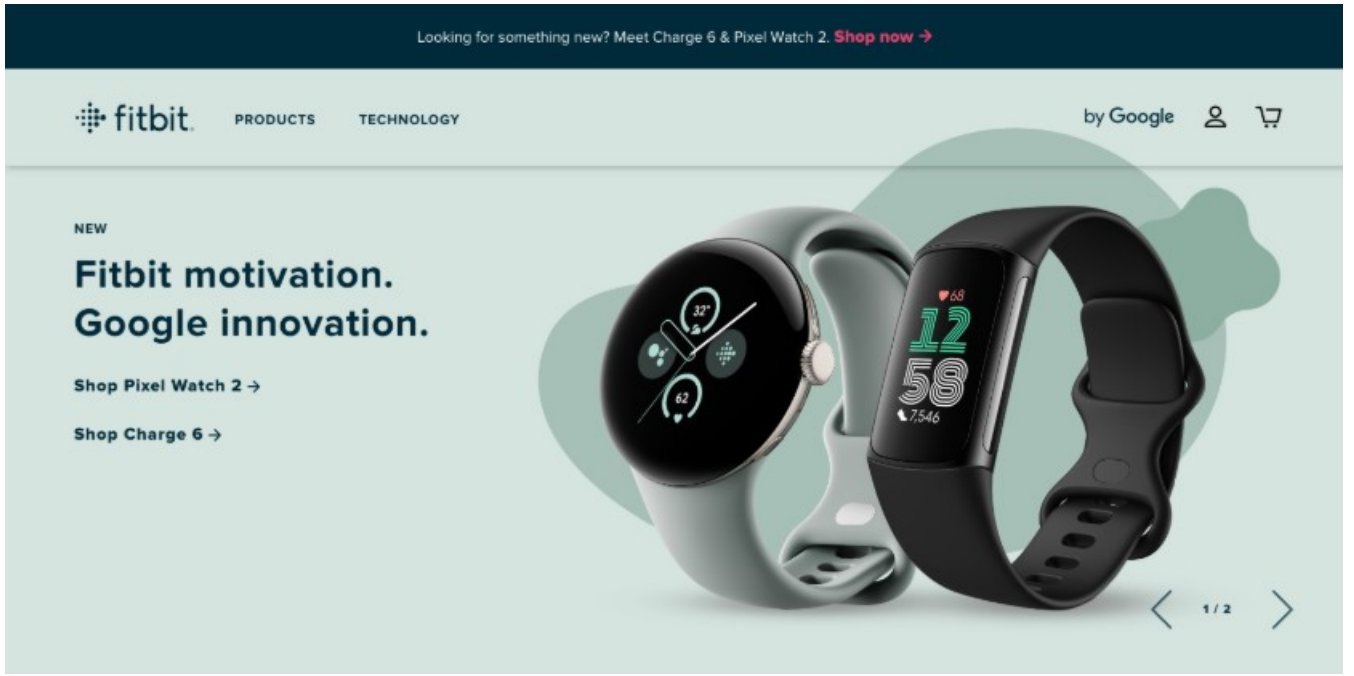
Aesthetic-Usability Effect





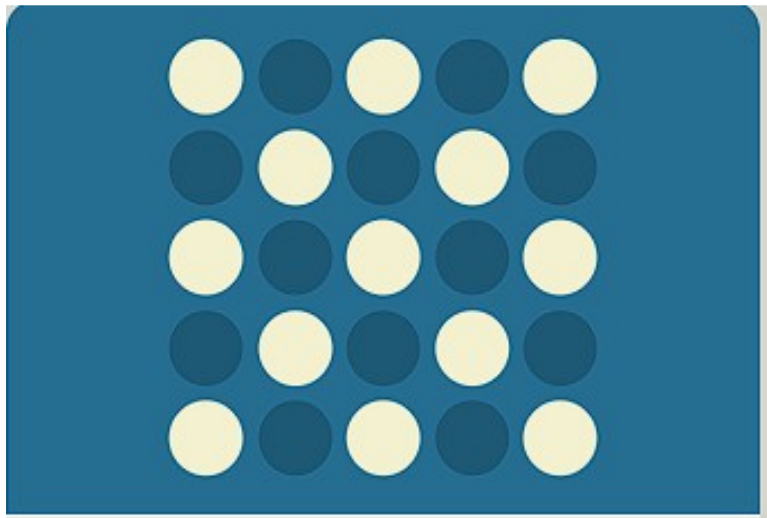
Aesthetic-Usability Effect





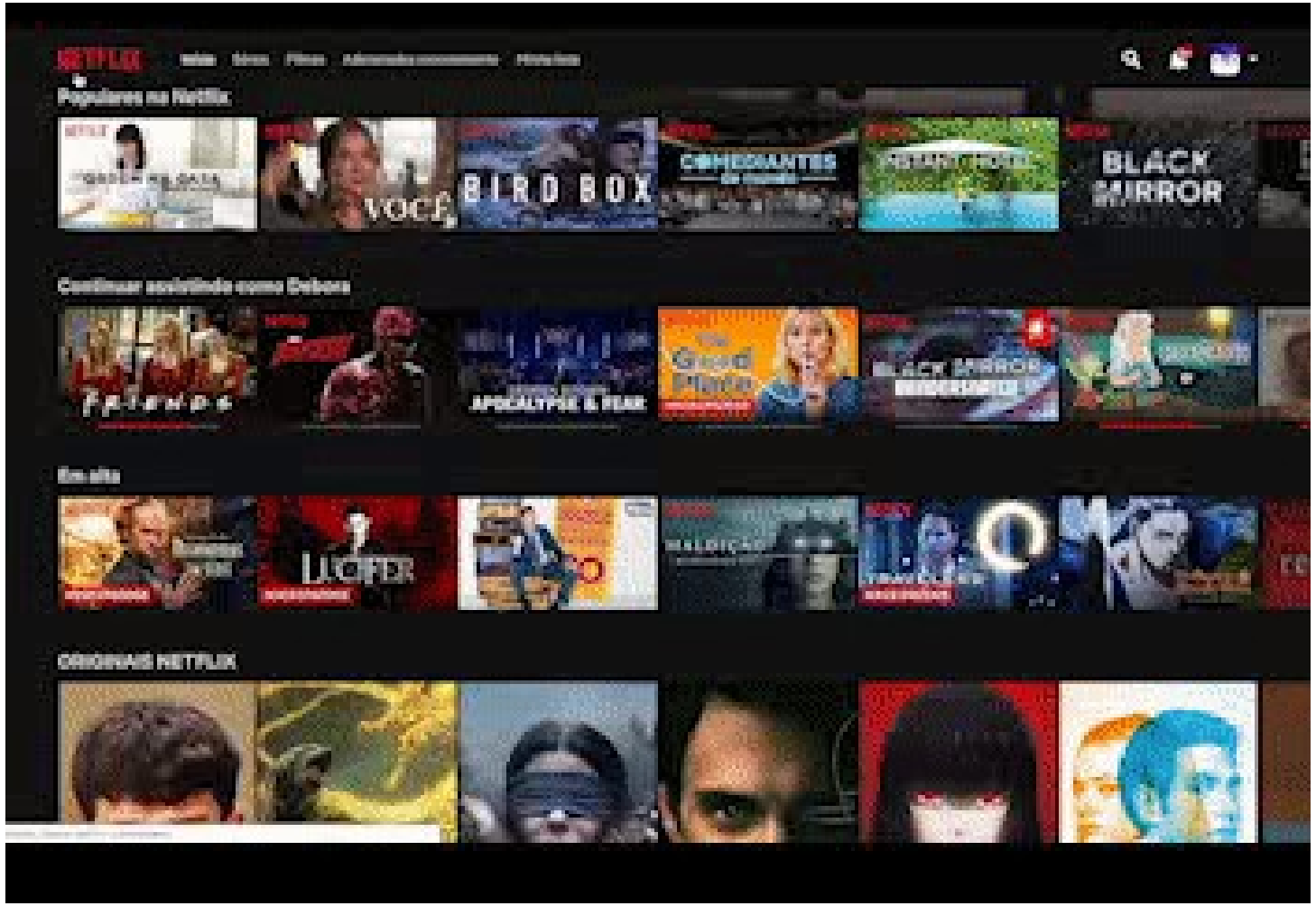
Despite encountering many problems while shopping on FitBit's site, a study participant still rated it highly. When asked for feedback, she commented only about its visual design — hindering the researcher from learning about the design's problems.

Choice Overload

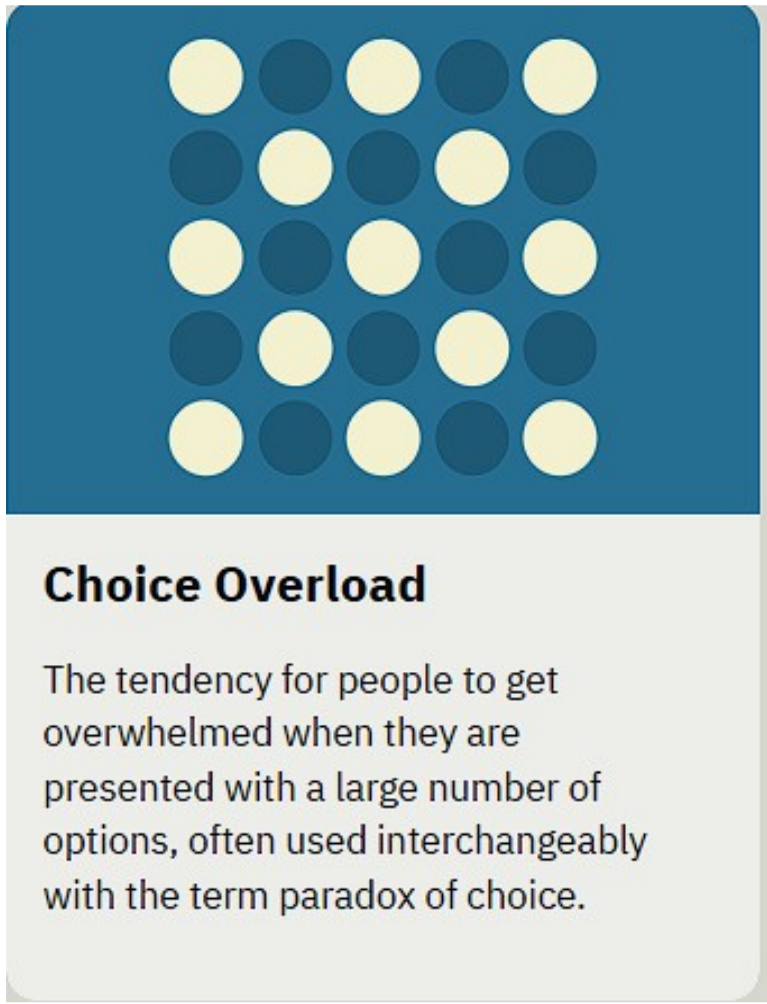


Choice Overload

The tendency for people to get overwhelmed when they are presented with a large number of options, often used interchangeably with the term paradox of choice.



Choice Overload



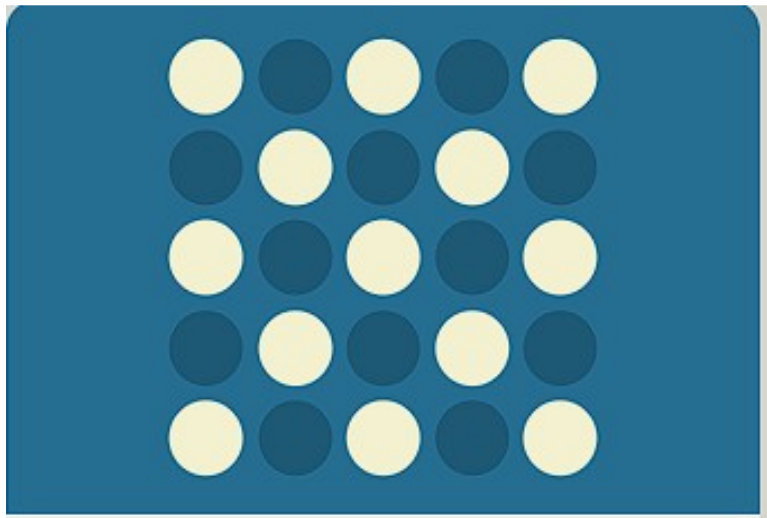
SCENARIO:

A user visits an online clothing store looking for a pair of jeans. The site presents them with 150 different styles, cuts, and brands.

SOLUTION:

Implement filtering and sorting options to help users narrow down choices based on their preferences. For example, filters for size, brand, color, and price can streamline the decision-making process.

Choice Overload



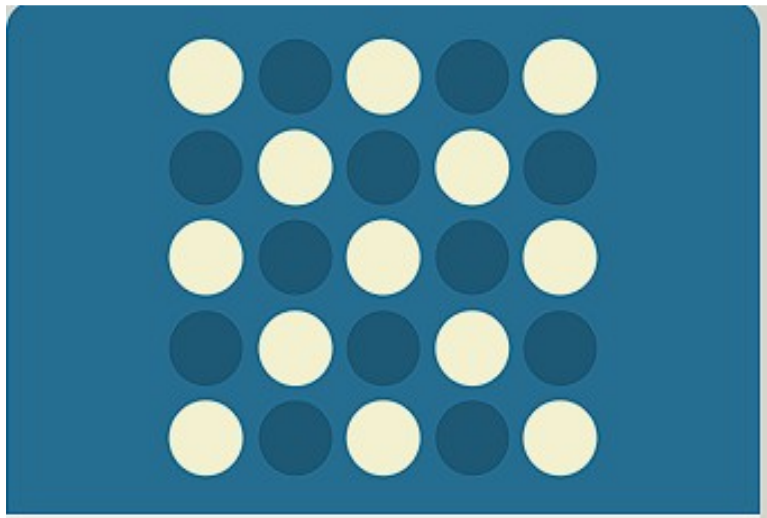
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BEST PRACTICES

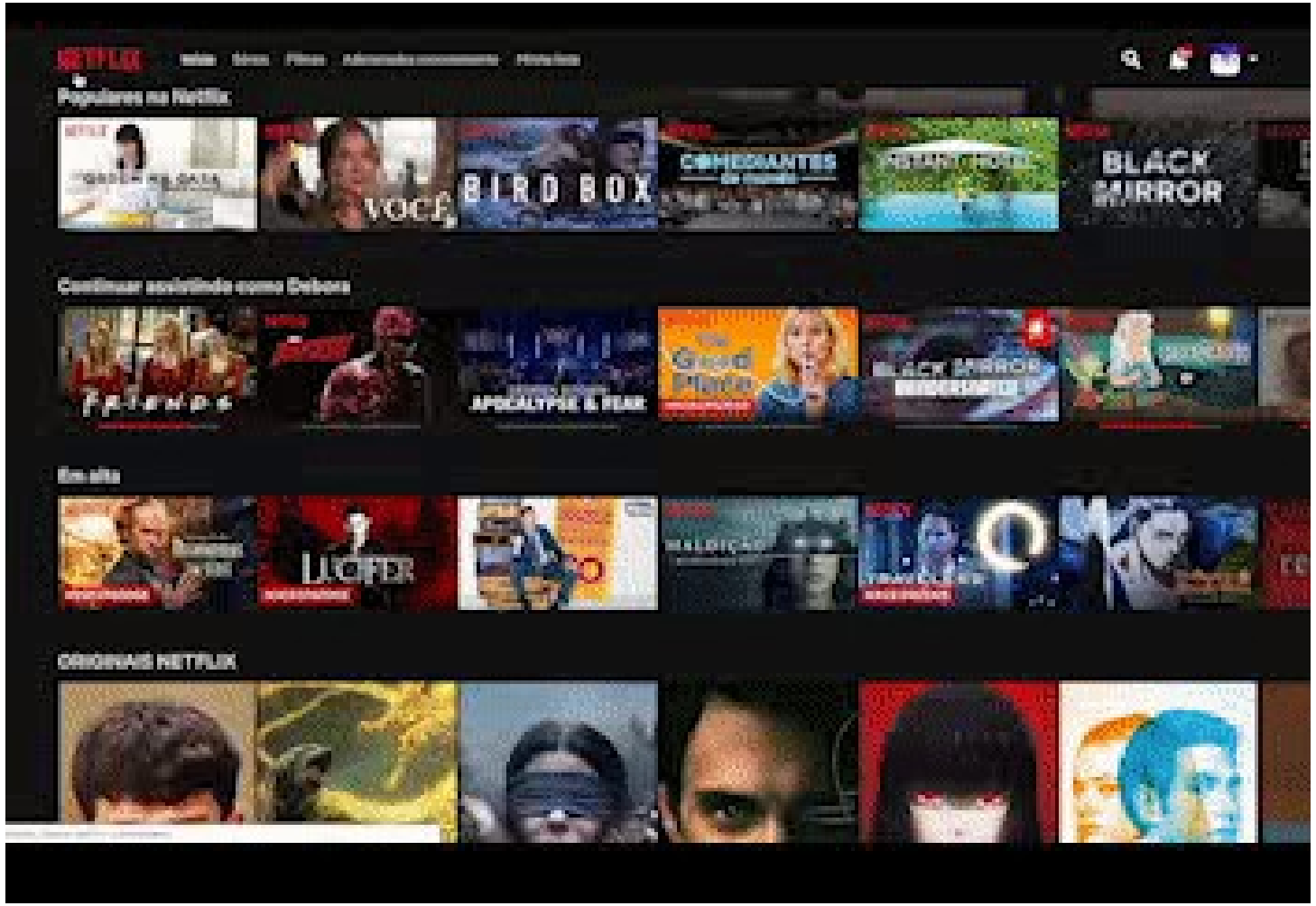
1. Limit the number of Options
2. Utilize Categorization
3. Implement Decision Aids
4. Personalization and Recommendation
5. Progressive Disclosure

Choice Overload

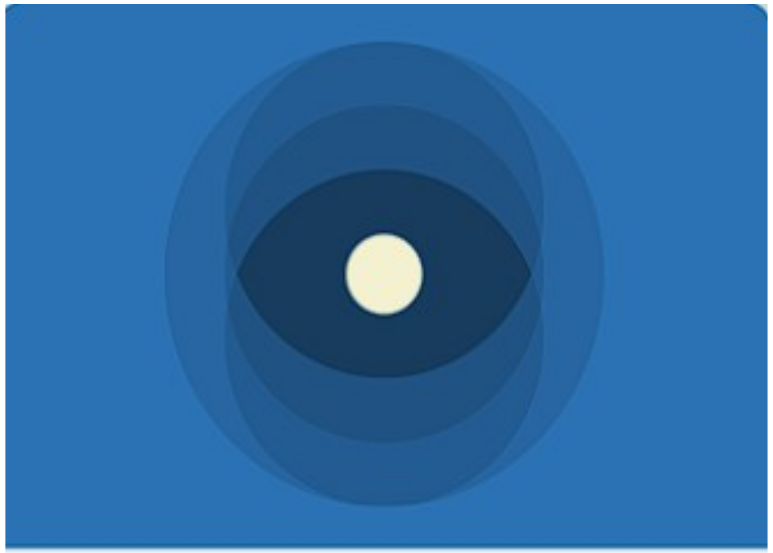


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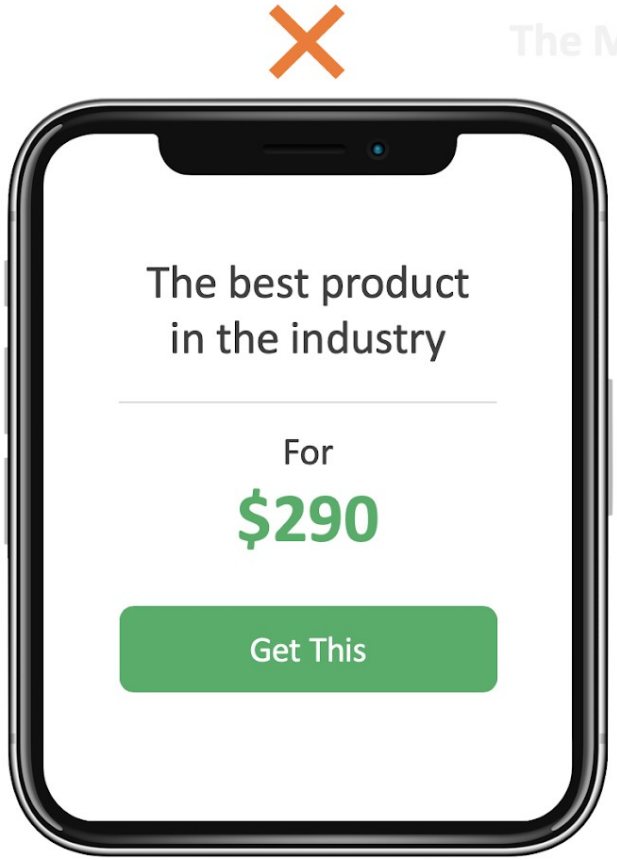
Cognitive Bias

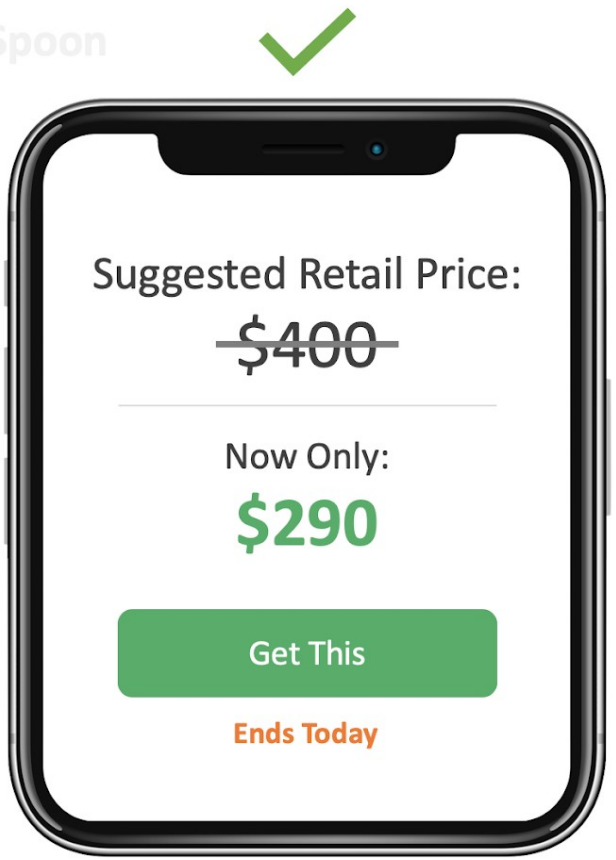


Cognitive Bias

A systematic error of thinking or rationality in judgment that influence our perception of the world and our decision-making ability.

The Mobile Spoon





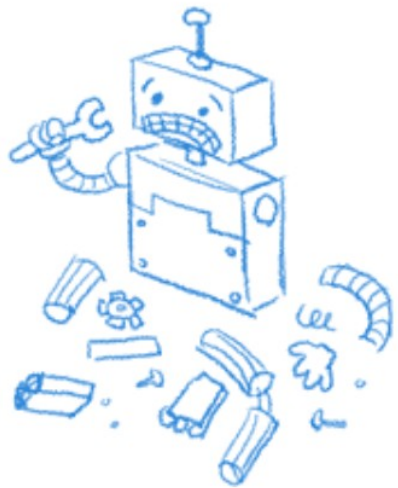
Error 500 (Server Error)!!1

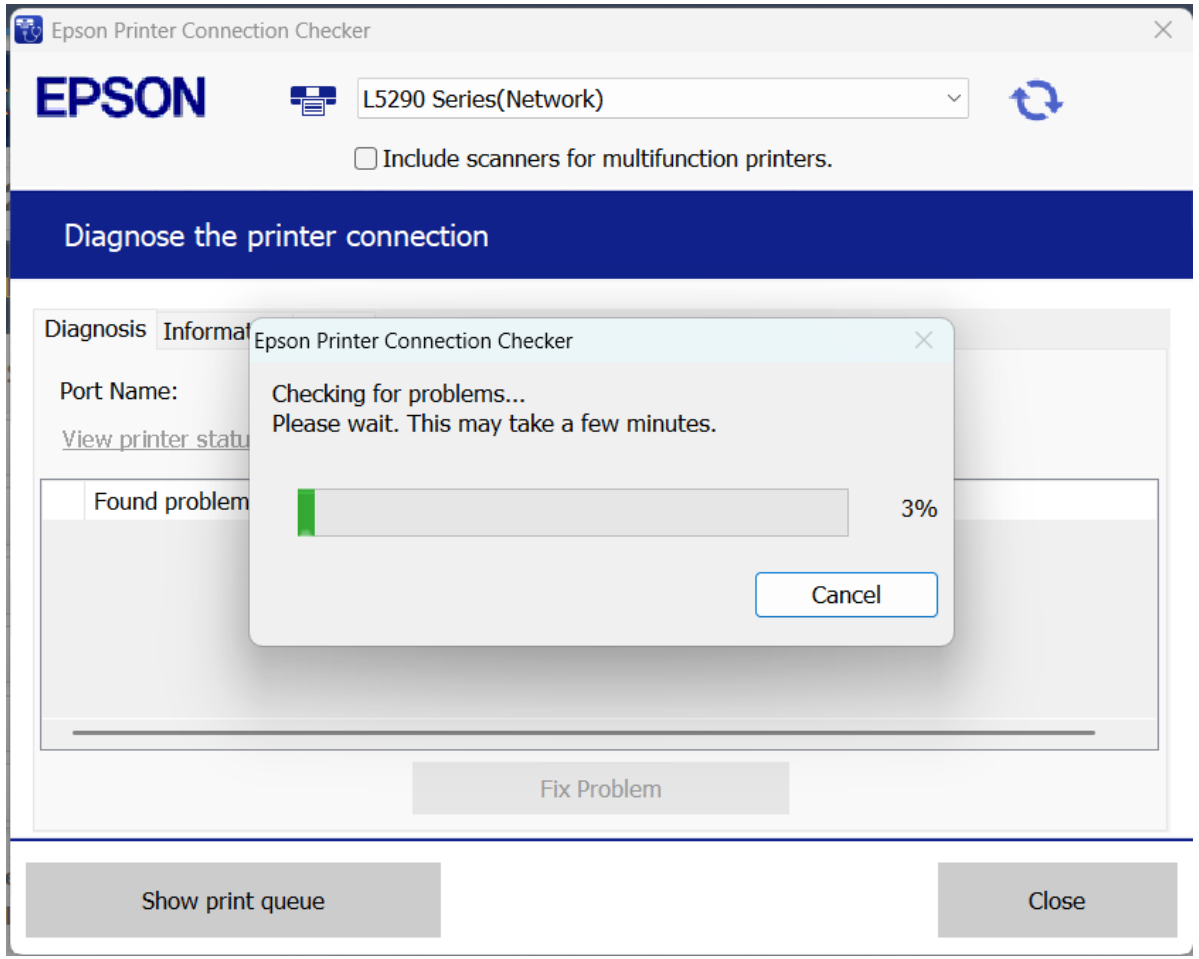
classroom.google.com

Google

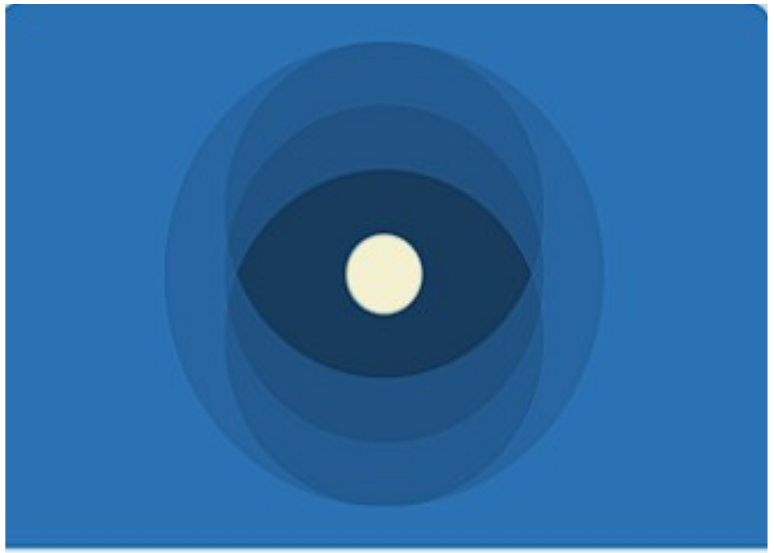
500. That's an error.

There was an error. Please try again later. That's all we know.




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Include scanners for multifunction printers.
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Epson Printer Connection Checker
Checking for problems...
Please wait. This may take a few minutes.
3%
Cancel

Cognitive Bias



Cognitive Bias

A systematic error of thinking or rationality in judgment that influence our perception of the world and our decision-making ability.

	Premium	Premium Lite
	₱189.00/month Cancel anytime Get YouTube Premium	New ₱109.00/month Cancel anytime Get Premium Lite
Ad-free YouTube videos	✓ Ad-free	✓ Most videos ad-free*
Ad-free YouTube Kids	✓ Ad-free	✓ Most videos ad-free*
Ad-free YouTube Music app	✓	—
Download and play offline	✓	—
Background play	✓	—

Cognitive Bias

- CONFIRMATION BIAS – We notice and trust information that agrees with what we already believe.
- FALSE CONSENSUS EFFECT – We assume most people think like us (“surely 90% of users agree with me”).
- FRAMING EFFECT – The framing effect is **a cognitive bias in which people make decisions based on how the information is presented**.

- Treatment A has a 90% survival rate
- Treatment A has a 10% mortality rate

Cognitive Load

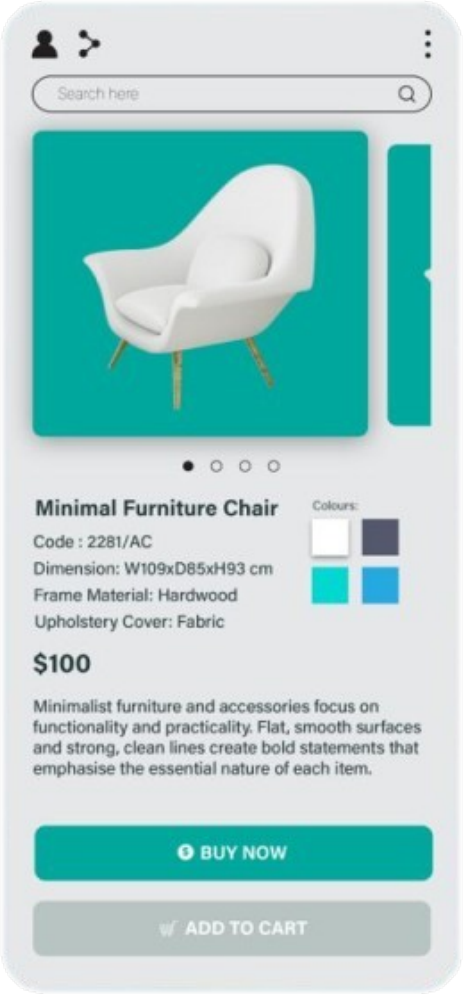


Cognitive Load

The amount of mental resources needed to understand and interact with an interface.

x





Reducing Cognitive Load: What to Do

- **Reduce Visual Clutter:** Remove unnecessary elements to prevent distraction.
- **Use Existing Mental Models:** Follow conventions like placing search bars at the top right or using familiar icons.
- **Chunk Information:** Break down long forms or large amounts of text into smaller, manageable sections.
- **Offload Tasks:** Allow the system to remember information (e.g., **autofill, remembering login sessions**).

Cognitive Load

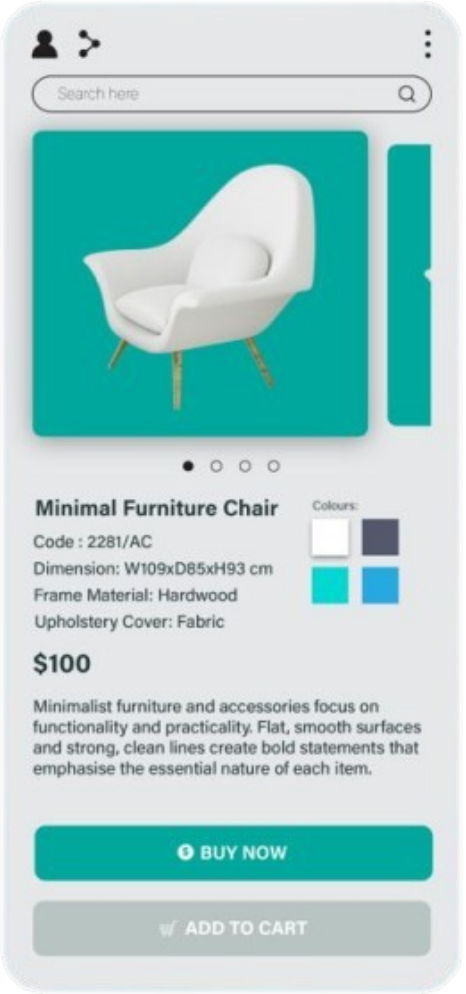


Cognitive Load

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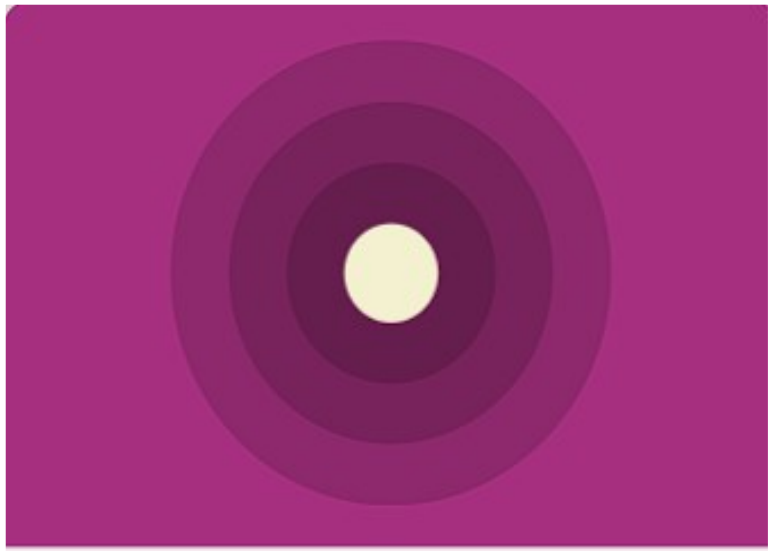


Doherty Threshold



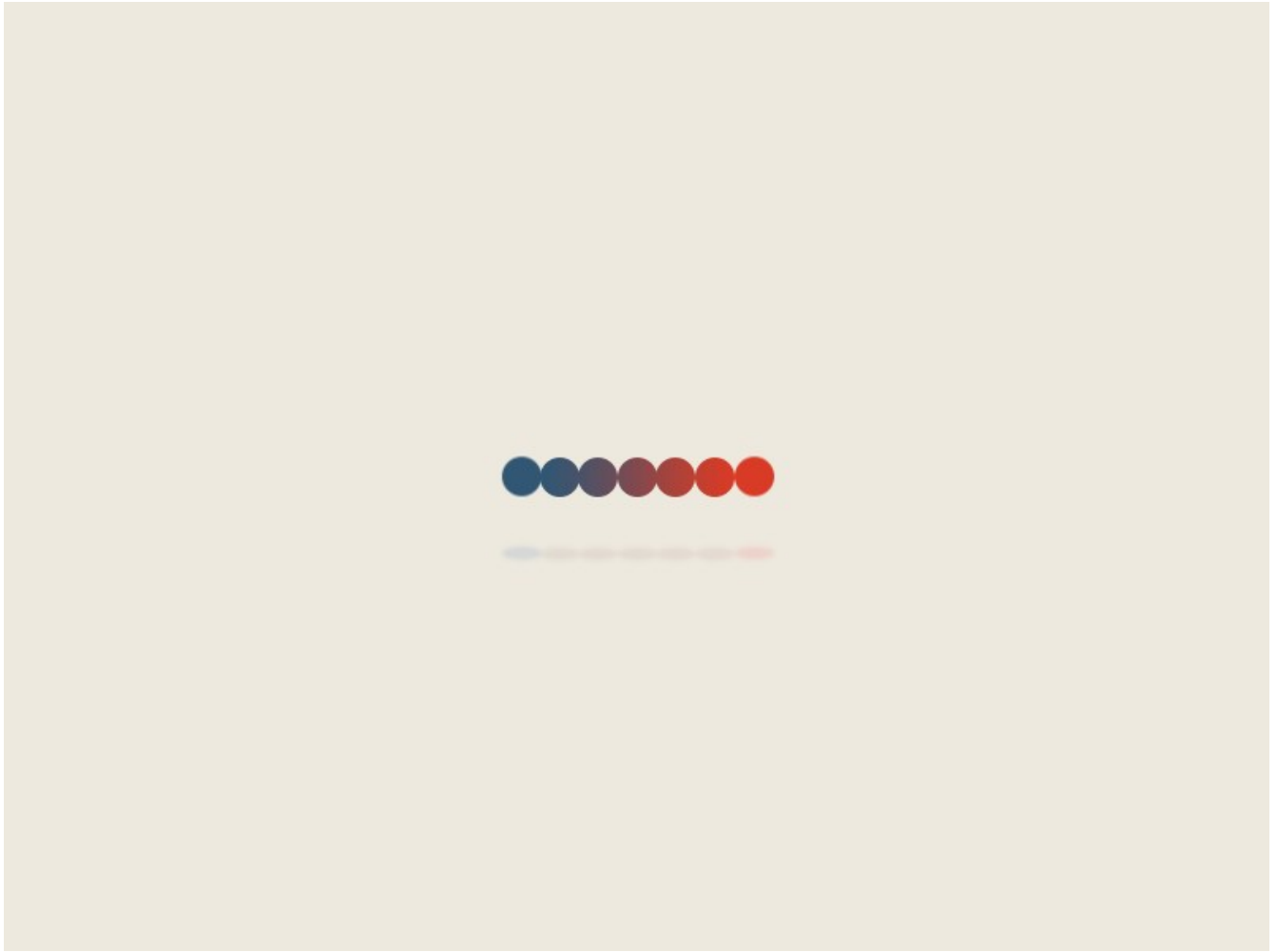


Doherty Threshold

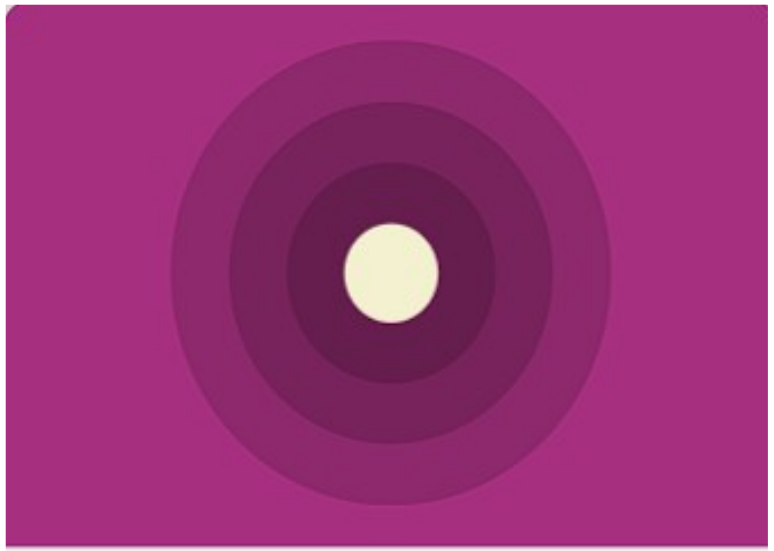


Doherty Threshold

Productivity soars when a computer and its users interact at a pace (<400ms) that ensures that neither has to wait on the other.



Doherty Threshold

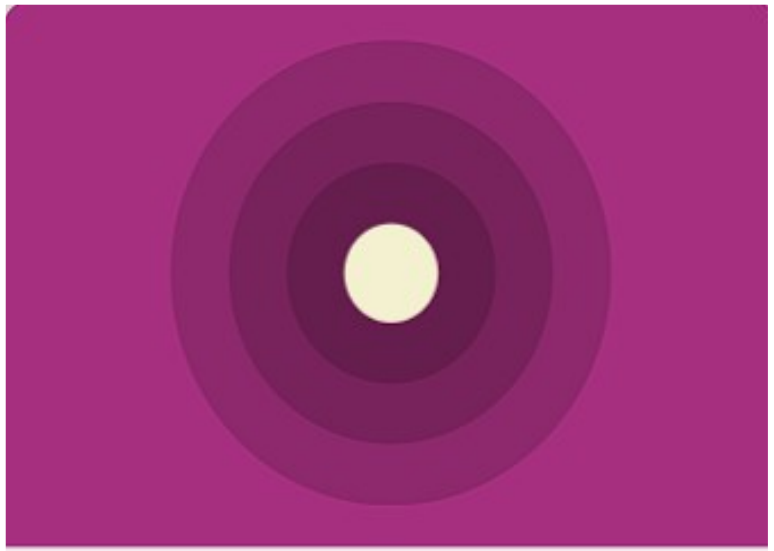


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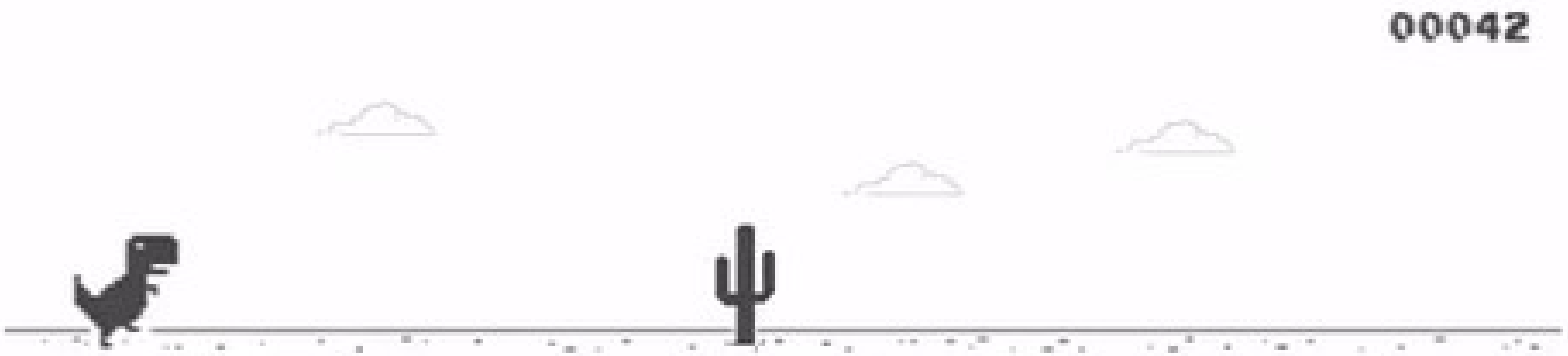


Doherty Threshold

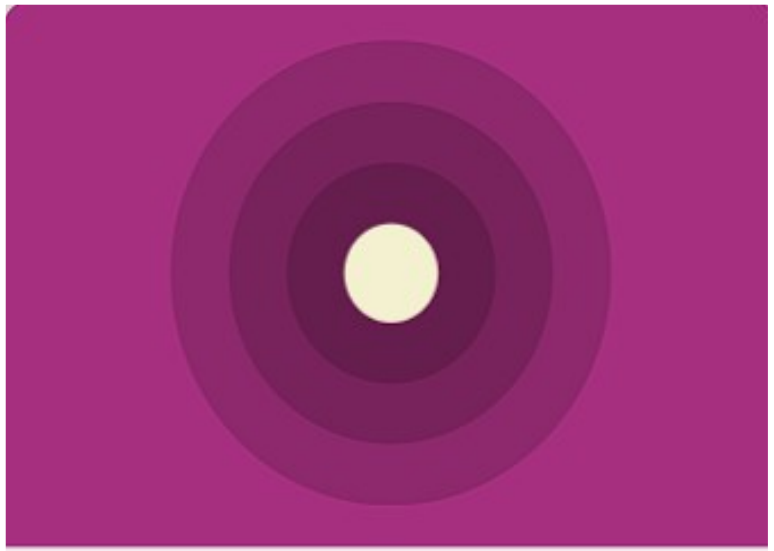


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Doherty Threshold



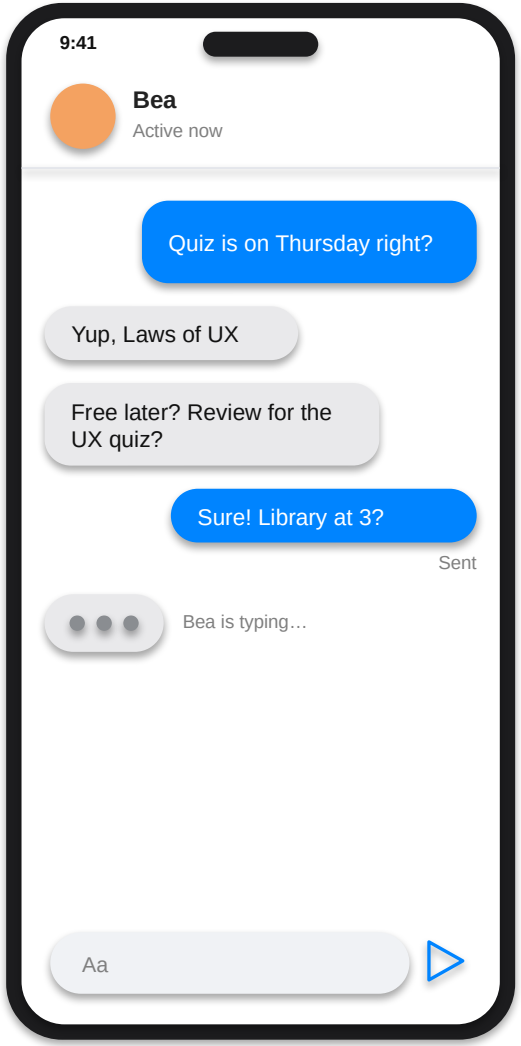
Doherty Threshold

Productivity soars when a computer and its users interact at a pace (<400ms) that ensures that neither has to wait on the other.

Buy now with 1-Click®

Doherty Threshold





Simplified illustration of a familiar app screen, not an actual screenshot.

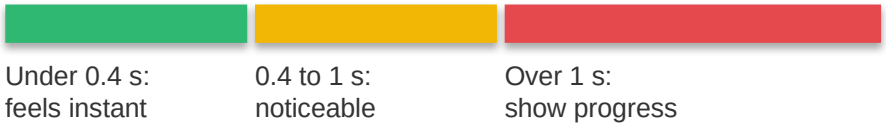
EVERYDAY EXAMPLE · MESSENGER

It answers right away

Your message shows up the moment you tap Send, and “Sent” follows a split second later.

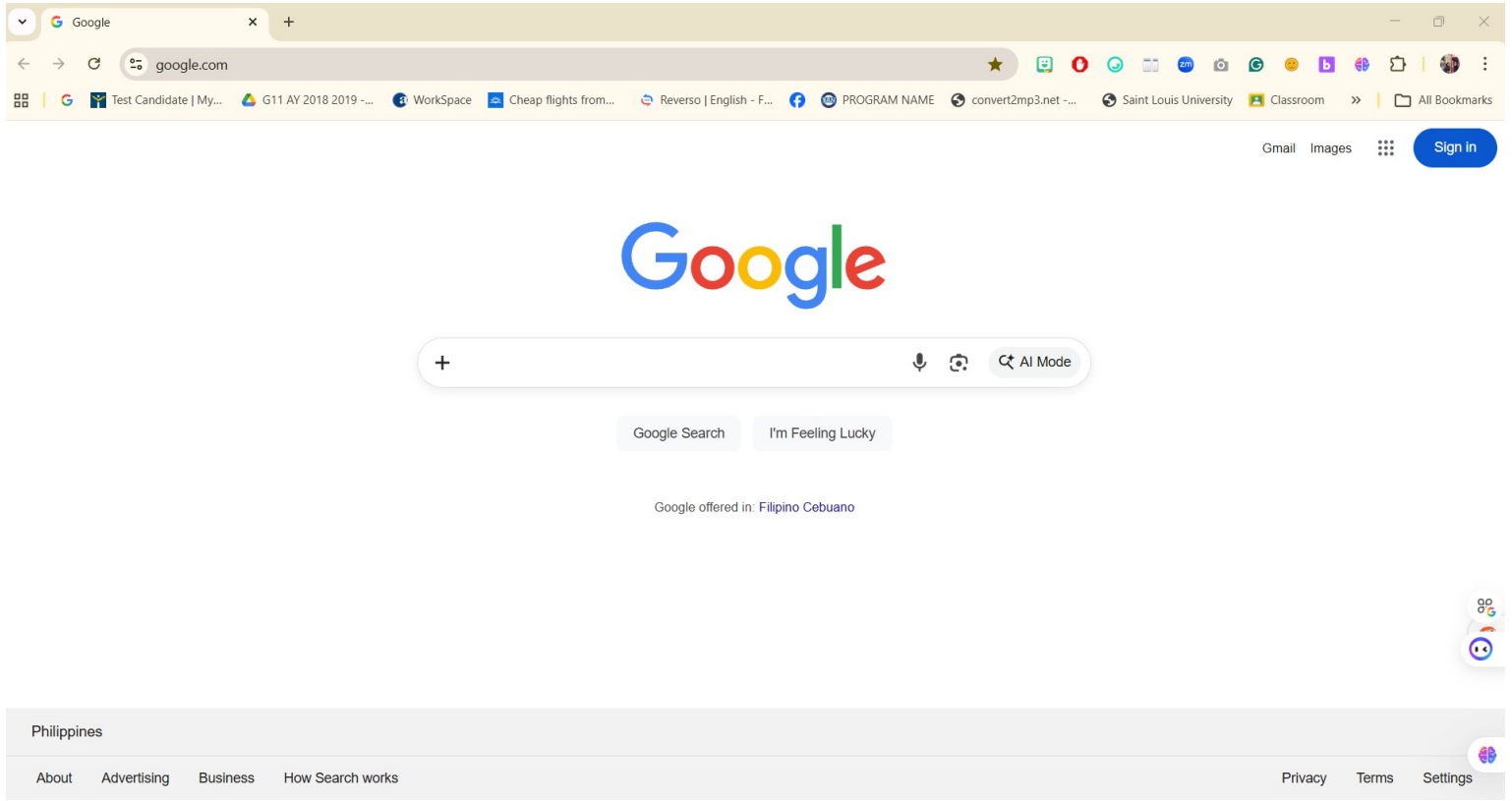
The typing dots tell you a reply is coming, so waiting never feels like lag.

How fast should an app respond?



Fitts’s Law


Fitts’s Law
The time to acquire a target is a function of the distance to and size of the target.

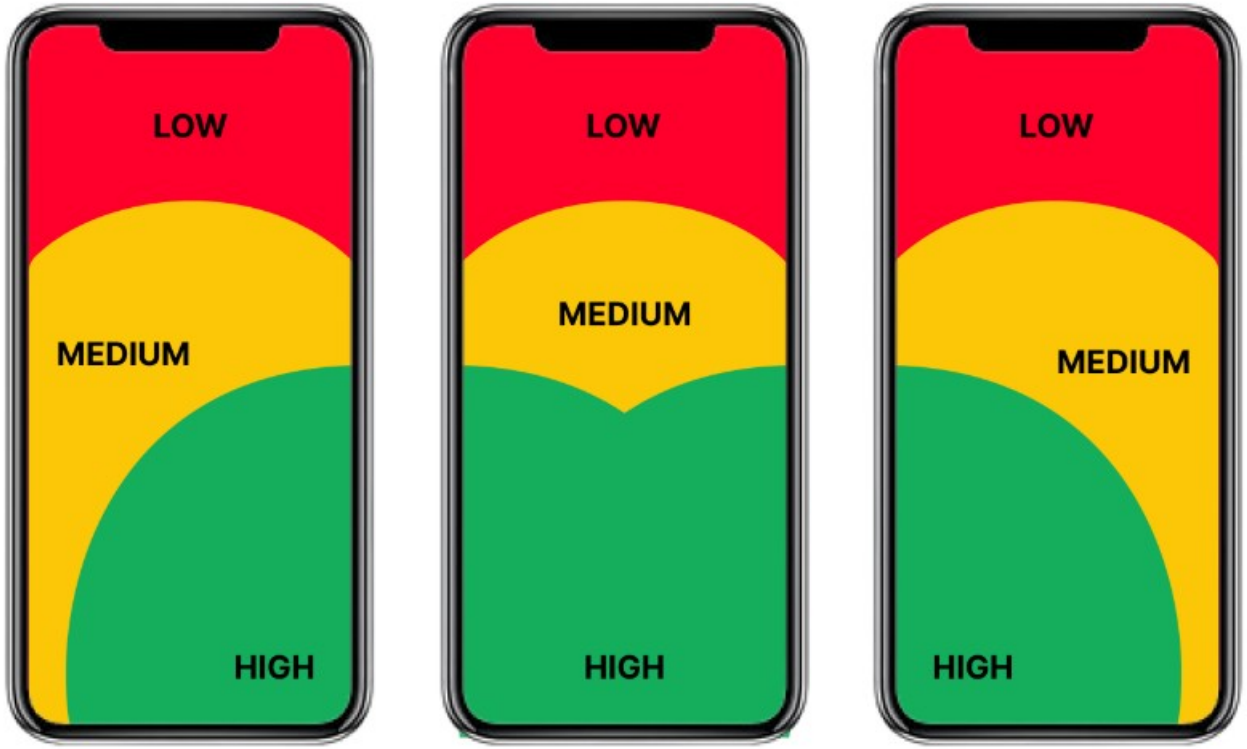

Google
google.com
Test Candidate | My...
G11 AY 2018 2019 -...
WorkSpace
Cheap flights from...
Reverso | English - F...
PROGRAM NAME
convert2mp3.net -...
Saint Louis University
Classroom
All Bookmarks
Gmail
Images
Sign in
Google
AI Mode
Google Search
I'm Feeling Lucky
Google offered in: Filipino Cebuano
Philippines
About
Advertising
Business
How Search works
Privacy
Terms
Settings

Fitts’s Law



Fitts’s Law

The time to acquire a target is a function of the distance to and size of the target.



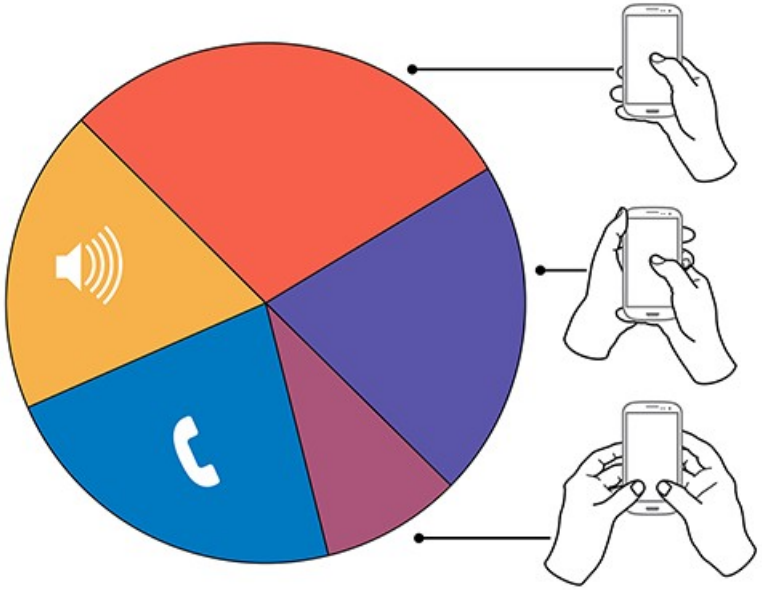
Fitts’s Law



Fitts’s Law

The time to acquire a target is a function of the distance to and size of the target.

- 49% of users hold it in one hand
- 36% of users use one hand to interact while cradling it in the other
- 15% of users use two hands to hold it as if they were using a small tablet



Source: UX Matters

Fitts’s Law





EVERYDAY EXAMPLE · INSTAGRAM

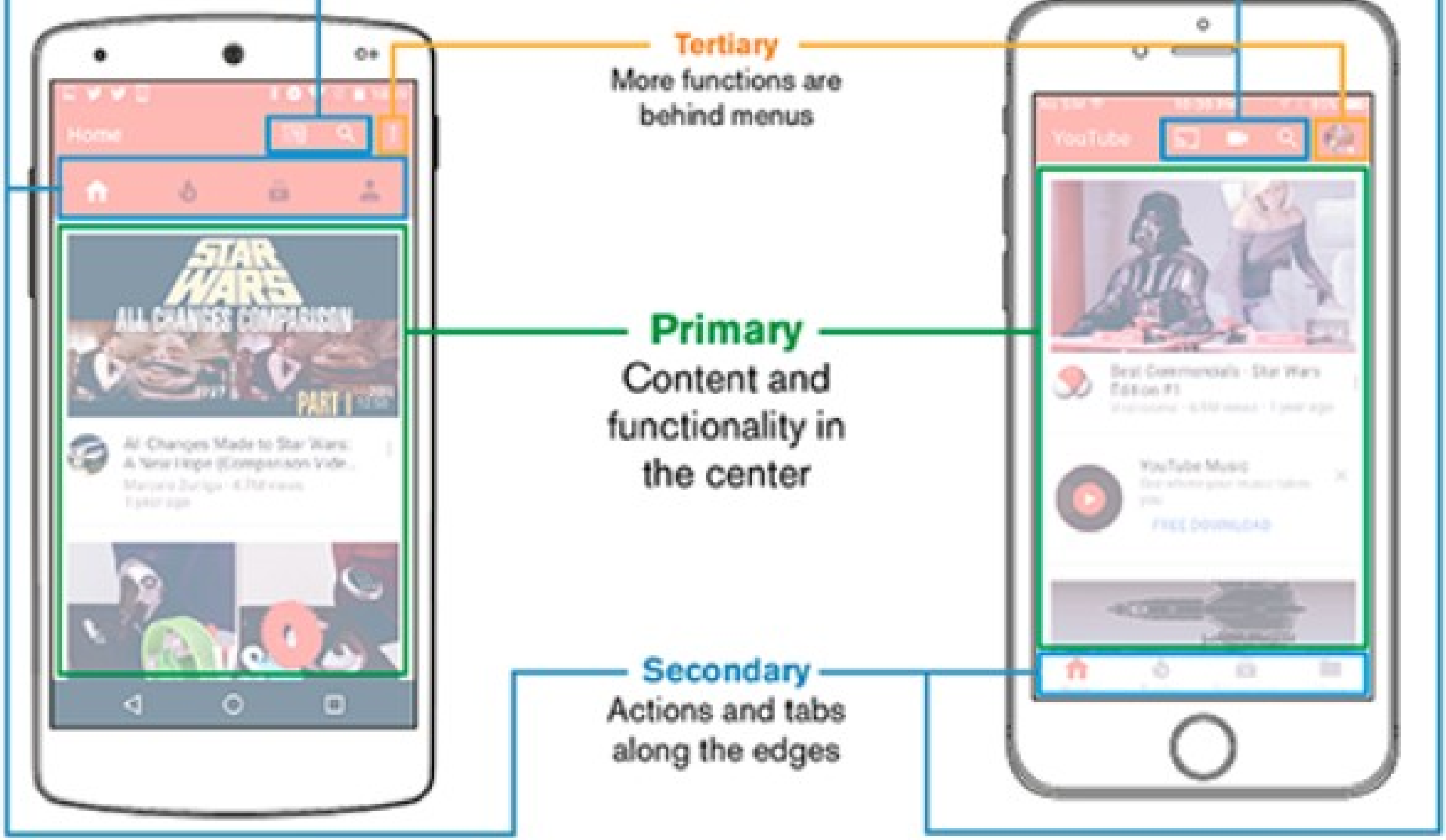
The whole photo is the button

To like a post, double-tap anywhere on the photo: a target far bigger than the small heart icon below it.

The main tabs sit in the bottom bar, right where your thumb already rests.

Bigger and closer targets are faster to hit.

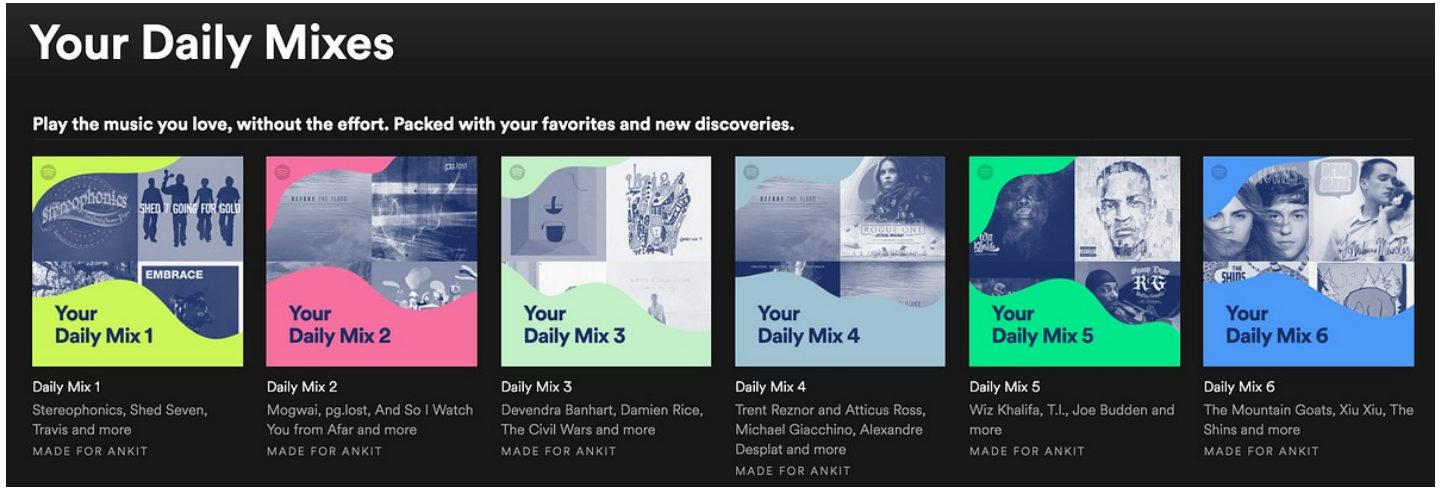
Simplified illustration of a familiar app screen, not an actual screenshot.


Tertiary
More functions are behind menus
Primary
Content and functionality in the center
Secondary
Actions and tabs along the edges
Home
YouTube

Flow

Flow

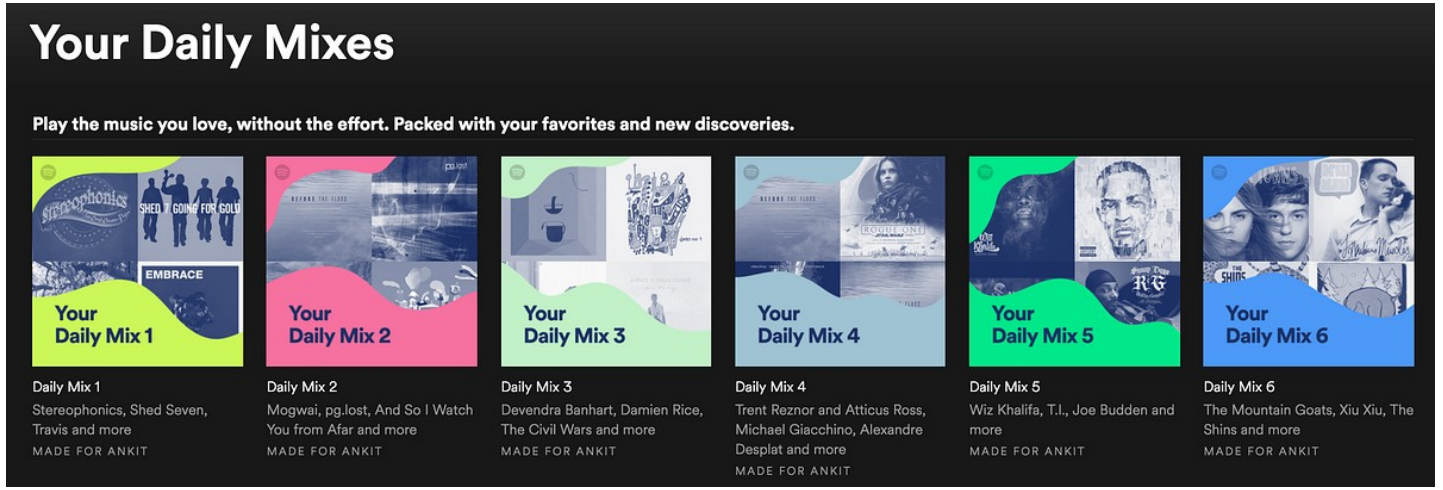
The mental state in which a person performing some activity is fully immersed in a feeling of energized focus, full involvement, and enjoyment in the process of the activity.



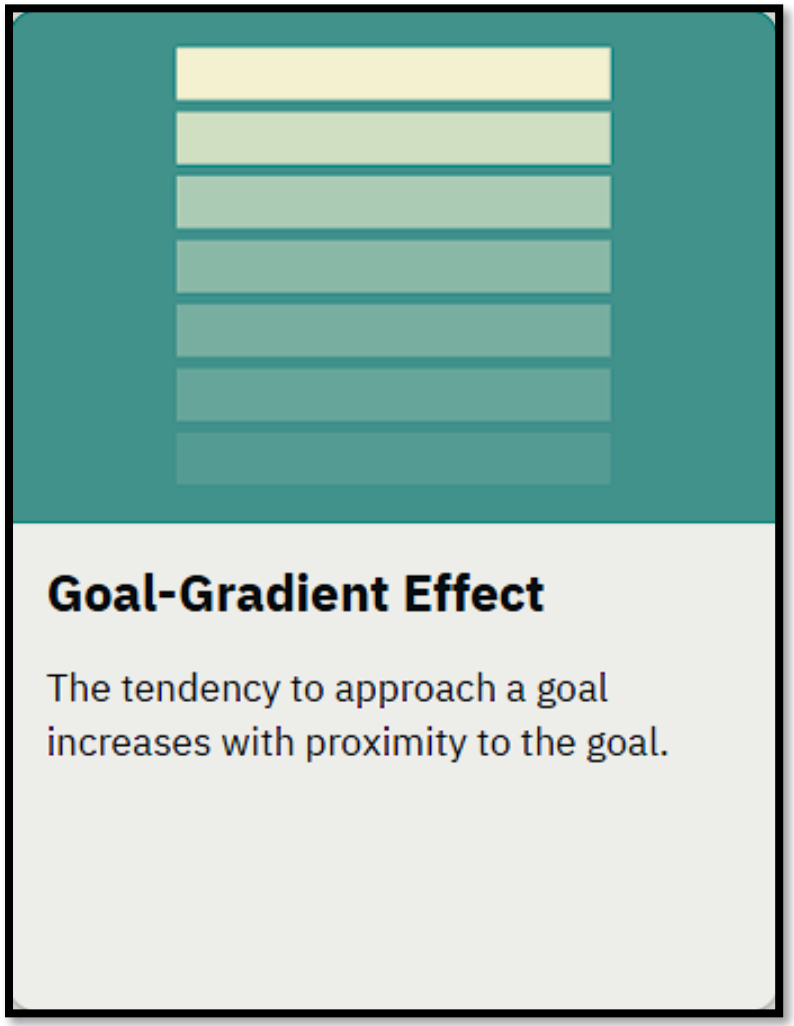
Flow

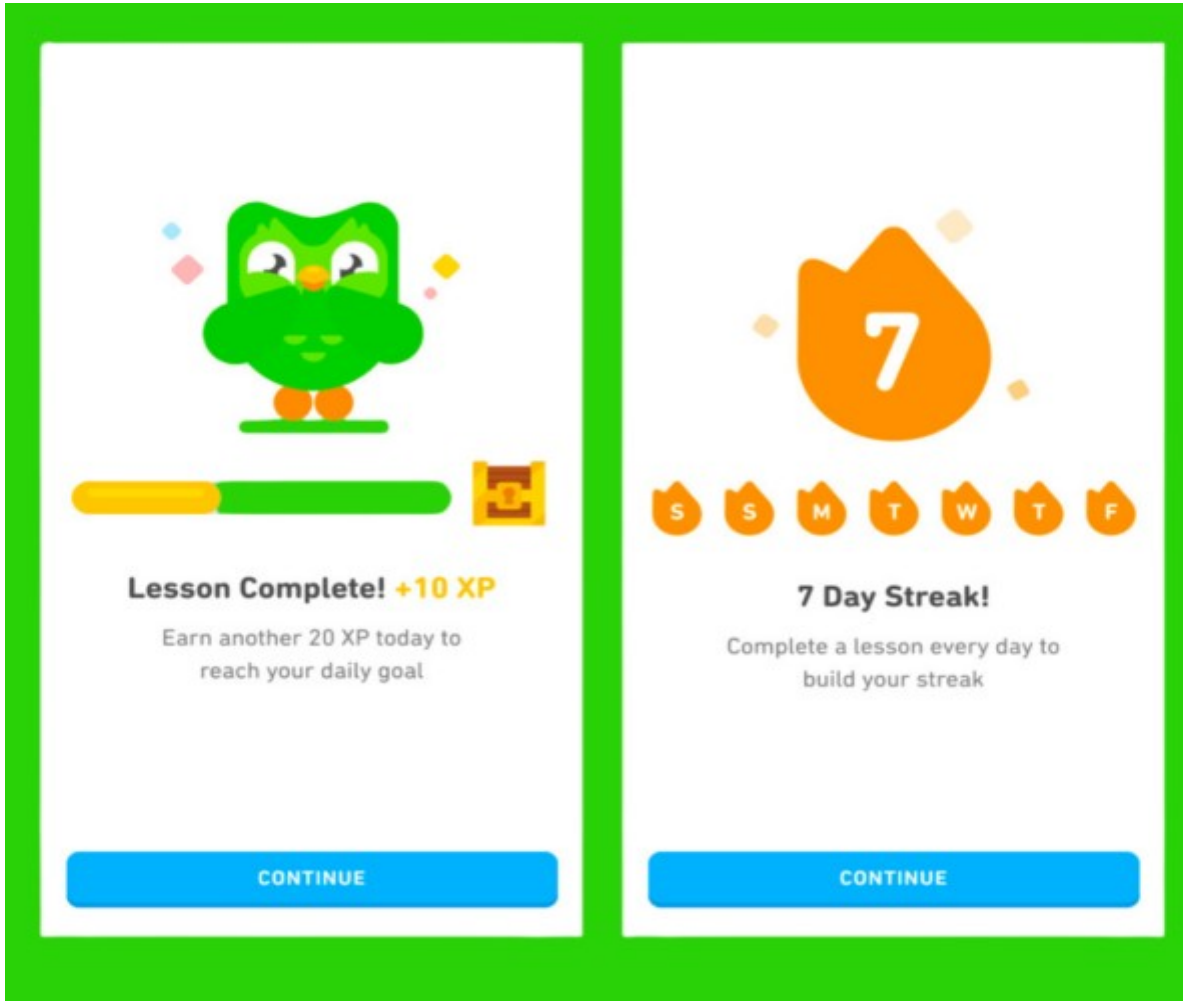
Flow

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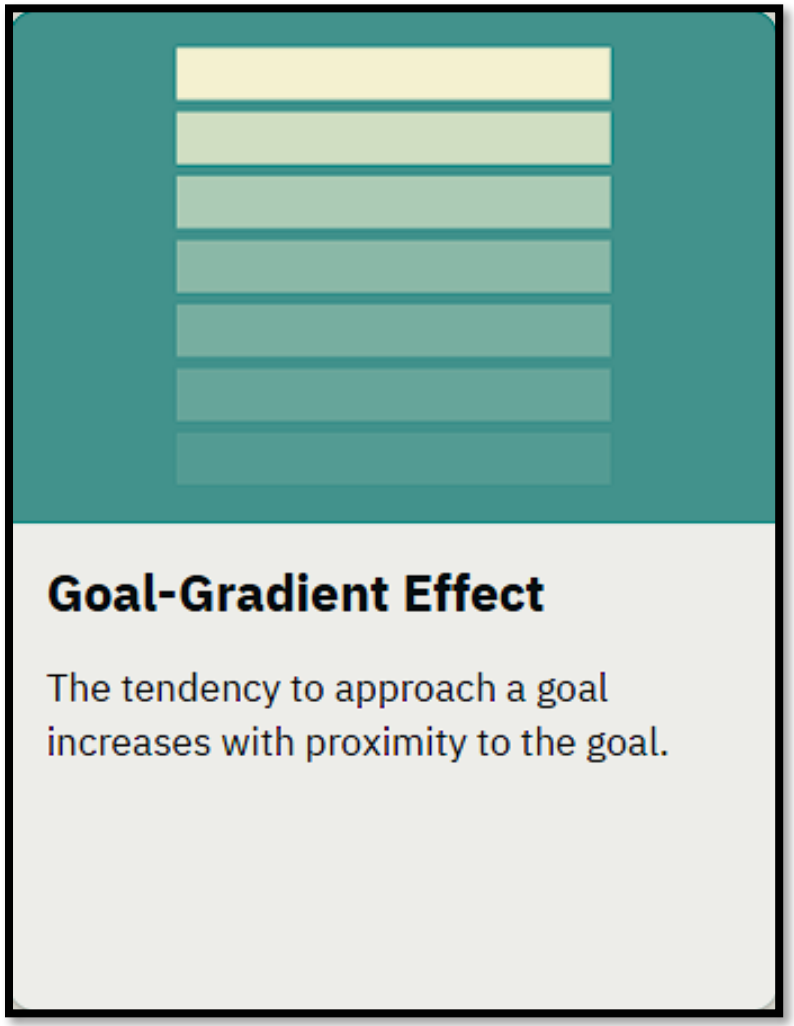


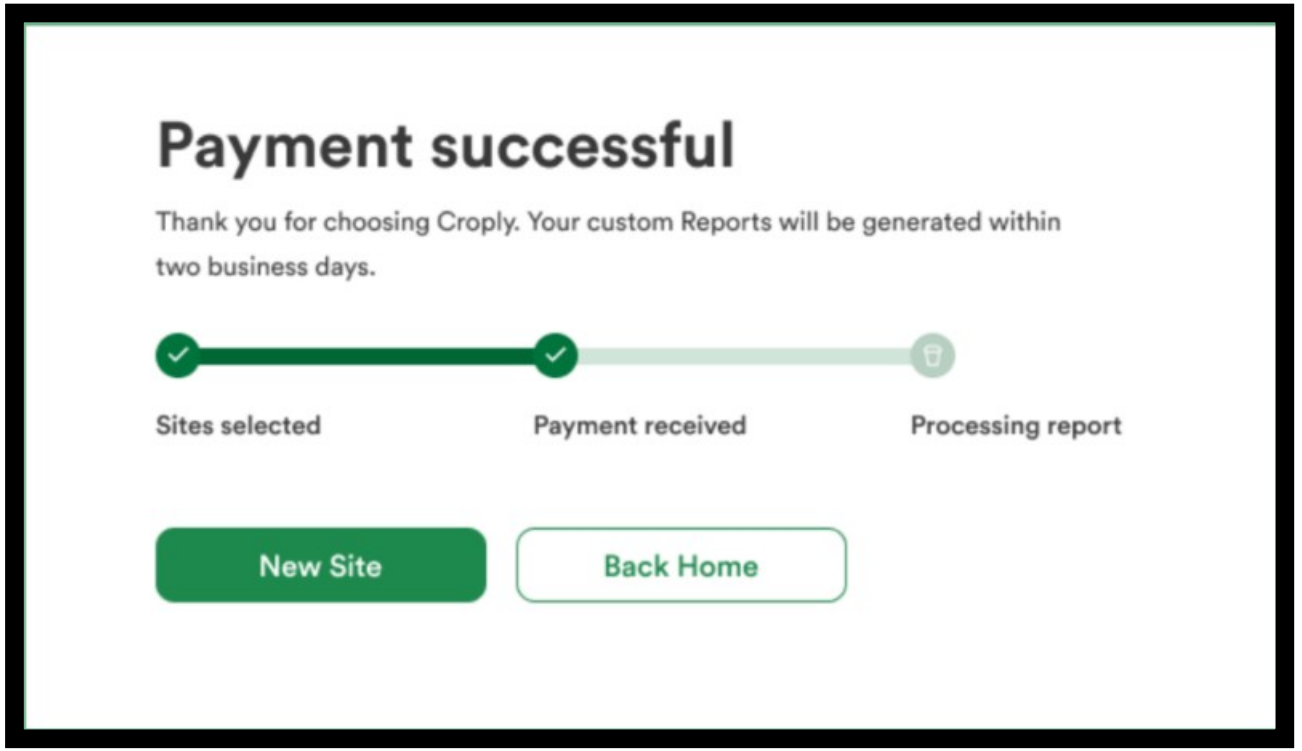
Goal-Gradient Effect



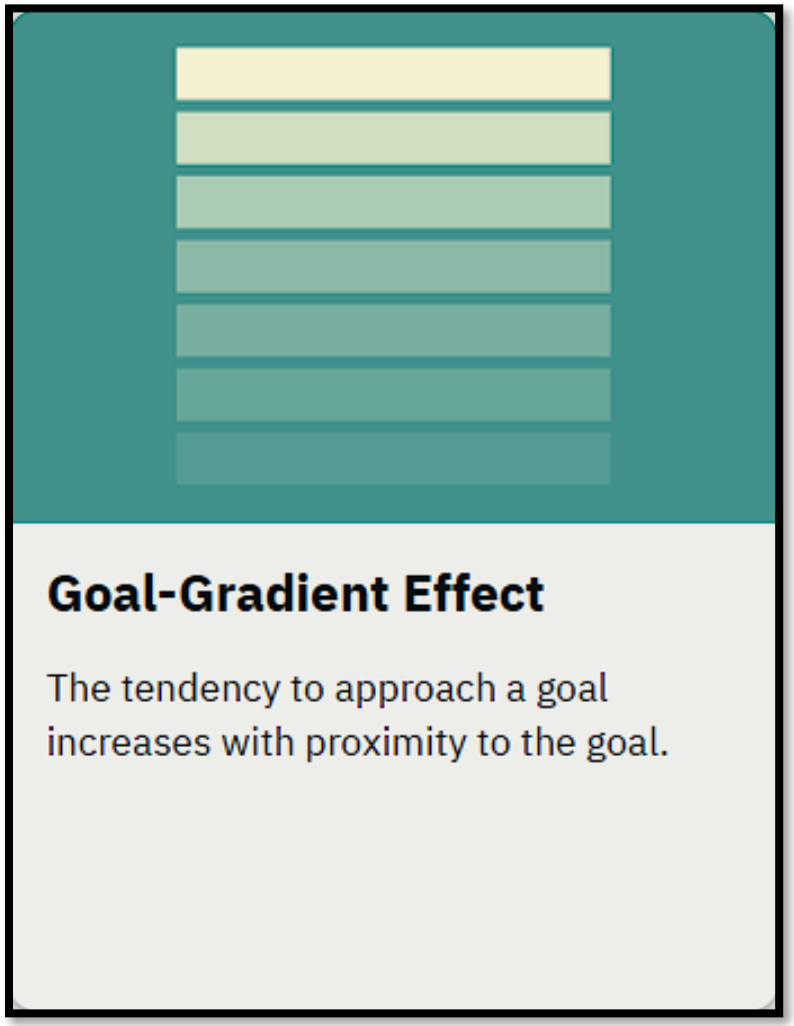


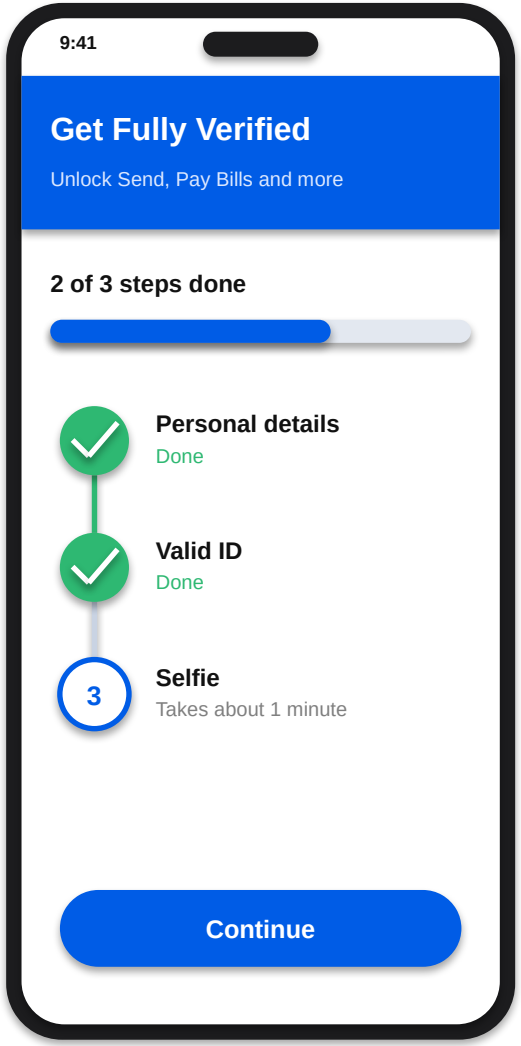
Goal-Gradient Effect





Goal-Gradient Effect





EVERYDAY EXAMPLE · GCASH

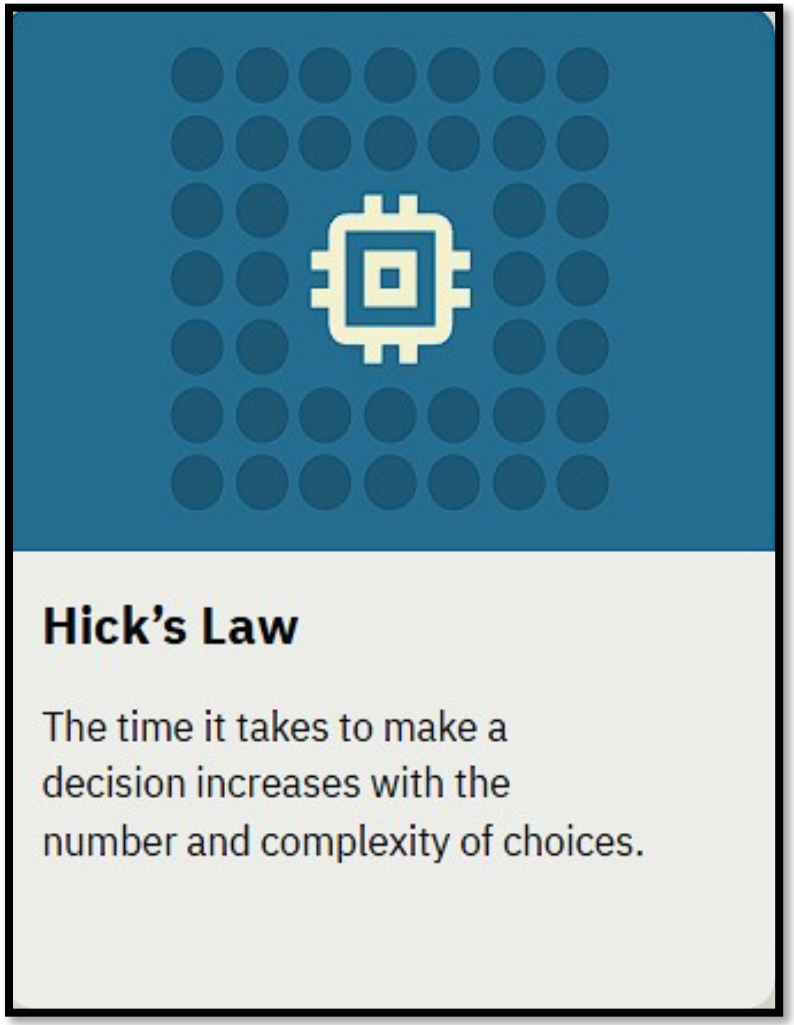
Almost there!

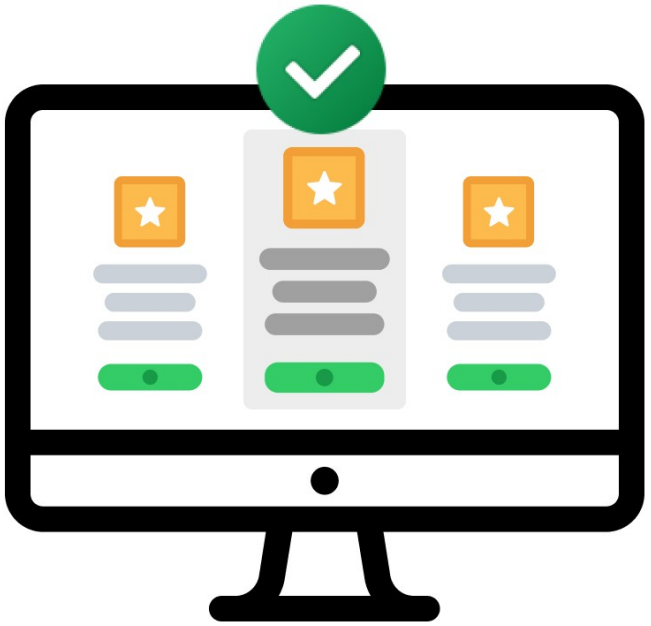
Splitting verification into 3 short steps and showing “2 of 3 steps done” makes the finish line feel close.

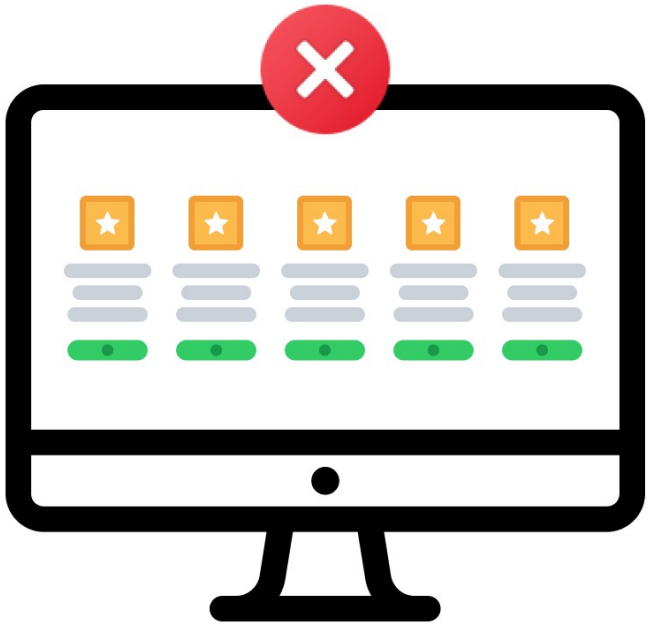
The closer people get to a goal, the harder they push to finish it.

Simplified illustration of a familiar app screen, not an actual screenshot.

Hick’s Law


Hick’s Law
The time it takes to make a decision increases with the number and complexity of choices.





Hick's Law

Select your plan

View plan comparison >

Lite

- Medical Expenses
  Up to ₱ 500,000
- Hospital Cash Allowance
  ₱ 2,000 per day, up to 10 days
- Emergency Local Follow-up Treatment
  Up to ₱ 50,000
- Medical Assistance in case of Pre-existing Illness
  Up to ₱ 10,000
- Burns Benefit

More benefits >

Essential

- Medical Expenses
  Up to ₱ 1,000,000
- Hospital Cash Allowance
  ₱ 2,000 per day, up to 10 days
- Emergency Local Follow-up Treatment
  Up to ₱ 100,000
- Medical Assistance in case of Pre-existing Illness
  Up to ₱ 10,000
- Burns Benefit

More benefits >

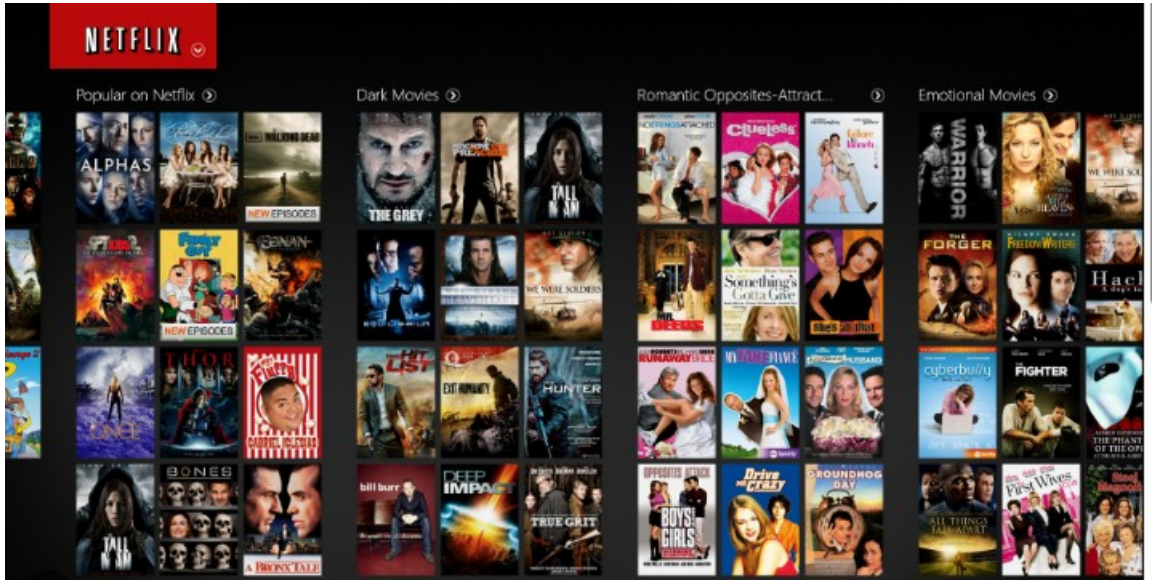
Deluxe

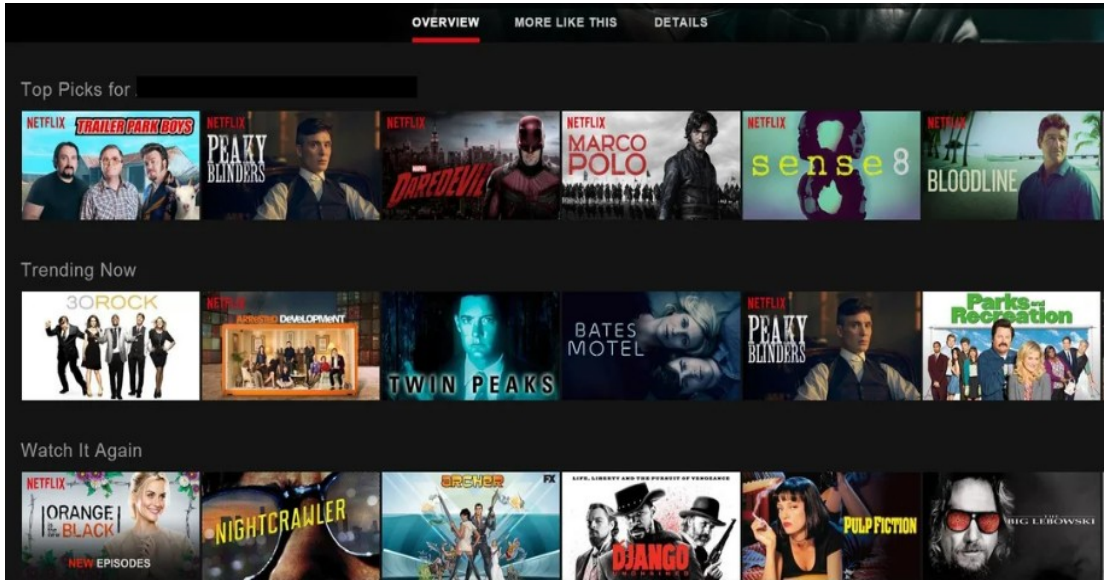
- Medical Expenses
  Up to ₱ 2,500,000
- Hospital Cash Allowance
  ₱ 2,000 per day, up to 10 days
- Emergency Local Follow-up Treatment
  Up to ₱ 250,000
- Medical Assistance in case of Pre-existing Illness
  Up to ₱ 10,000
- Burns Benefit

More benefits >

Buy Now

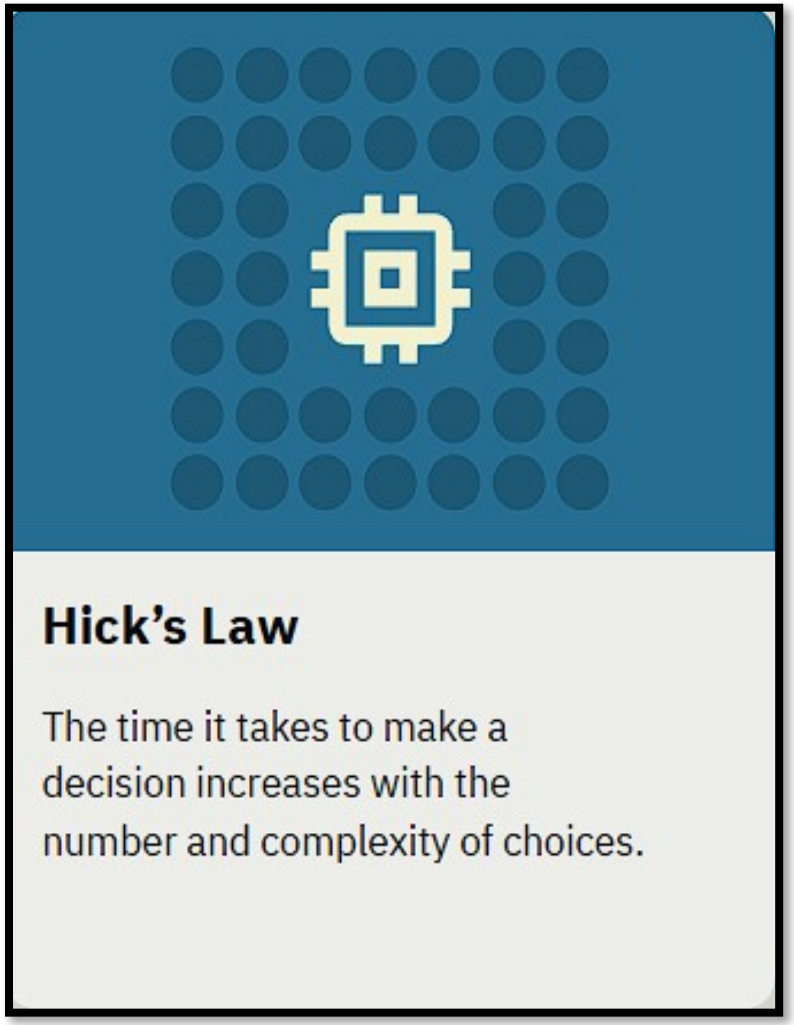
Hick's Law (Decision Time)

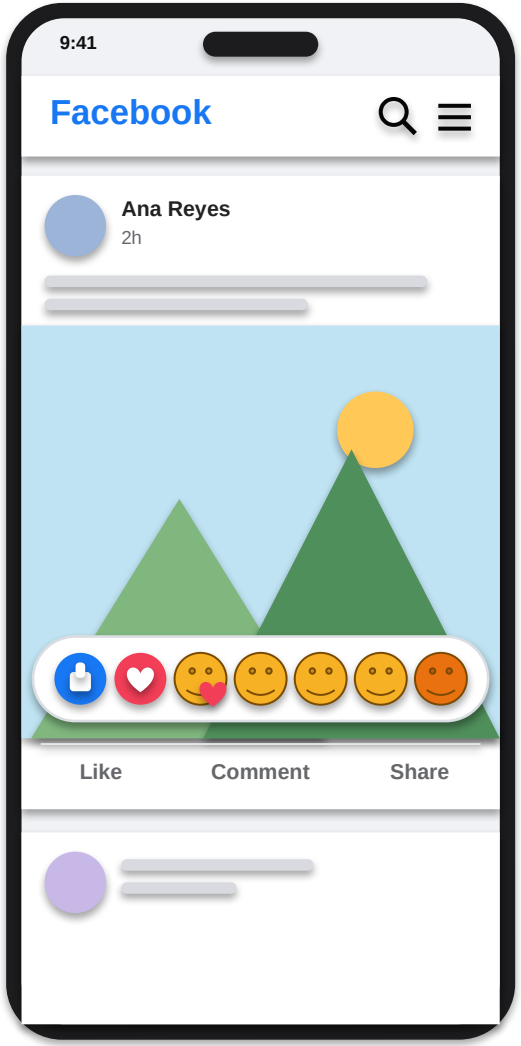





TOOLS
RETURN
EXIT
A
B
C
D
SEARCH
STATUS
P.SIZE
E-MANUAL
HISTORY

Hick’s Law





EVERYDAY EXAMPLE · FACEBOOK

Only 7 reactions

Press and hold Like and you get just 7 choices, not the hundreds of emoji on your keyboard.

Fewer, clearer options let people decide in about a second.

7 reactions



Quick to decide

Full emoji keyboard

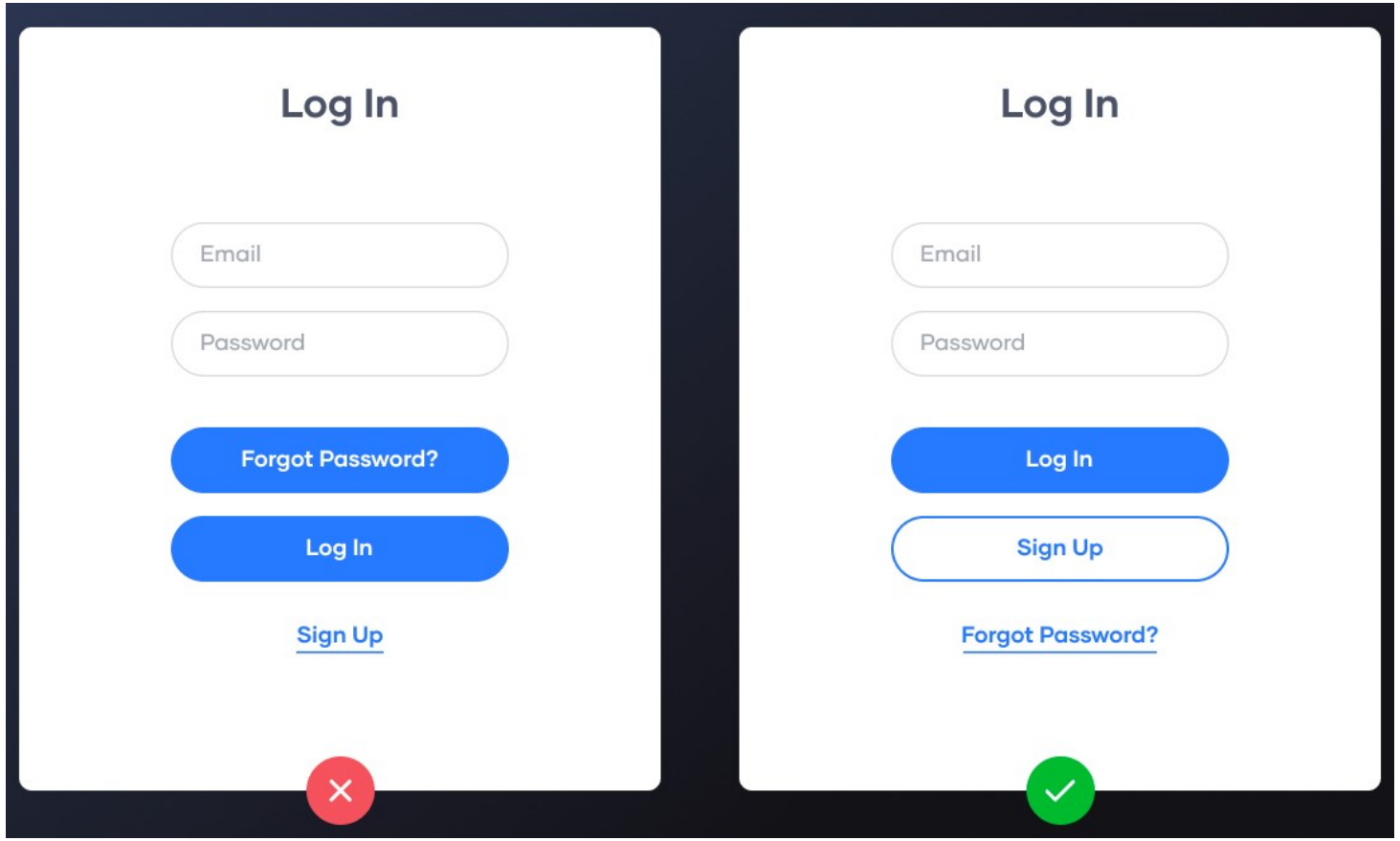


Slow to decide

Simplified illustration of a familiar app screen, not an actual screenshot.

Jakob’s Law


Jakob’s Law
Users spend most of their time on other sites. This means that users prefer your site to work the same way as all the other sites they already know.


Log In
Email
Password
Forgot Password?
Log In
Sign Up
Log In
Email
Password
Log In
Sign Up
Forgot Password?

Jakob’s Law





EVERYDAY EXAMPLE · INSTAGRAM, FACEBOOK, GCASH, DISCORD

Same spot, every app

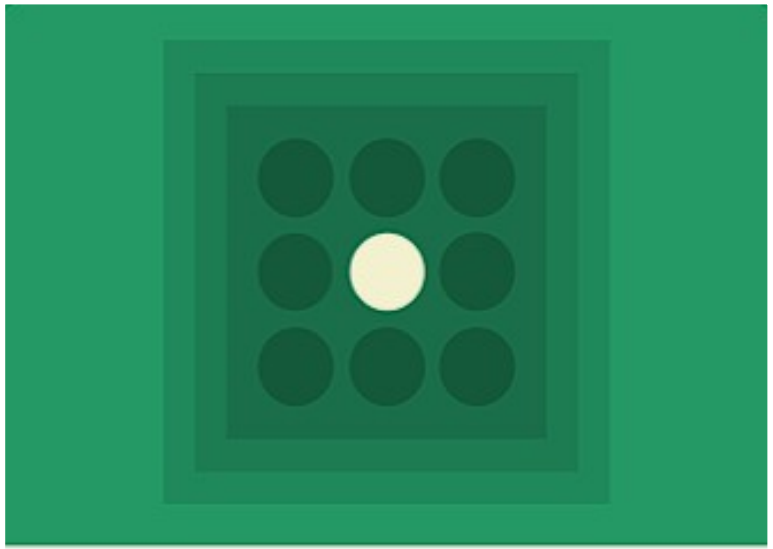
Each app looks different, but Home is always the first tab at the bottom and your profile or menu sits at the far end (circled). You learned this pattern from other apps, so a new app feels familiar on day one.

Simplified illustration of a familiar app screen, not an actual screenshot.

Jakob’s Law: Key Ideas

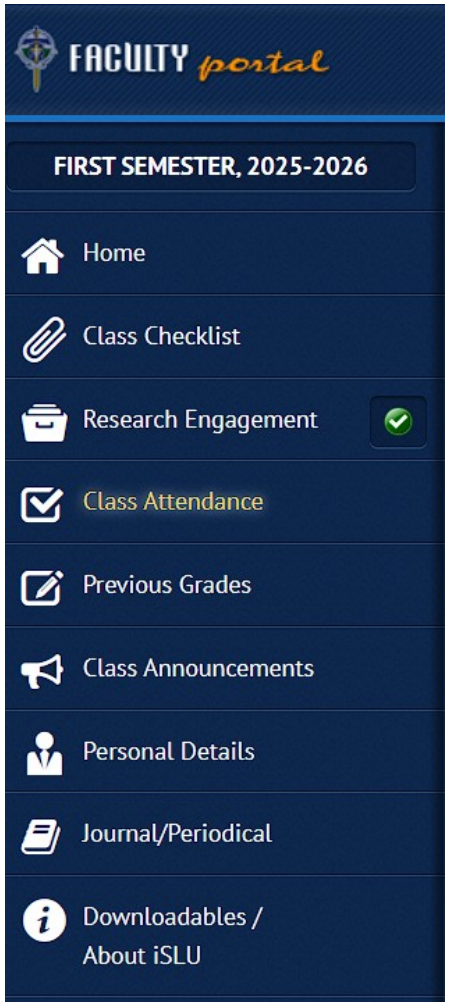
- **Expectation-Driven Design:** Users bring expectations from other sites; violating these patterns causes frustration.
- **Familiarity Over Novelty:** Users prefer familiar, predictable, and consistent layouts (e.g., logo on the top left, shopping cart icon, search bar).
- **Reduced Cognitive Load:** By using established patterns, you reduce the mental effort required to learn a new interface.
- **Context:** Applicable to websites, apps, and voice interfaces.

Law of Common Region

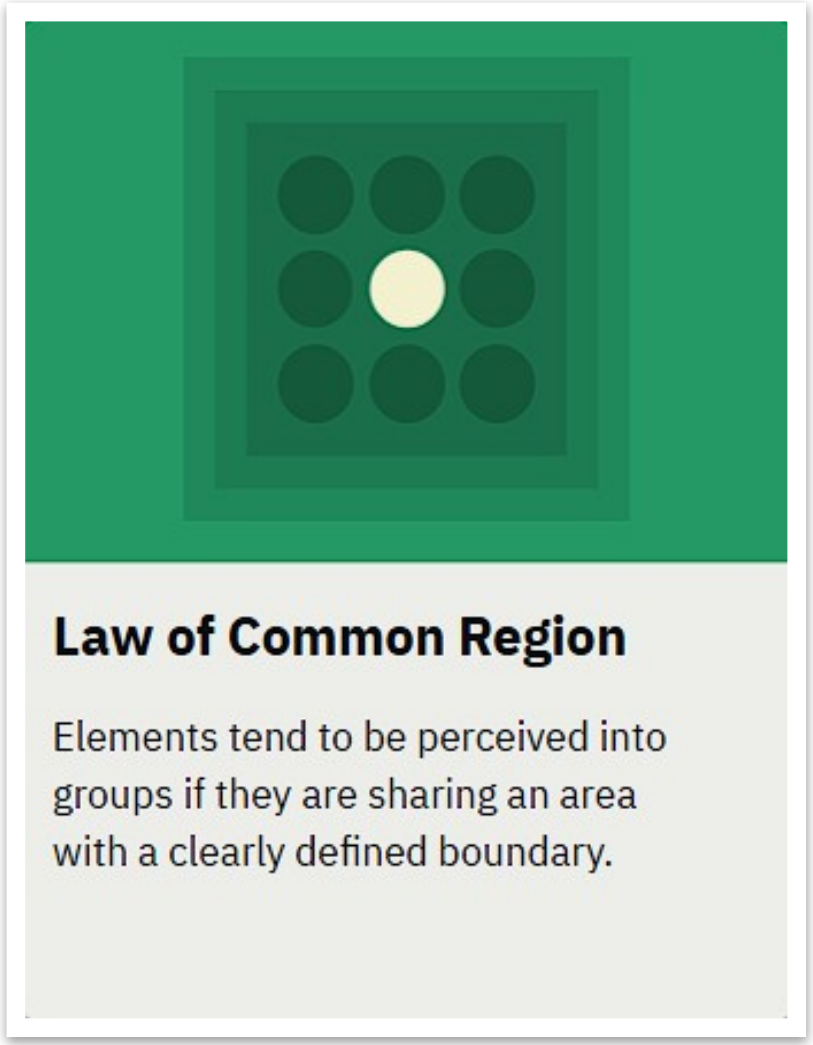


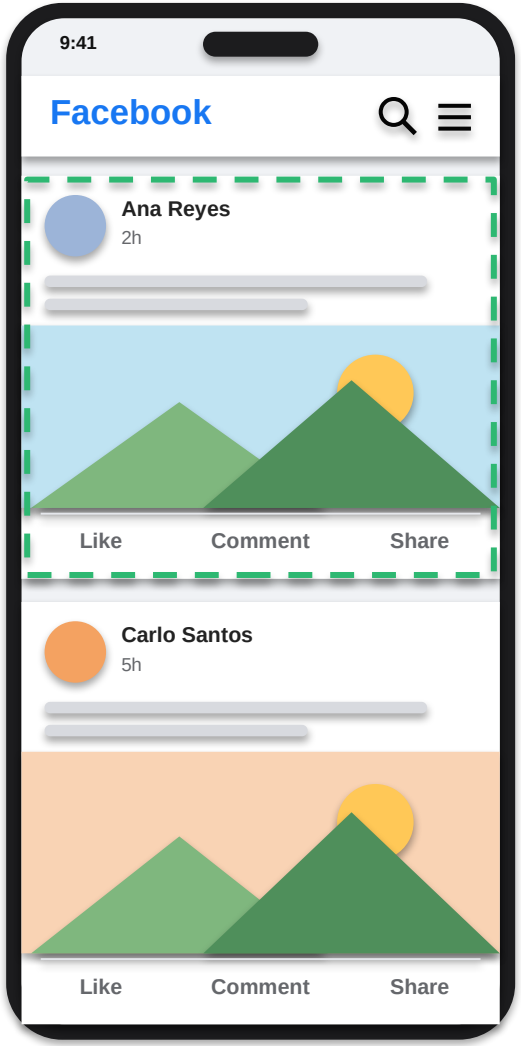
Law of Common Region

Elements tend to be perceived into groups if they are sharing an area with a clearly defined boundary.



Law of Common Region





Simplified illustration of a familiar app screen, not an actual screenshot.

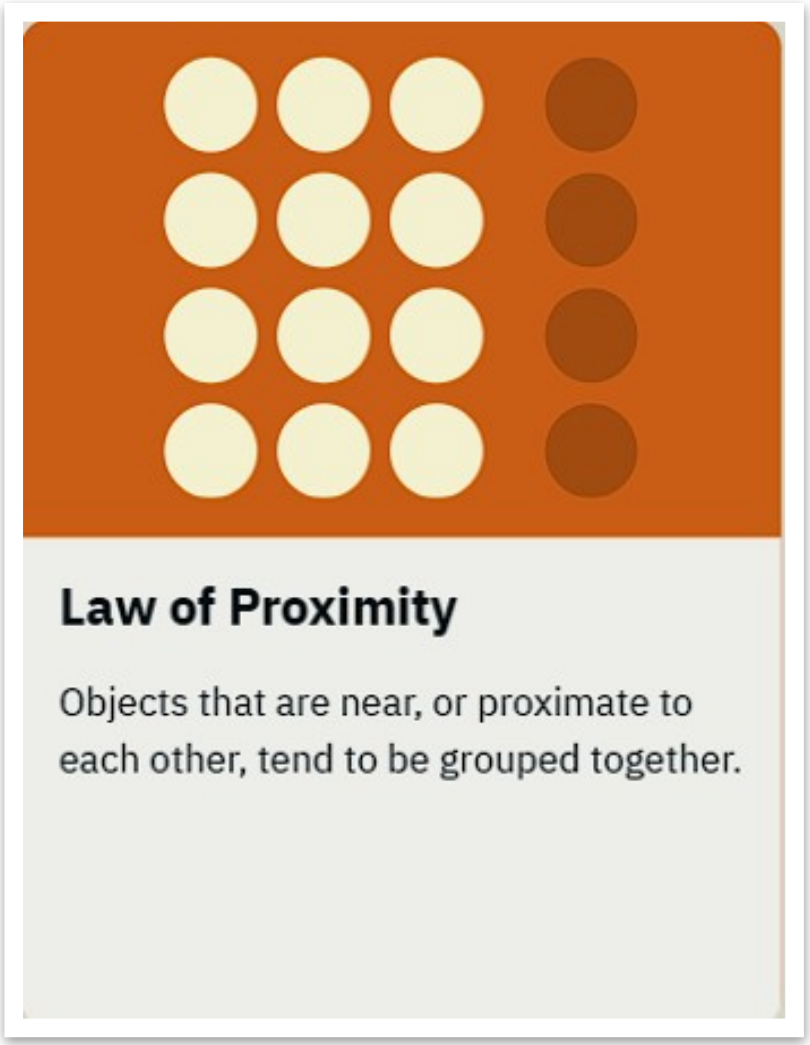
EVERYDAY EXAMPLE · FACEBOOK

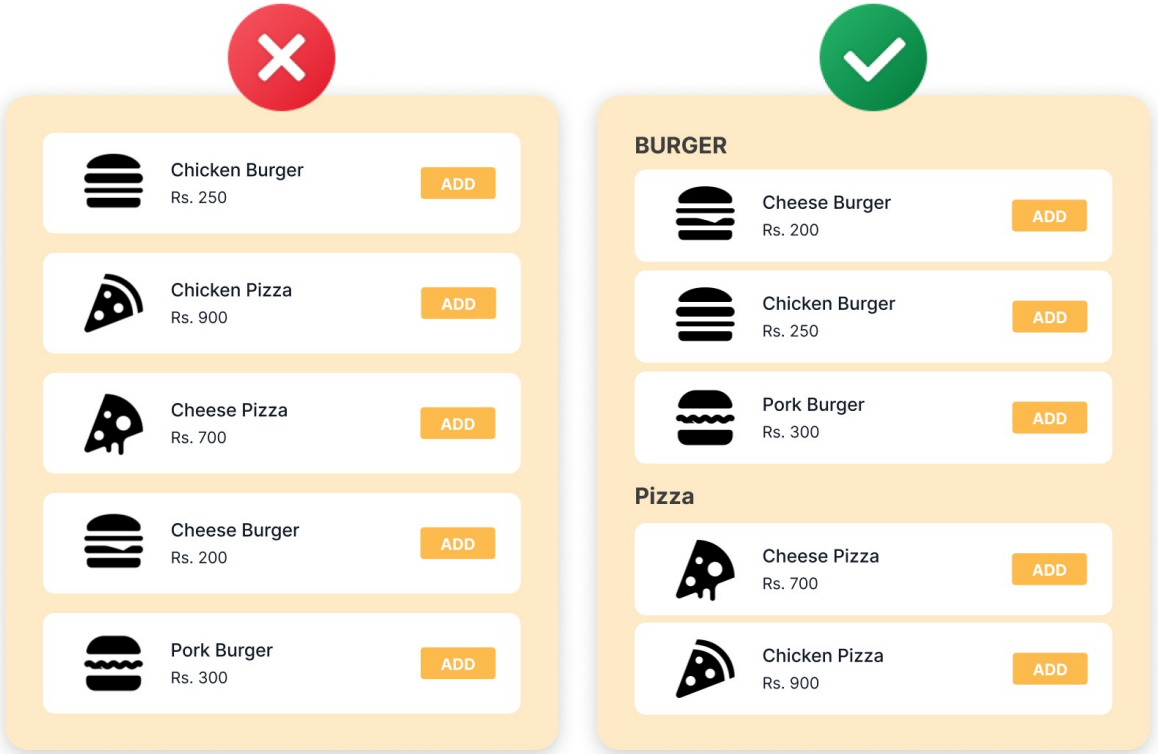
Every post is a card

Each post sits on its own white card against a grey background.

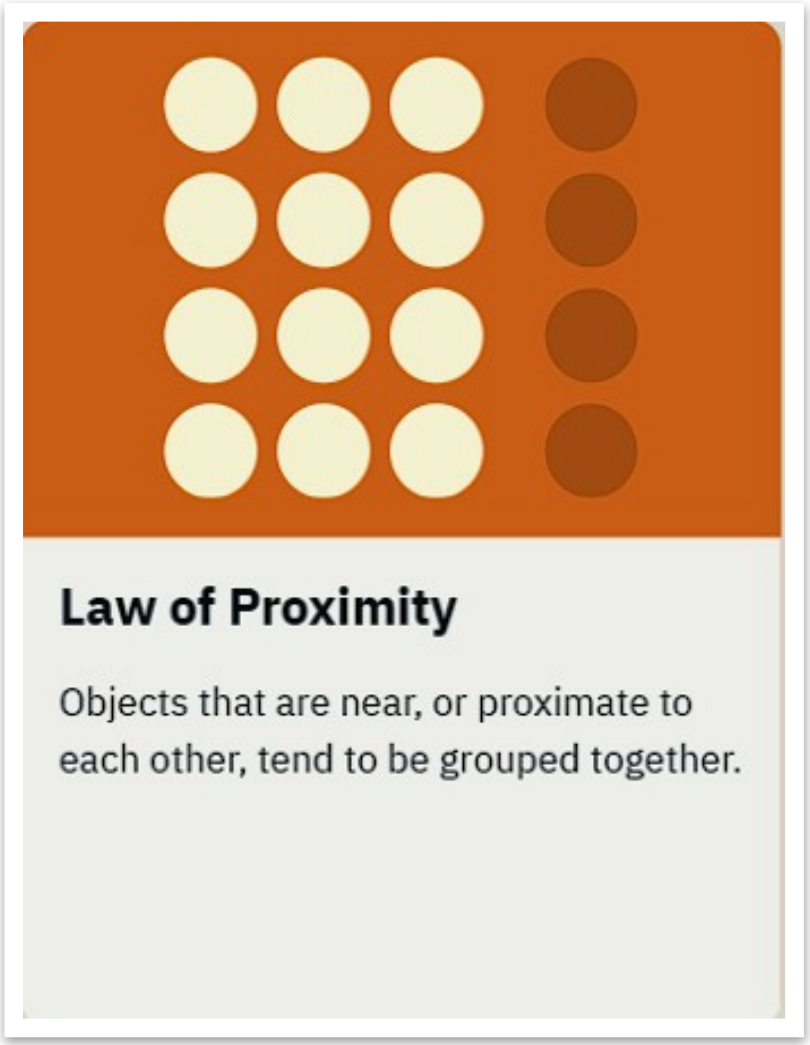
Posts are stacked close together, yet the card edges tell you which photo, likes and comments belong together.

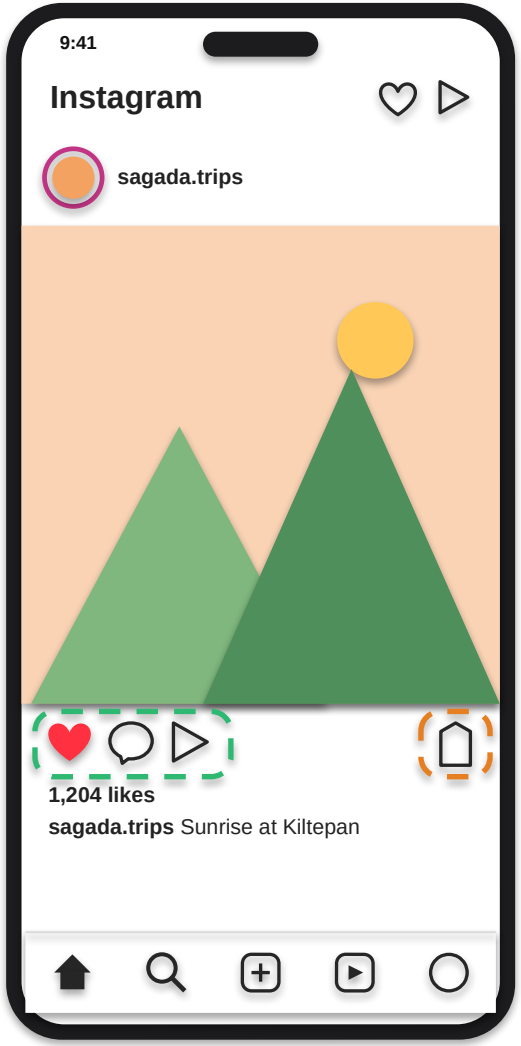
Law of Proximity





Law of Proximity





EVERYDAY EXAMPLE · INSTAGRAM

Spacing does the labeling

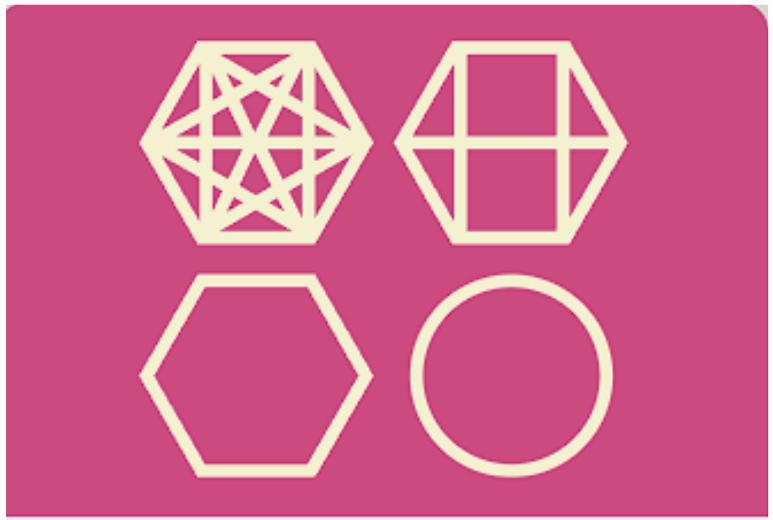
Like, Comment and Share sit together on the left (green): all ways to interact with other people.

Save sits alone on the right (orange) because it is just for you.

No labels needed. The gaps tell you what belongs together.

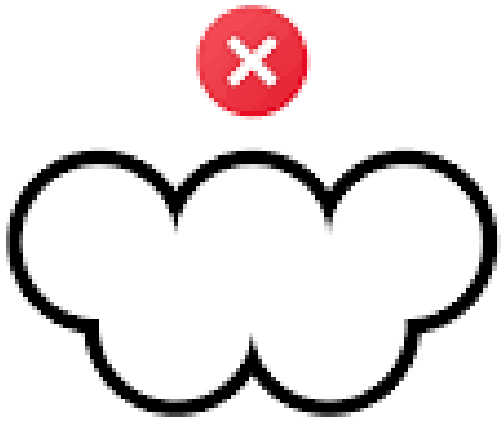
Simplified illustration of a familiar app screen, not an actual screenshot.

Law of Prägnanz



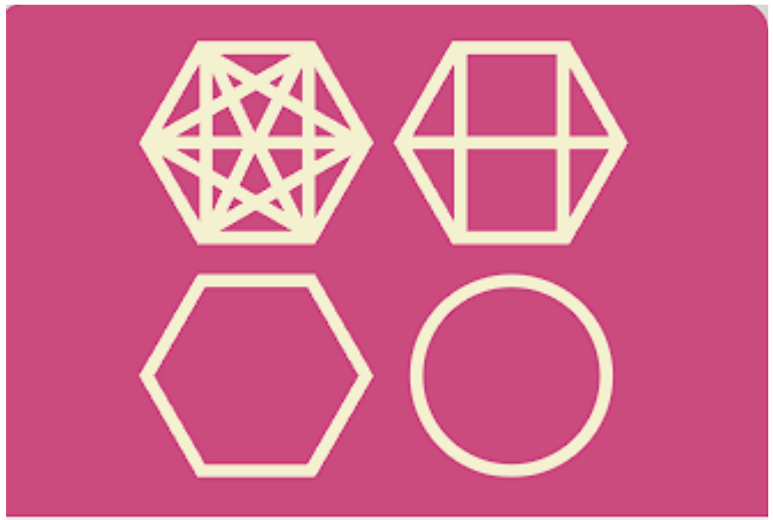
Law of Prägnanz

People will perceive and interpret ambiguous or complex images as the simplest form possible, because it is the interpretation that requires the least cognitive effort of us.



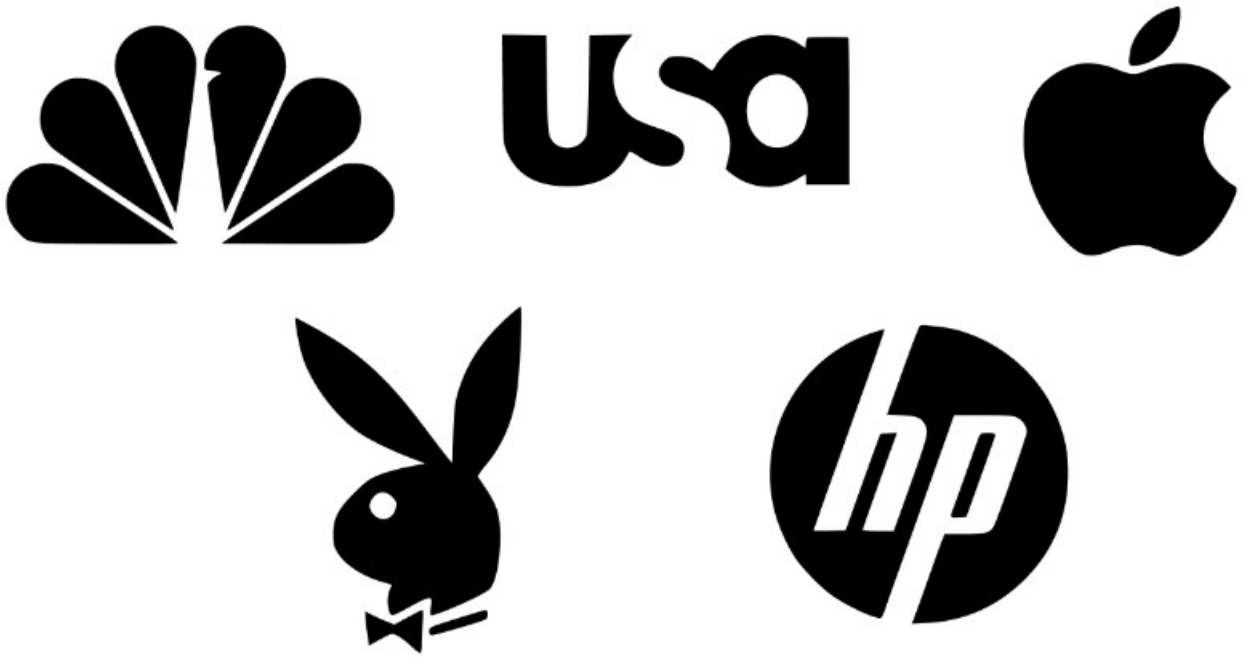


Law of Prägnanz



Law of Prägnanz

People will perceive and interpret ambiguous or complex images as the simplest form possible, because it is the interpretation that requires the least cognitive effort of us.



Prägnanz and Related Gestalt Principles

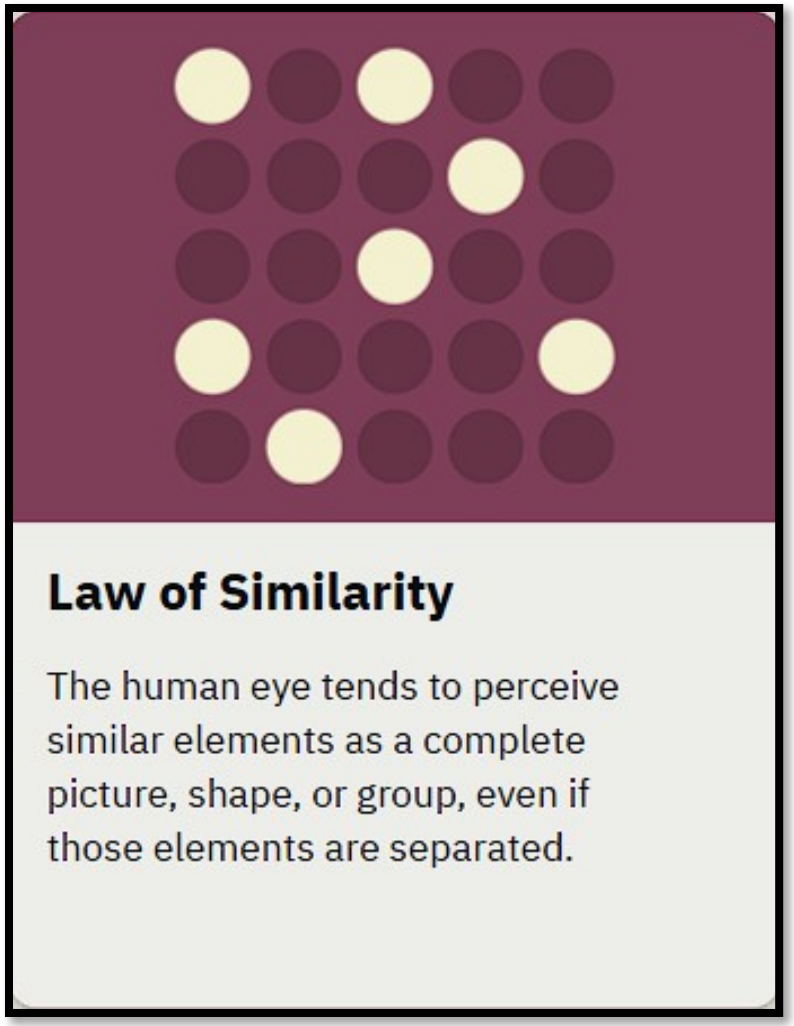
- **Closure:** The mind fills in the blanks in shapes, making a whole even when some components are absent. This is seen in design when a part of an image is left up to the viewer's meaning in logos or icons.
- **Symmetry:** People tend to view symmetrical forms as more harmonious and stable. Symmetry is a tool that UX designers frequently use to balance layouts and appeal to users' sense of order.
- **Continuity:** Objects arranged in a flowing or continuous line are thought to be more related. This idea is applied when creating user flows or navigation menus to guarantee a seamless combined experience.
- **Similarity:** We group elements that look alike. This principle highlights related elements' similarities and sets them apart from other elements in UX design, helping to create a visual hierarchy.

Law of Prägnanz: Key Ideas

Key Aspects and Applications:

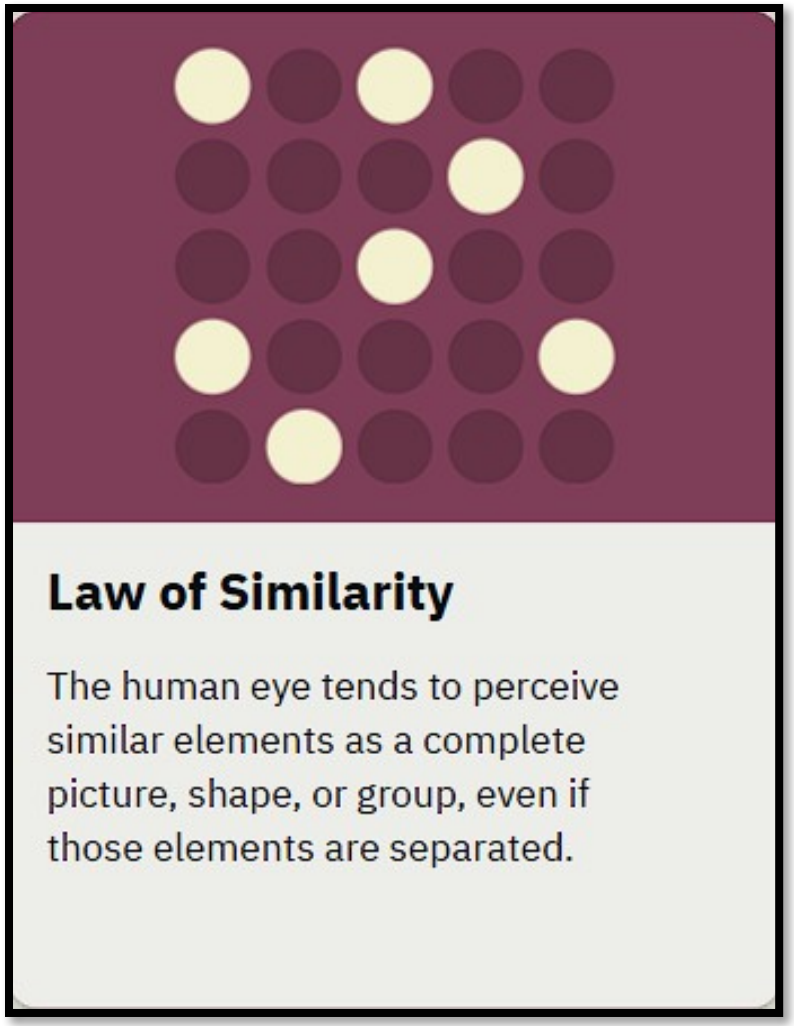
- **Fundamental Principle:** It serves as a foundation for other Gestalt laws, including closure, similarity, and proximity.
- **Definition:** "Prägnanz" is German for "pithiness" or "conciseness," implying that the mind organizes visual input into "good figures" (e.g., shapes, patterns) that are easy to interpret.
- **UX/UI Design:** Designers use this to create minimalist, intuitive interfaces (e.g., Google’s homepage is little more than a logo and a search bar), reducing user confusion, and improving memorability.
- **Examples:**
    - **Logos:** Nike's swoosh or Apple's logo are instantly recognized because they are simple, organized shapes.
    - **Visuals:** Perceiving five separate Olympic rings as a unified, simple symbol rather than a complex arrangement of lines.
- **Core Benefit:** It minimizes the mental effort required to interpret visual information.

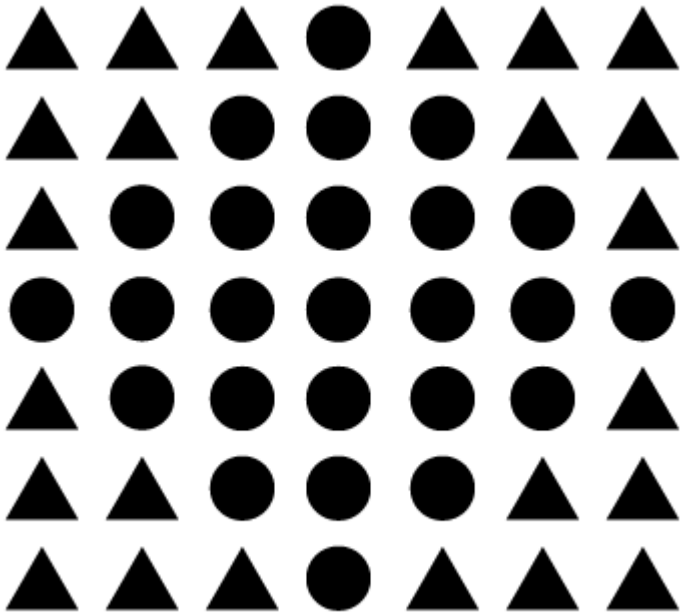
Law of Similarity



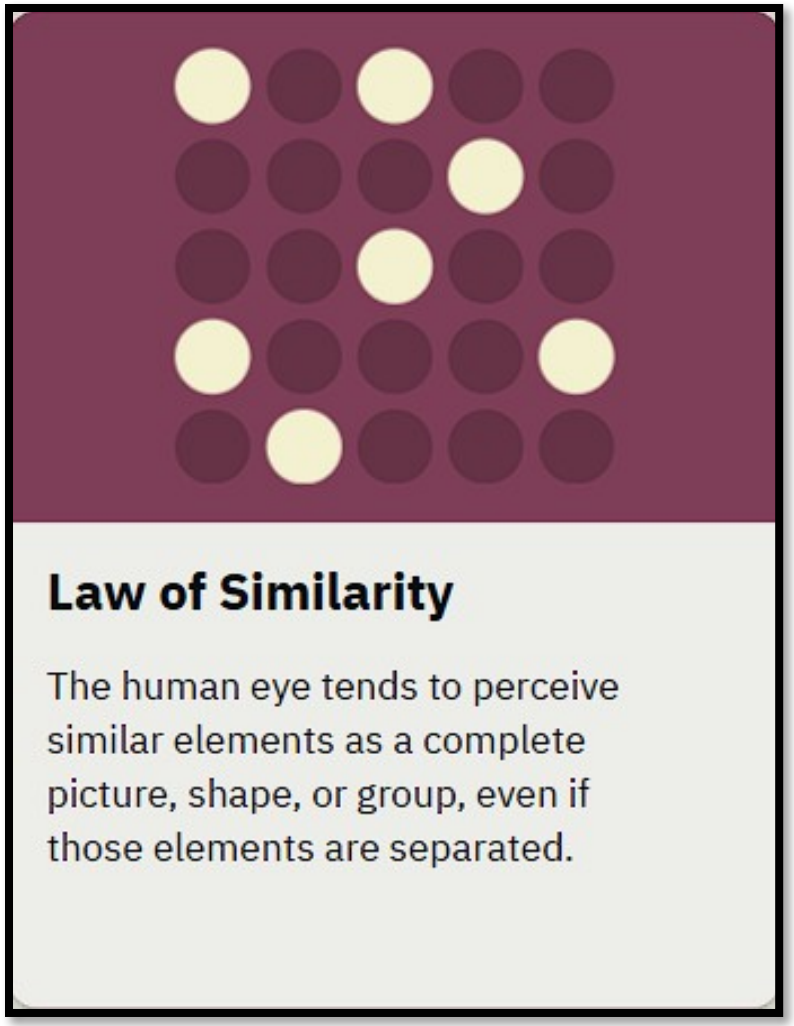


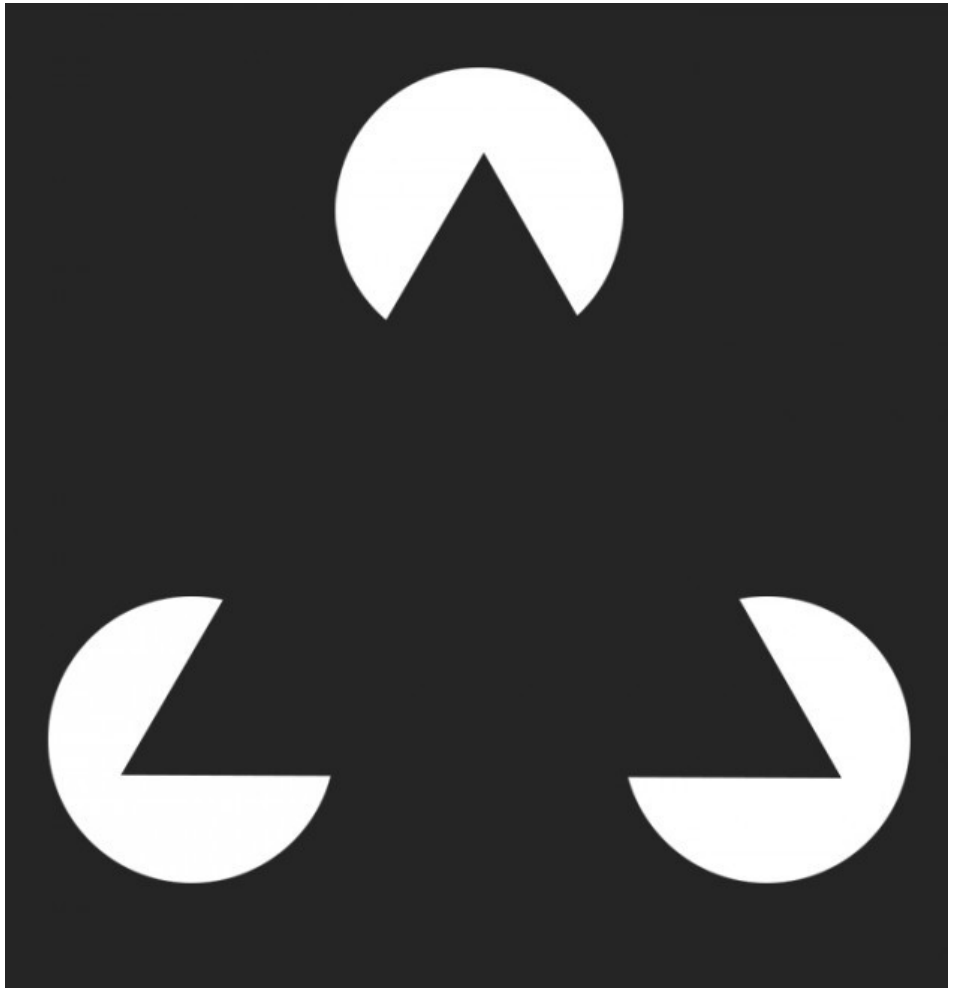
Law of Similarity



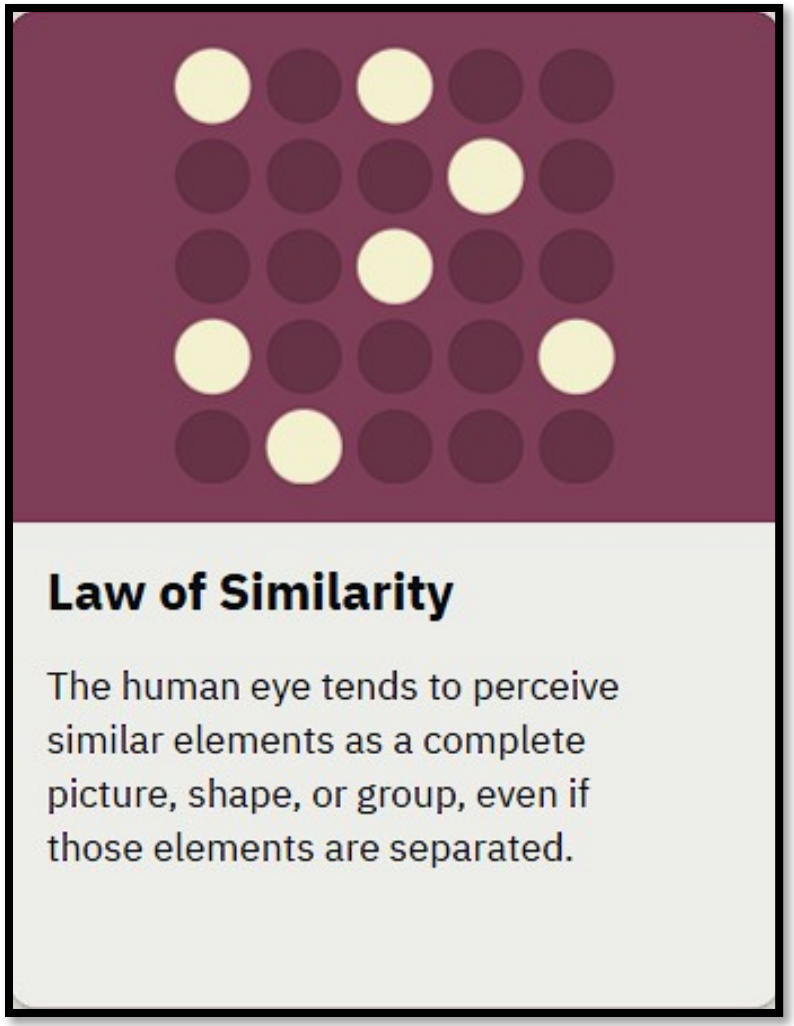


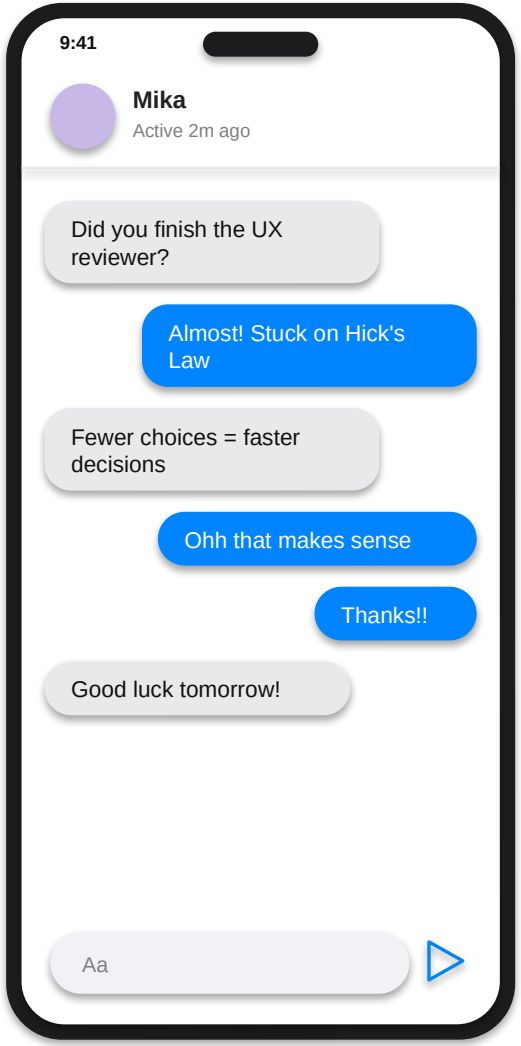
Law of Similarity





Law of Similarity





Simplified illustration of a familiar app screen, not an actual screenshot.

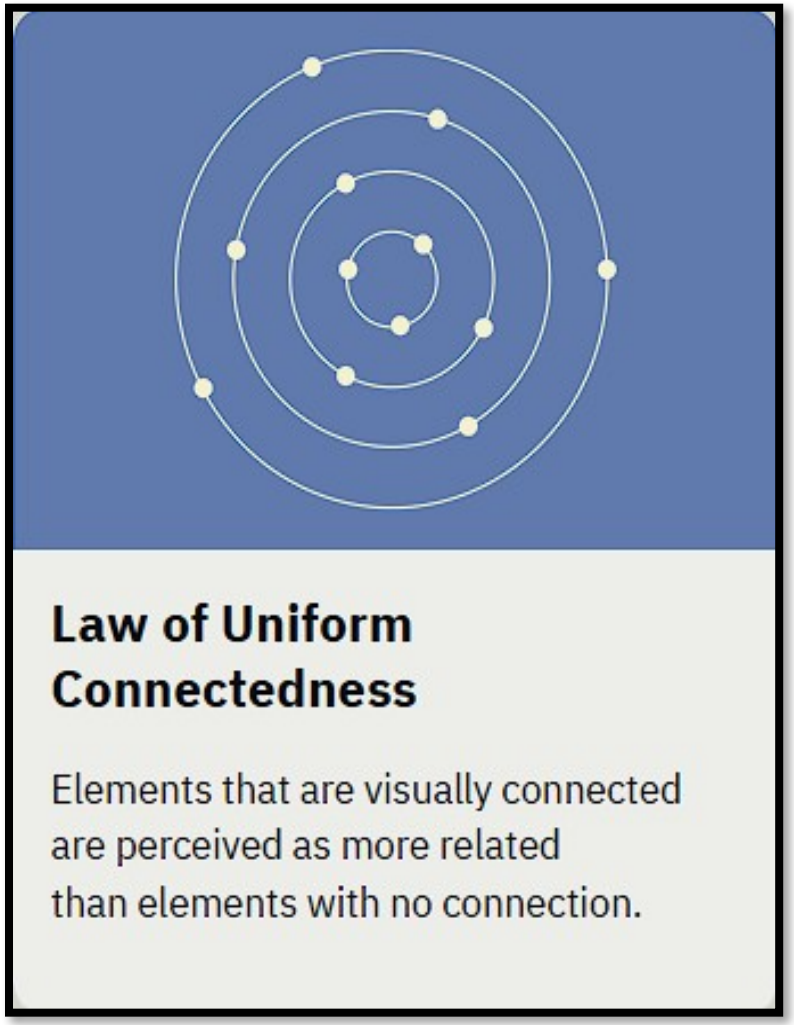
EVERYDAY EXAMPLE · MESSENGER

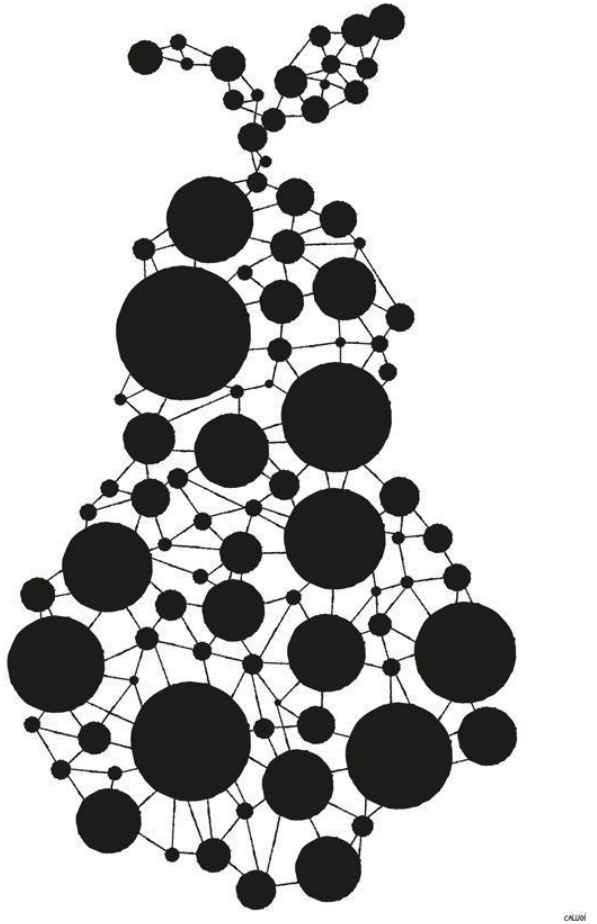
Blue = you, grey = them

You never read names in a chat. The color and side of each bubble tell you who said what.

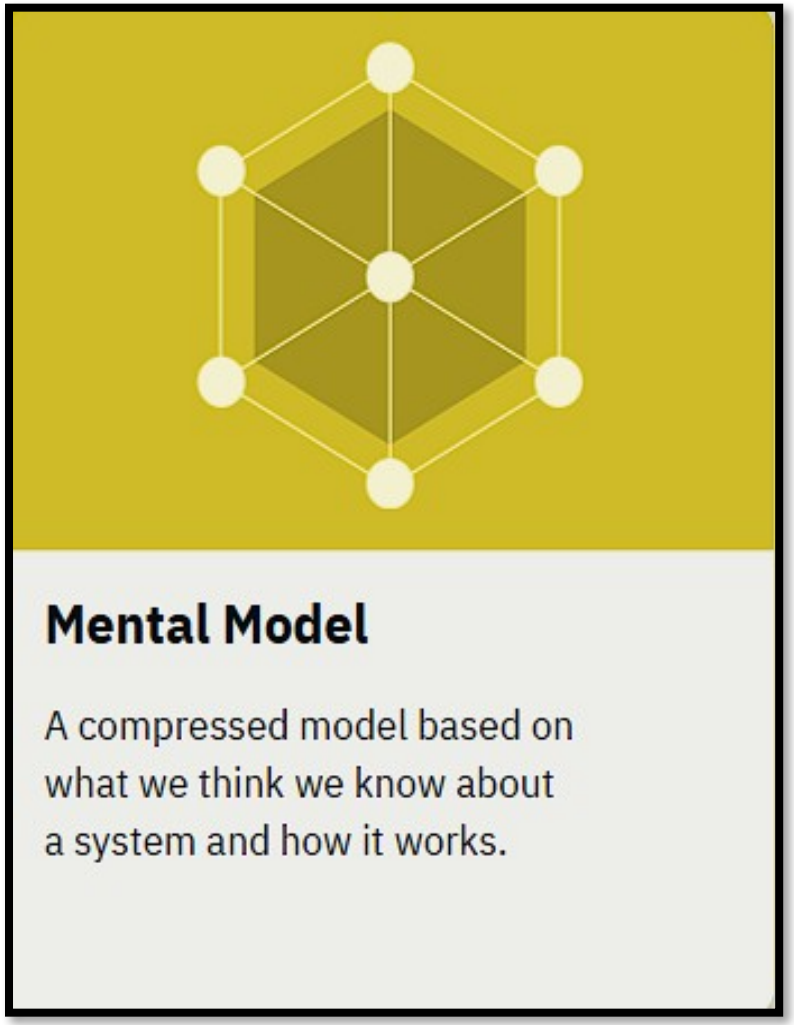
Things that look alike are seen as one group, even when they are far apart.

Law of Uniform Connectedness



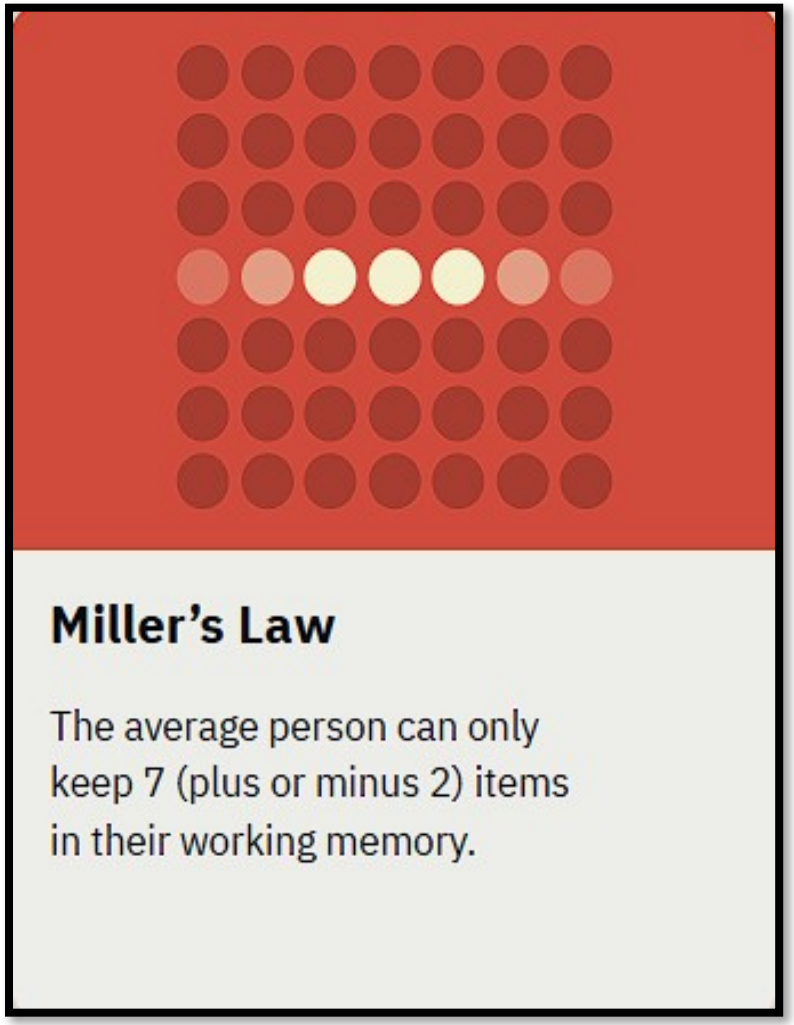


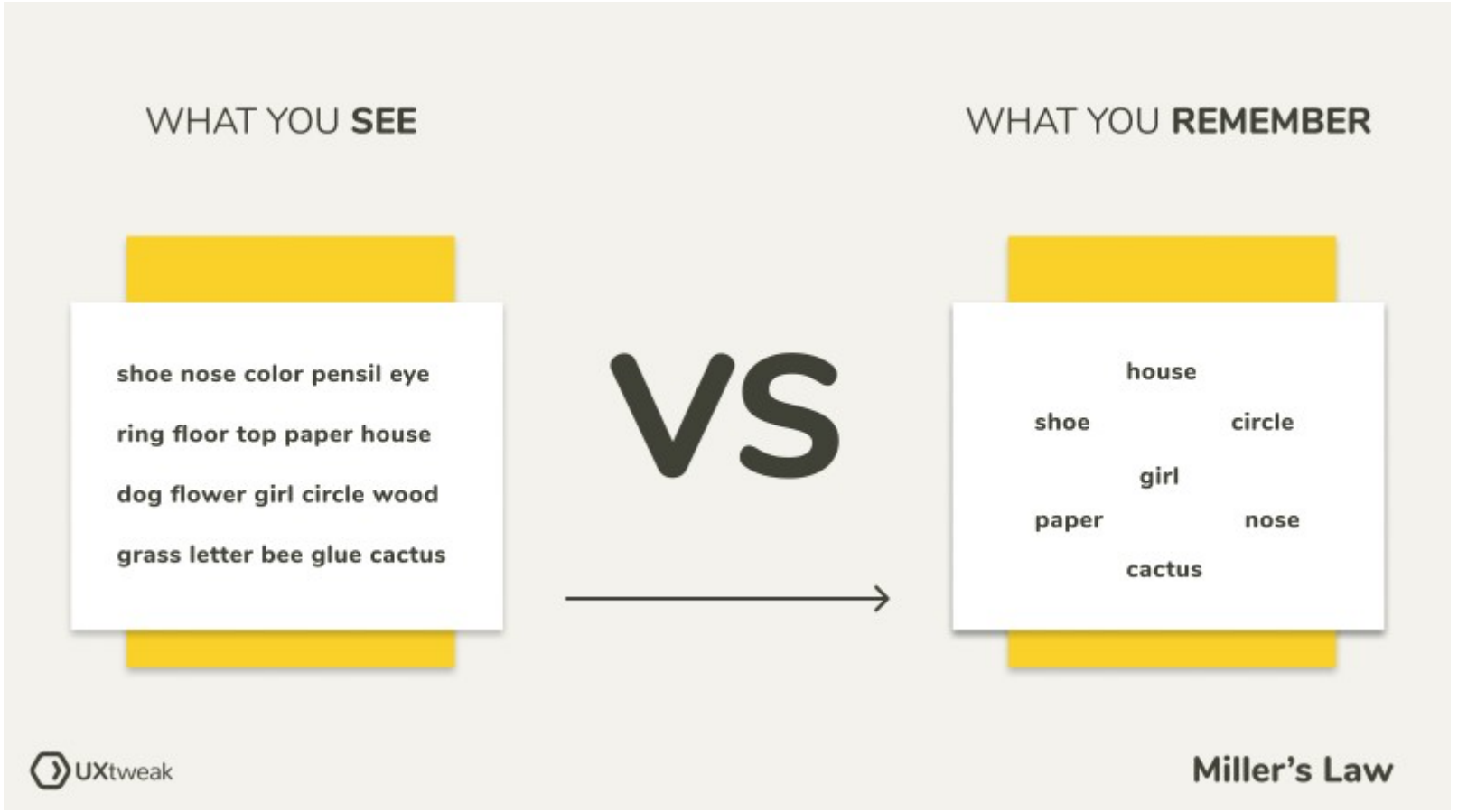
Mental Model



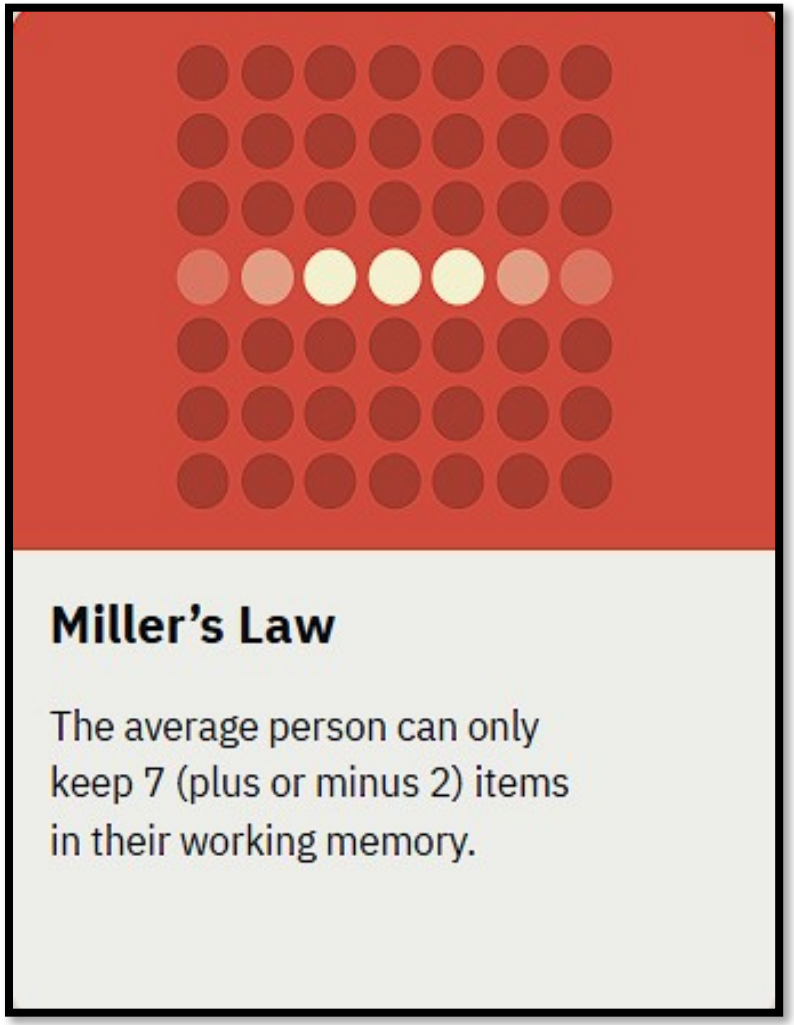


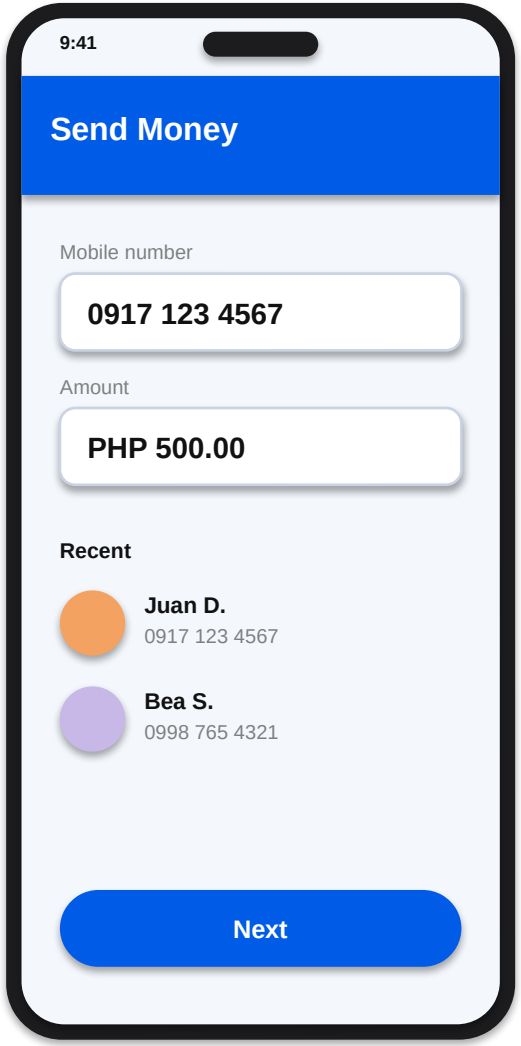
Miller’s Law


Miller’s Law
The average person can only keep 7 (plus or minus 2) items in their working memory.


WHAT YOU SEE
shoe nose color pensil eye
ring floor top paper house
dog flower girl circle wood
grass letter bee glue cactus
VS
WHAT YOU REMEMBER
house
shoe
circle
girl
paper
nose
cactus
UXtweak
Miller’s Law

Miller’s Law





EVERYDAY EXAMPLE · GCASH

Numbers in small chunks

A long number is easier to read, check and remember when it is split into small groups.

Chunking helps people spot a wrong digit before they send money to the wrong person.

Hard to check

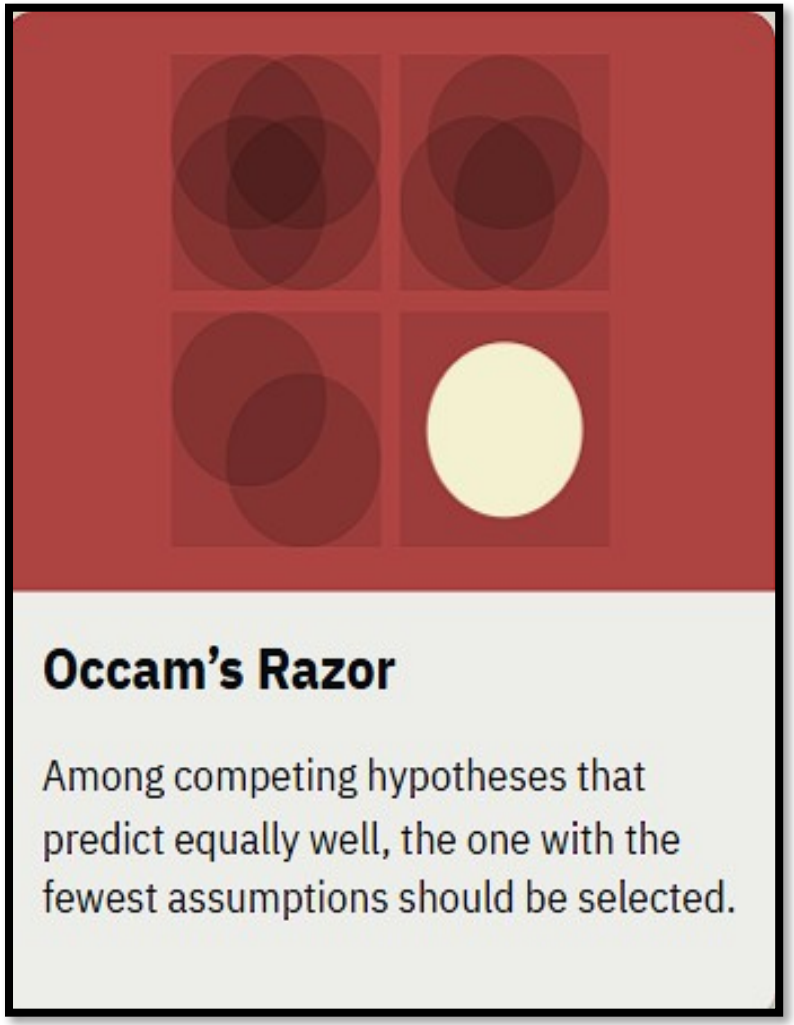
09171234567

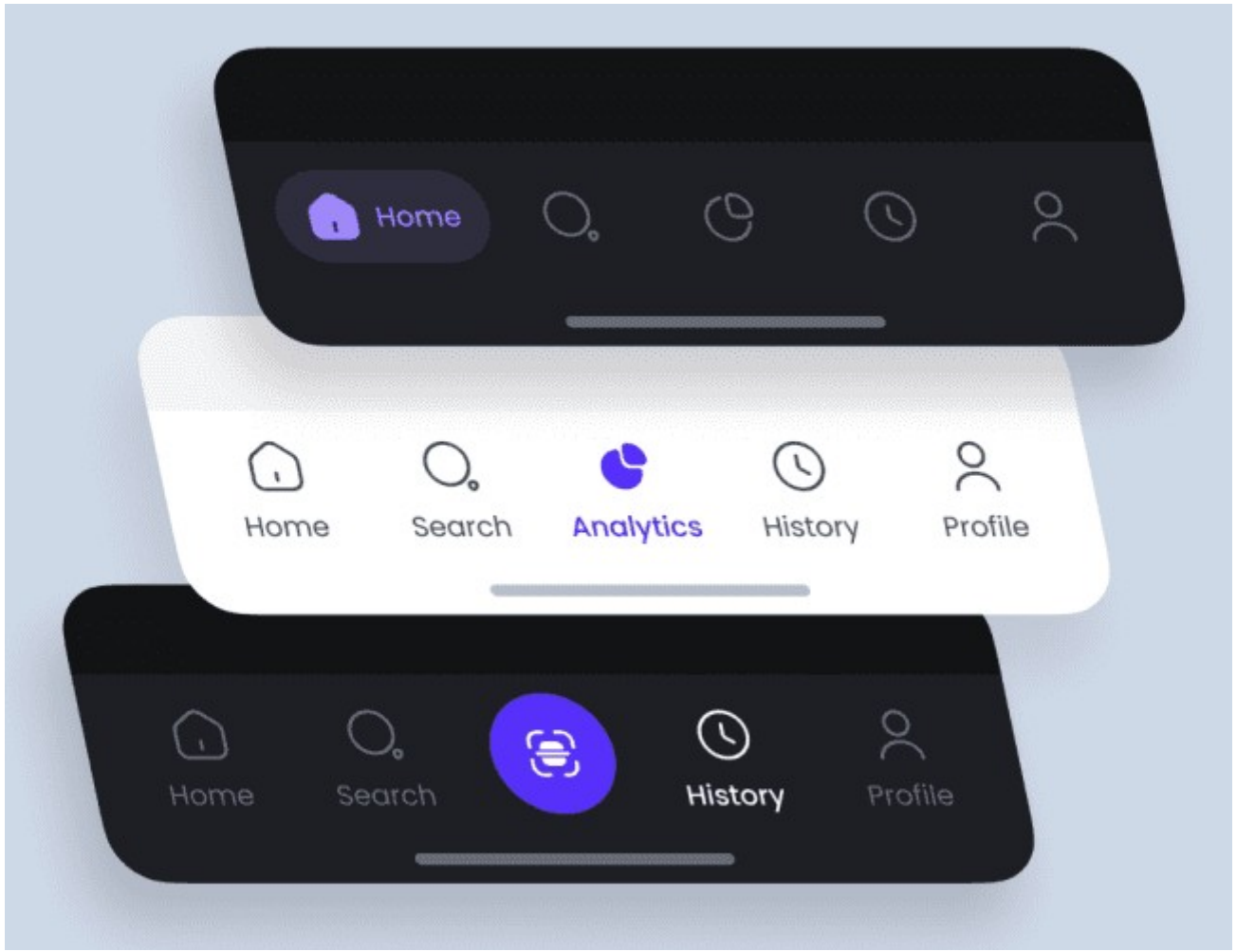
Easy to check

0917 123 4567

Simplified illustration of a familiar app screen, not an actual screenshot.

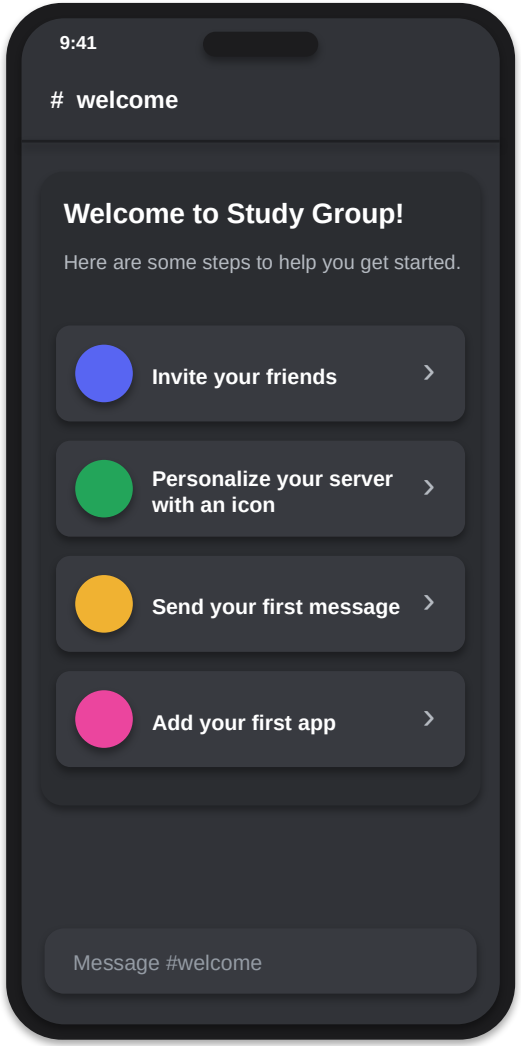
Occam's Razor





Paradox of the Active User





EVERYDAY EXAMPLE · DISCORD

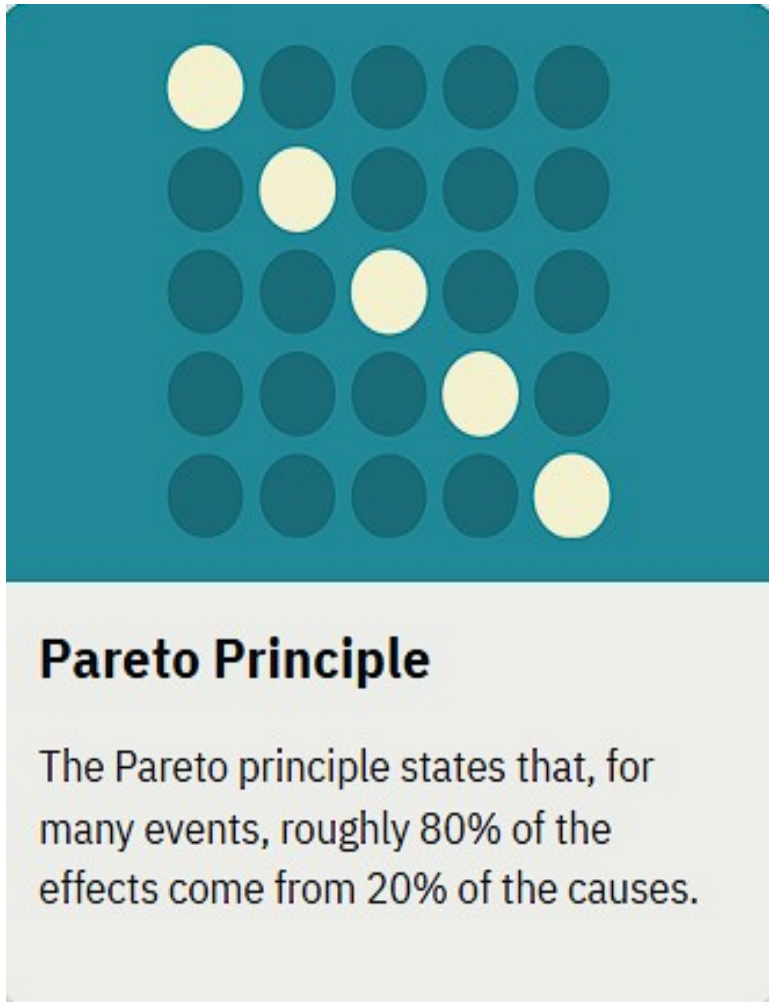
Help right where you are

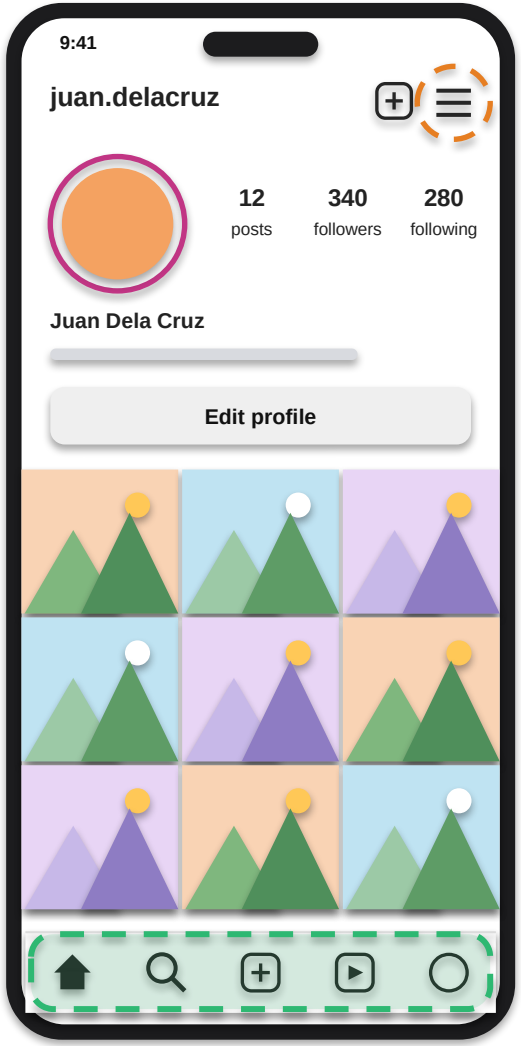
Nobody reads a manual before they start chatting. A new Discord server shows a short checklist inside the server itself.

People learn by doing, with help placed exactly where they need it.

Simplified illustration of a familiar app screen, not an actual screenshot.

Pareto Principle





Simplified illustration of a familiar app screen, not an actual screenshot.

EVERYDAY EXAMPLE · INSTAGRAM

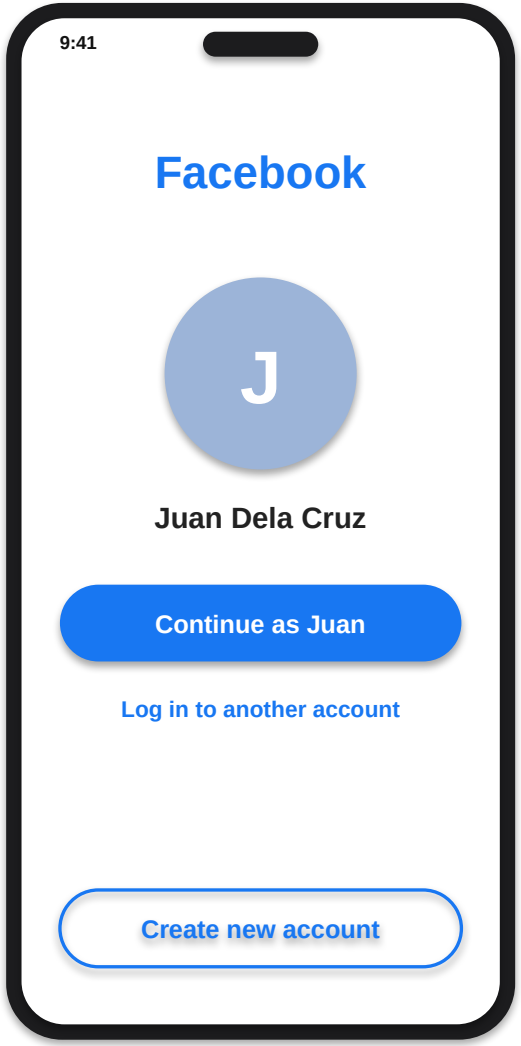
Focus on what most people do

Most time on Instagram goes to a few things: scrolling, searching, posting, Reels and your profile. Those five get the bottom tabs (green).

Dozens of settings matter to fewer people, so they sit behind the menu (orange).

Parkinson’s Law





Simplified illustration of a familiar app screen, not an actual screenshot.

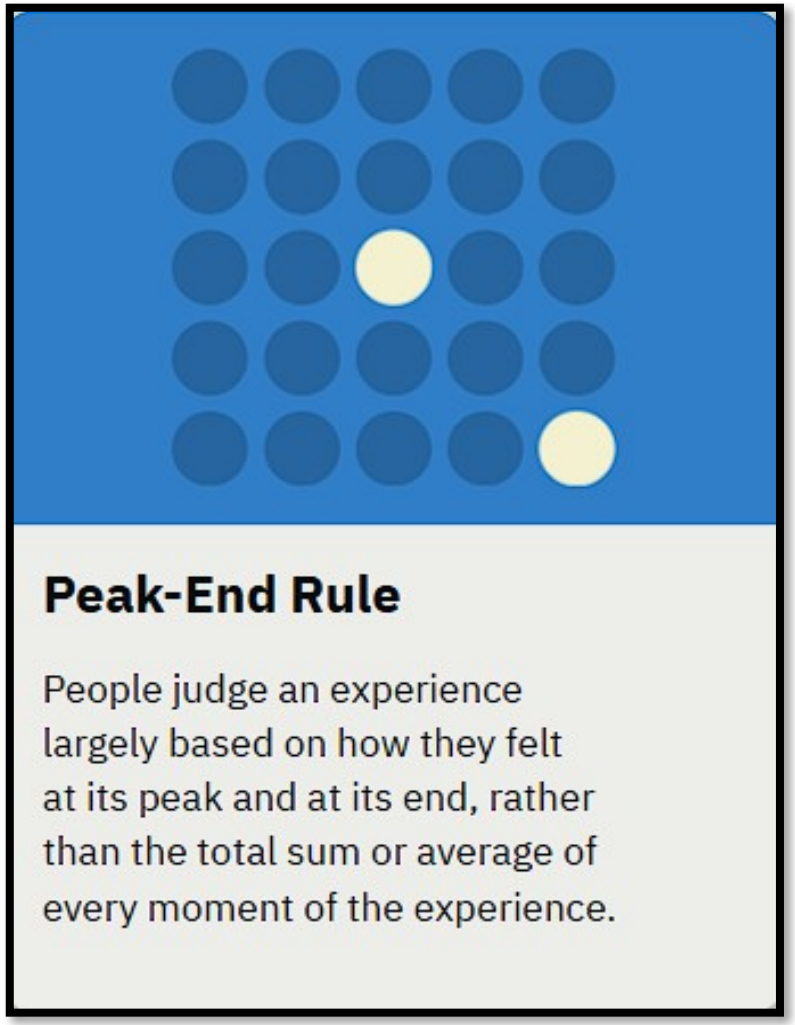
EVERYDAY EXAMPLE · FACEBOOK

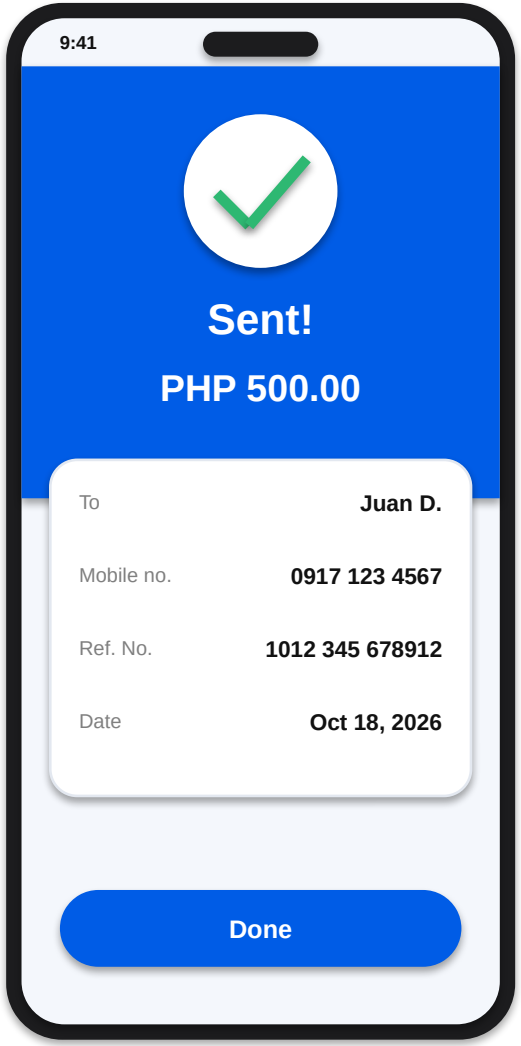
One tap instead of a form

People expect logging in to take a while: an email, a password, maybe a code.

Remembering the account and autofilling details turns it into one tap, so the task never drags on.

Peak-End Rule





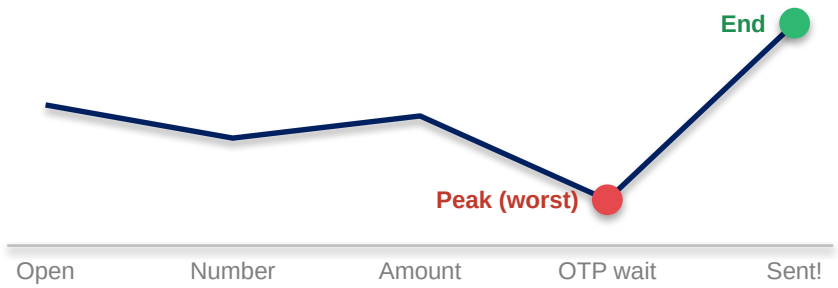
EVERYDAY EXAMPLE · GCASH

End on a high note

People remember a payment by its most intense moment and by how it ended.

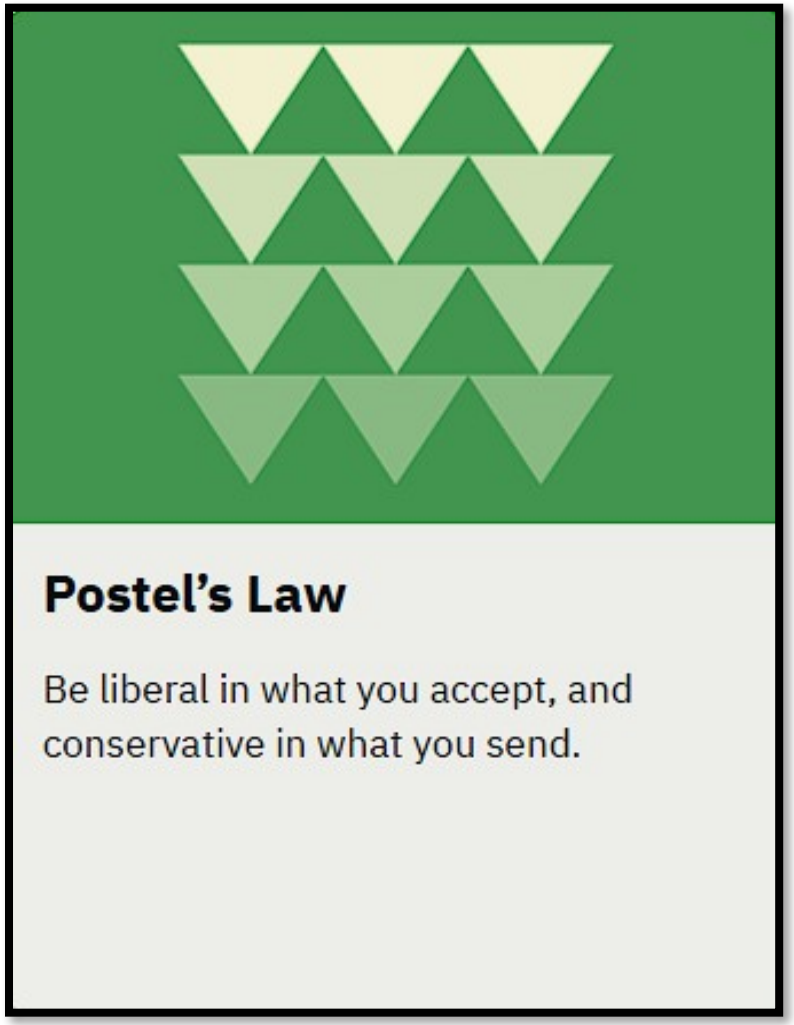
A clear, friendly success screen with the amount, recipient and reference number makes the whole transaction feel smooth.

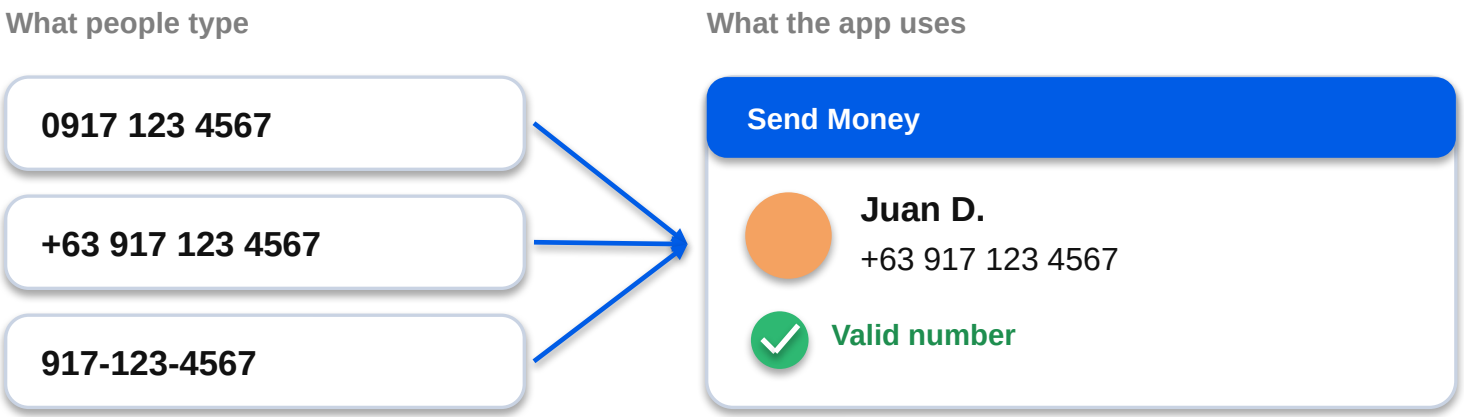
How the payment felt



Simplified illustration of a familiar app screen, not an actual screenshot.

Postel’s Law





EVERYDAY EXAMPLE · GCASH-STYLE SEND FORM

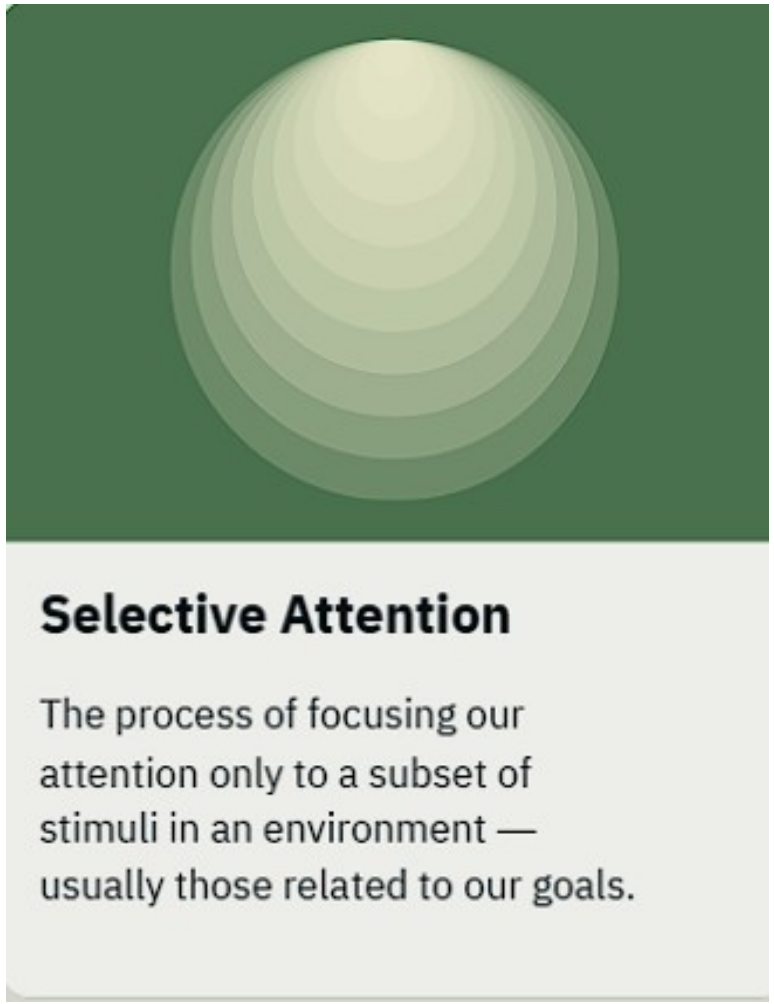
Accept any format

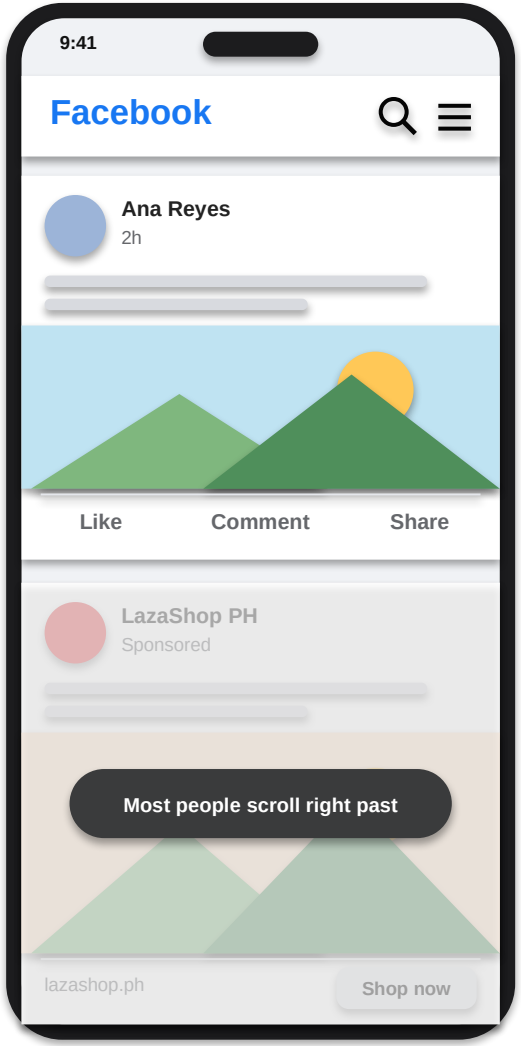
People type mobile numbers in many ways. A forgiving form accepts all of them and quietly converts them to one standard format.

Be liberal in what you accept: show “Invalid number” only when the number truly can’t be understood.

Simplified illustration, not an actual app screenshot.

Selective Attention





Simplified illustration of a familiar app screen, not an actual screenshot.

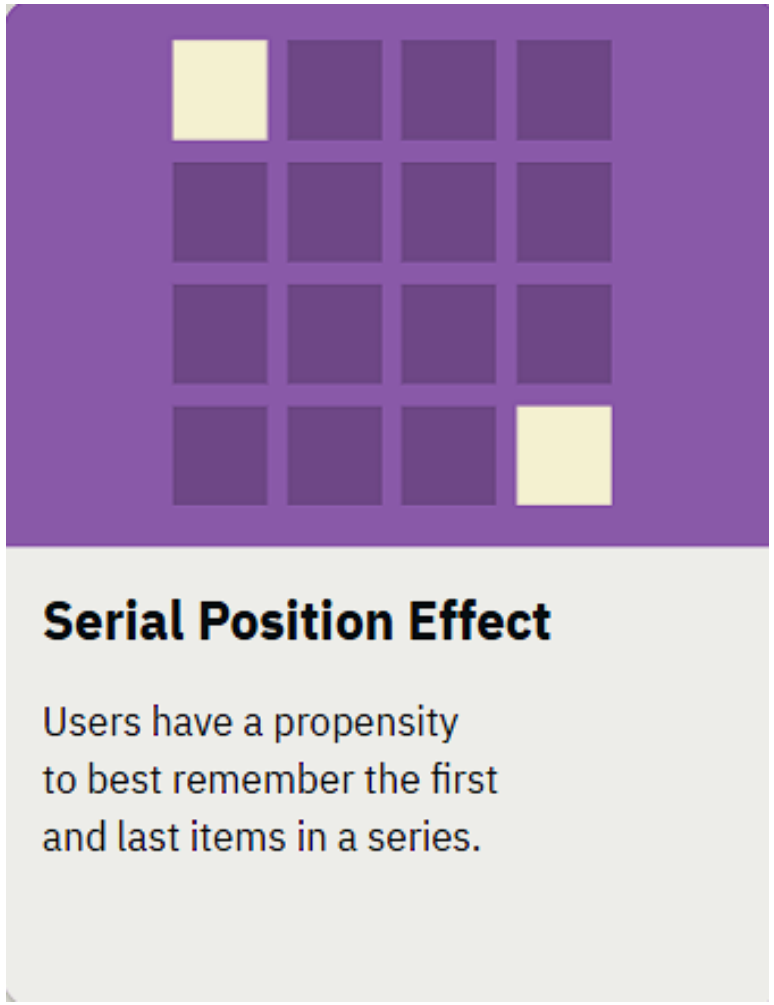
EVERYDAY EXAMPLE · FACEBOOK

Banner blindness

We learn to ignore anything that looks like an ad. Posts marked “Sponsored” often get scrolled past without being read.

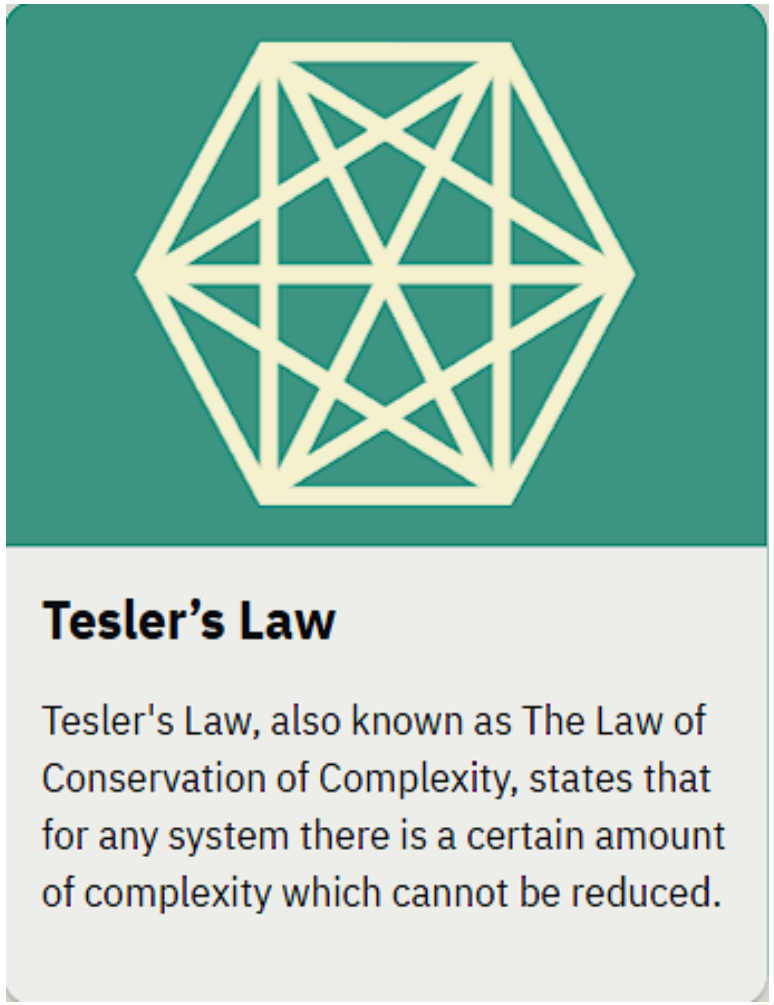
Attention locks onto what serves our goal (friends’ posts) and filters out the rest. Never make important information look like an ad.

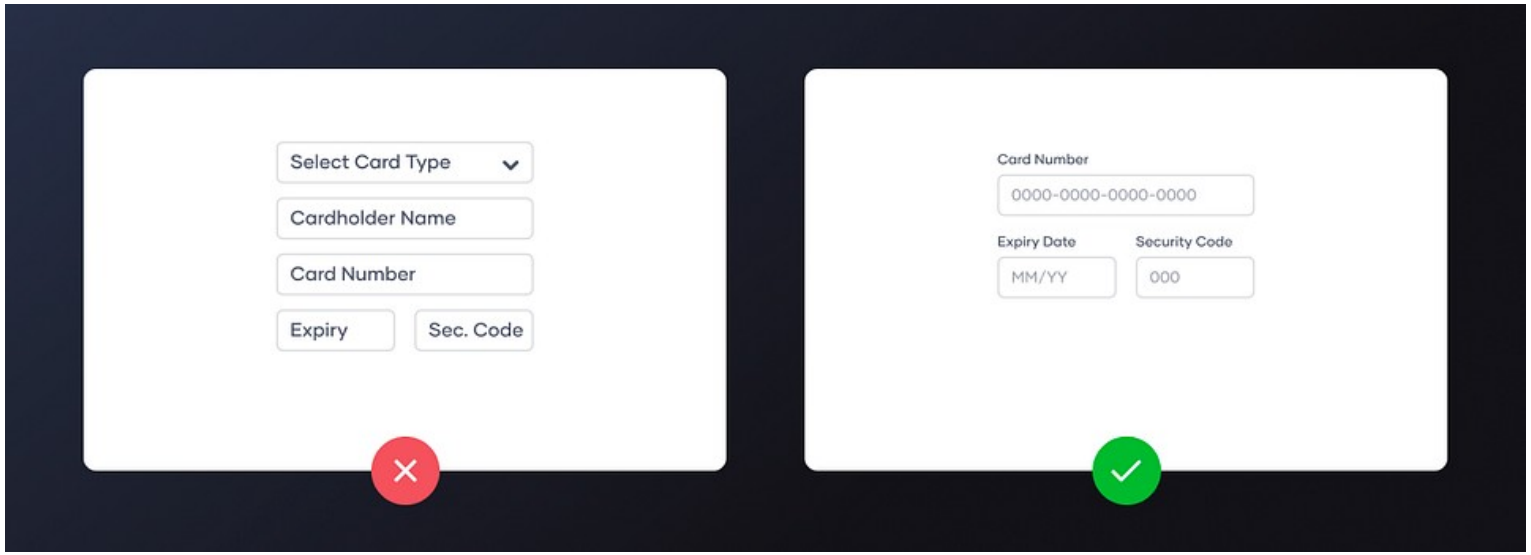
Serial Position Effect



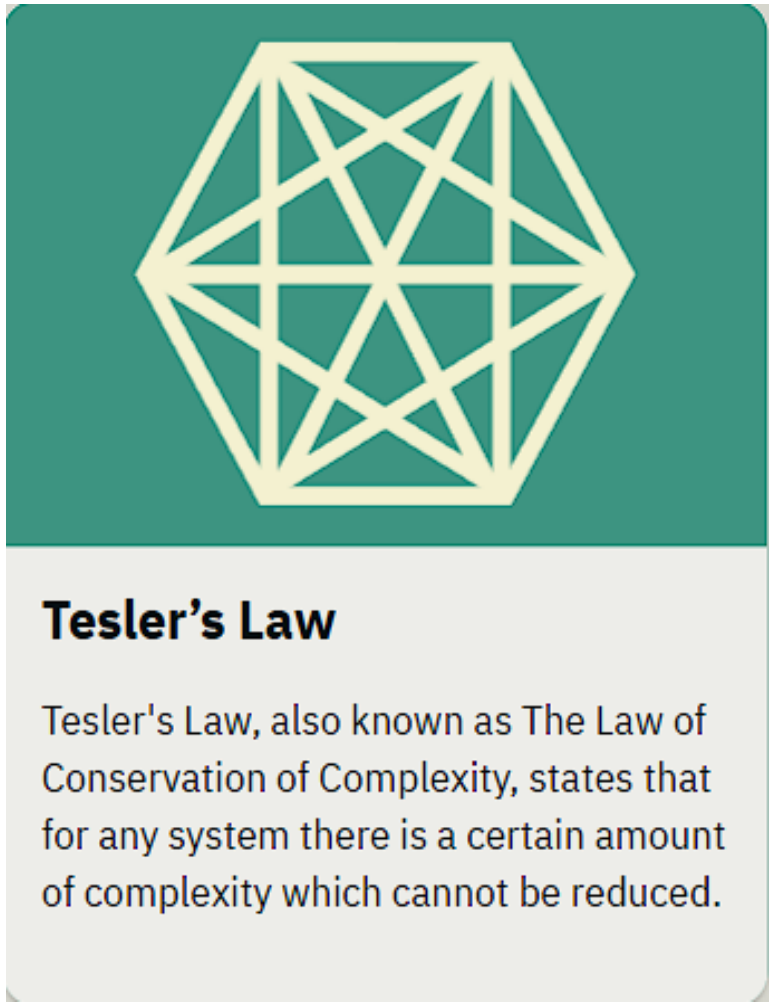
Memorize this list of items.
Apple
Churro
Racket
Lawn mover
Basket
Pen
Flask

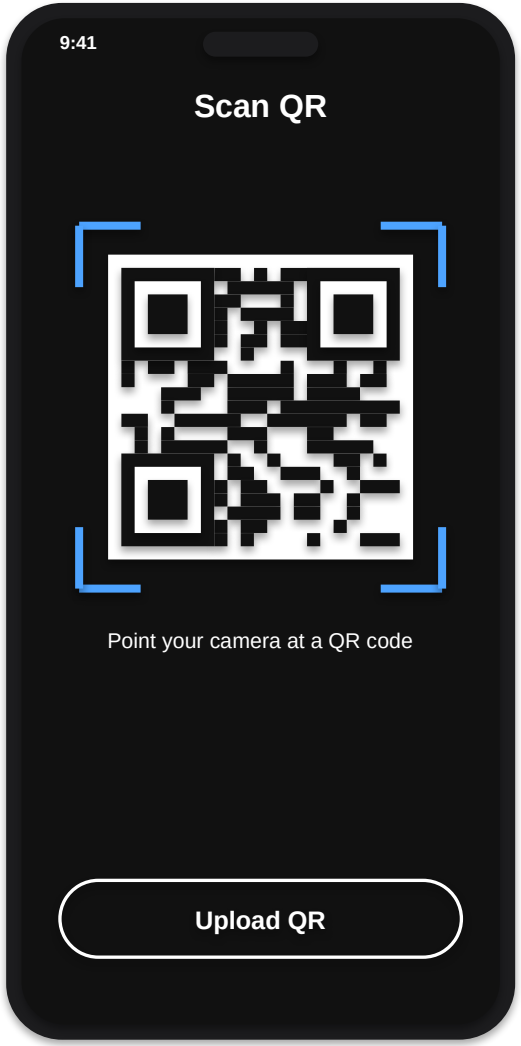
Tesler’s Law


Tesler’s Law
Tesler's Law, also known as The Law of Conservation of Complexity, states that for any system there is a certain amount of complexity which cannot be reduced.


Select Card Type
Cardholder Name
Card Number
Expiry
Sec. Code
Card Number
0000-0000-0000-0000
Expiry Date
Security Code
MM/YY
000

Tesler’s Law





EVERYDAY EXAMPLE · GCASH

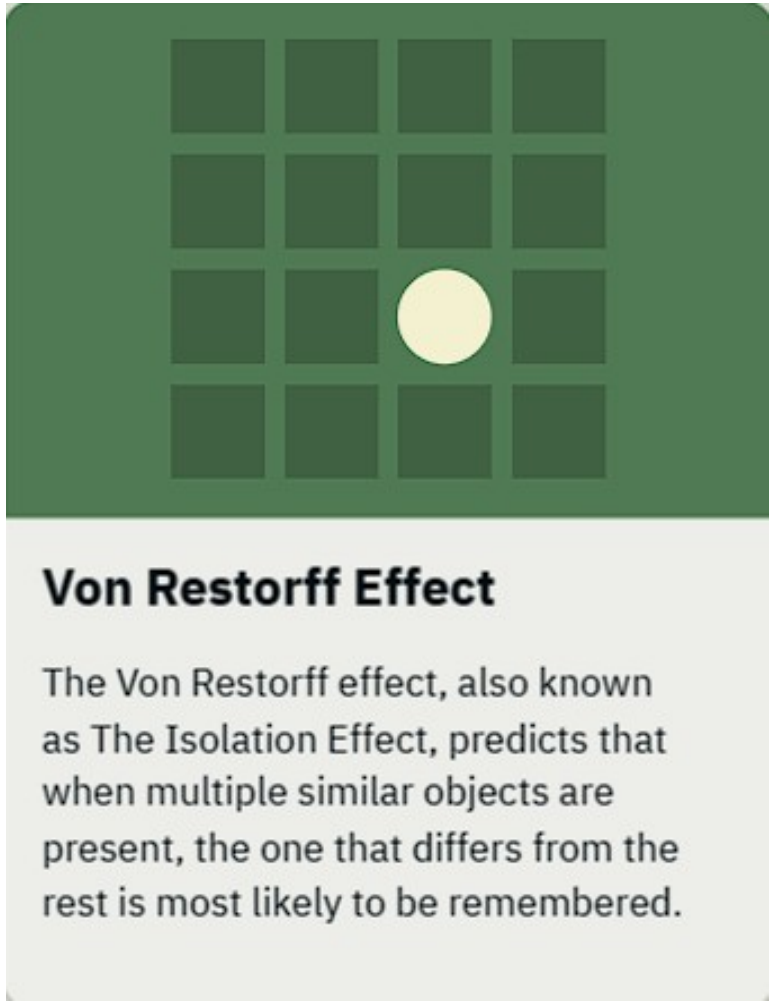
The app takes on the hard part

Paying a store by typing its account number, bank and exact name is slow and easy to get wrong.

A QR code still carries all of that, but the app reads it for you. The complexity did not vanish; it moved from the user to the system.

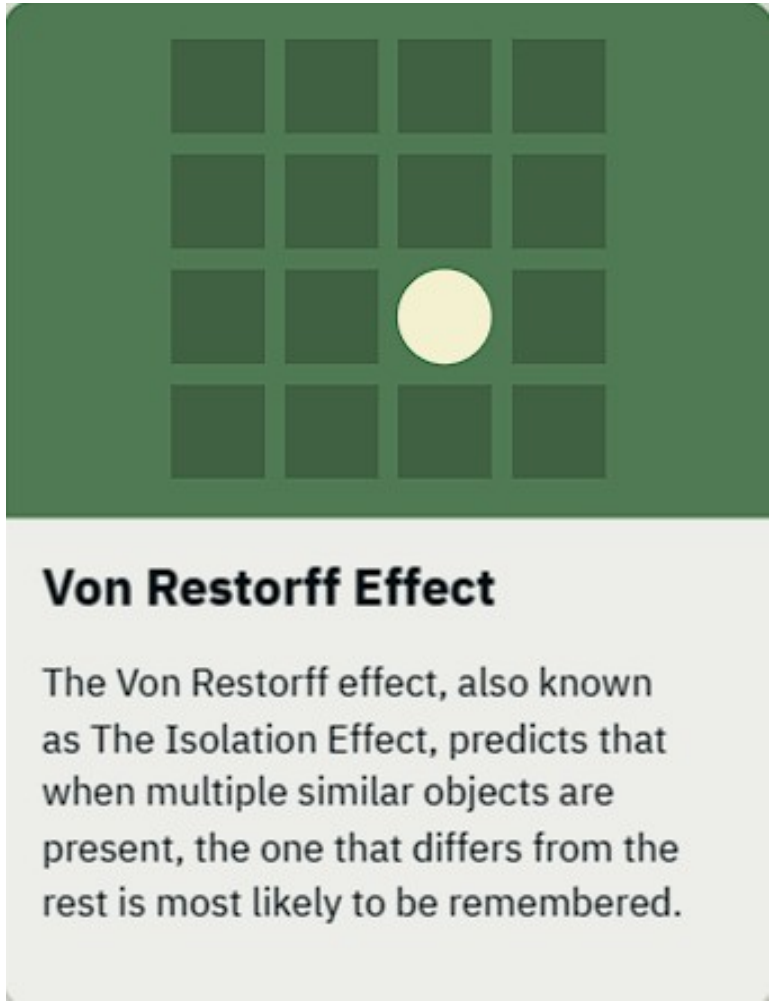
Simplified illustration of a familiar app screen, not an actual screenshot.

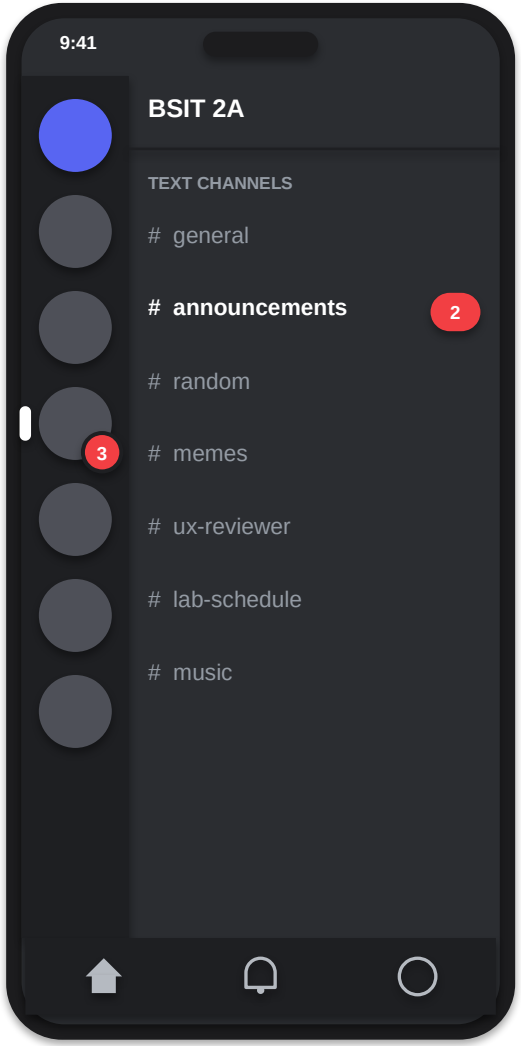
Von Restorff Effect





Von Restorff Effect





Simplified illustration of a familiar app screen, not an actual screenshot.

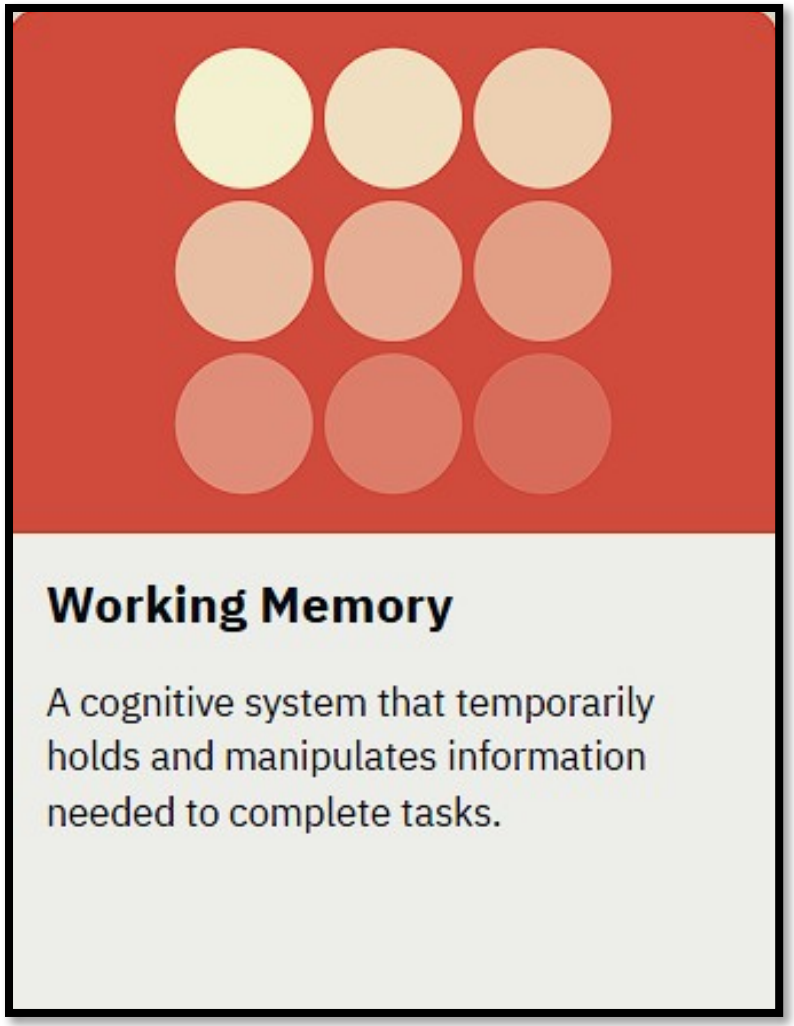
EVERYDAY EXAMPLE · DISCORD

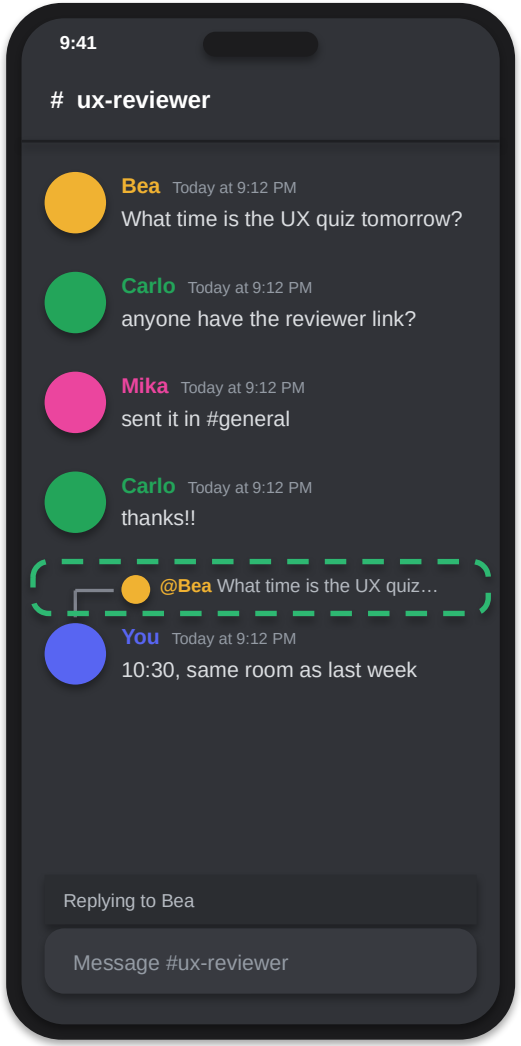
The red badge stands out

In a column of grey servers and channels, the red badge is what your eye goes to first, and what you remember.

It works because Discord uses red sparingly. If everything were red, nothing would stand out.

Working Memory





Simplified illustration of a familiar app screen, not an actual screenshot.

EVERYDAY EXAMPLE · DISCORD

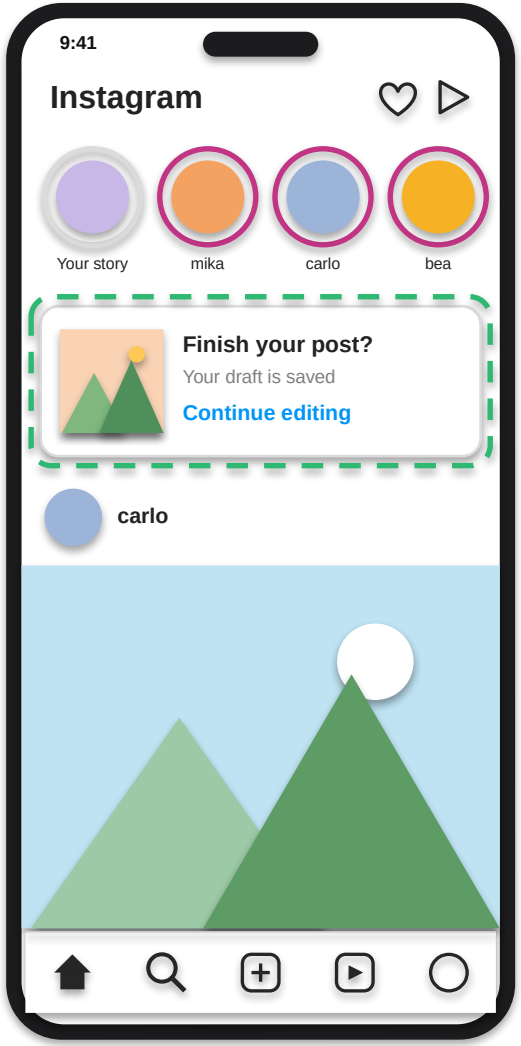
Let the screen remember

By the time you reply, the question may have scrolled away. Discord shows it right above your answer (green).

Working memory is small and fades in seconds, so show people what they need instead of making them recall it.

Zeigarnik Effect





Simplified illustration of a familiar app screen, not an actual screenshot.

EVERYDAY EXAMPLE · INSTAGRAM

Unfinished tasks stick with us

Leave a post halfway and Instagram keeps it as a draft, waiting for you.

An unfinished task nags at us more than a finished one, so people come back to complete it.

SUMMARY

Law	Definition	Design Implication	Concrete Example (UI/UX)
Aesthetic-Usability Effect	Users perceive attractive interfaces as easier to use.	Make design visually appealing to build trust.	Apple’s product pages: clean, beautiful layouts that feel “usable” even before interaction.
Doherty Threshold	Productivity improves when system response time is under 400ms.	Keep interactions snappy; show feedback if delay.	Google search autocomplete updates instantly as you type.
Fitts’s Law	The time to reach a target depends on size and distance.	Bigger, closer targets = easier interaction.	Large “Send” button on mobile at thumb’s reach.

SUMMARY

Law	Definition	Design Implication	Concrete Example (UI/UX)
Goal-Gradient Effect	Users work harder to complete a goal as they get closer.	Use progress bars or completion trackers.	Duolingo shows % progress toward daily streak goal.
Hick's Law	Decision time increases with number/complexity of choices.	Reduce visible options; use categories.	Netflix shows top categories ("Trending Now") instead of dumping all movies.
Jakob's Law	Users expect your design to work like familiar ones.	Follow common conventions.	Shopping cart icon universally used for e-commerce.

SUMMARY

Law	Definition	Design Implication	Concrete Example (UI/UX)
Law of Common Region	Elements within a boundary are perceived as related.	Use cards, boxes, containers for grouping.	Google Drive groups files within shaded cards.
Law of Prägnanz (Simplicity)	People perceive complex images in the simplest form possible.	Use simple, clear visuals; avoid clutter.	Logo designs like WWF panda — minimal yet recognizable.
Law of Proximity	Objects close together are seen as related.	Place related buttons and labels near each other.	Form fields grouped with their labels right above them.

SUMMARY

Law	Definition	Design Implication	Concrete Example (UI/UX)
Law of Similarity	Similar-looking items are perceived as related.	Use consistent styles for categories.	Spotify playlists use same card style, different images.
Law of Uniform Connectedness	Elements connected by lines/visuals are related.	Use connectors to guide relationships.	Flowcharts in UI builders (Figma auto-connect lines).
Miller's Law	People can hold ~7±2 items in working memory.	Don't overload with options; chunk info.	Menu with 5–7 items max on navigation bar.

SUMMARY

Law	Definition	Design Implication	Example (UI/UX)
Occam's Razor	Simpler solutions are preferable to complex ones.	Don't over-design features.	Google homepage; one search box, nothing else.
Pareto Principle (80/20 Rule)	80% of effects come from 20% of causes.	Focus on key user tasks.	Instagram emphasizes posting & scrolling, not settings.
Parkinson's Law	Work expands to fill time available.	Give realistic deadlines/time limits.	Time-limited promo banners on e-commerce ("Ends in 2h").

SUMMARY

Law	Definition	Design Implication	Example (UI/UX)
Peak-End Rule	People judge experiences by peak & final moments.	Make last steps pleasant; reduce checkout friction.	Uber: smooth booking flow + positive “Thanks” screen.
Postel’s Law (Robustness Principle)	Be liberal in what you accept, strict in what you send.	Accept varied user input, return clear output.	Search box allows typos (“iphon” → iPhone).
Serial Position Effect	Users remember first & last items best.	Place key options at start/end.	Amazon puts “Sign In” and “Cart” at edges of nav bar.

SUMMARY

Law	Definition	Design Implication	Example (UI/UX)
Tesler's Law (Conservation of Complexity)	Complexity can't be eliminated, only shifted.	Hide unnecessary complexity from user.	Tax software hides calculations, just asks simple Q&A.
Von Restorff Effect (Isolation Effect)	Unique items stand out & are remembered.	Highlight key actions.	Red "Delete Account" button stands out from grey options.
Zeigarnik Effect	People remember incomplete tasks better.	Show progress to encourage return.	LinkedIn profile "You're 80% complete; add skills!"

SUMMARY

Law	Definition	Design Implication	Example (UI/UX)
Cognitive Load	Too much info strains user processing.	Keep tasks simple, reduce steps.	Auto-fill address forms in checkout.
Chunking	Break down info into smaller units.	Easier comprehension & memory.	Credit card numbers grouped into 4-digit blocks.
Cognitive Bias	Users' judgments are skewed by biases.	Design ethically, avoid manipulative tactics.	Social proof ("1,000+ bought today") affects perception.
Flow	Users enjoy uninterrupted immersion in tasks.	Reduce friction, distractions.	Medium's reading mode hides UI clutter while reading.

Applying UX Laws

- Use these laws to evaluate and improve interfaces.
- Checklist: Reduce choices, group items, ensure visual clarity.

Interactive Activity

- Analyze an app you use every day (e.g., GCash, Instagram, Facebook, Discord) and identify at least 5 UX laws in action.
- Suggest improvements.

Discussion Questions

- Which UX law surprised you most?
- How can these laws be applied to our university portal or e-commerce apps?
- Can aesthetics sometimes compromise usability?

References / Further Reading

- https://lawsofux.com/
- Jon Yablonski, Laws of UX Book
- UX design blogs and research articles